

Operlytics: Development and Evaluation of a Business Analytics and Monitoring System for Improving Operational Decision-Making.

In partial fulfillment for the requirements in Capstone 1 and Research Subject

ARCIDO, JOHN MICHAEL BSIT 3 - 2

NAVARRO, MAY ANN JOY BSIT 3 - 2

PENIERO, SEAN BSIT 3 - 2

SALE, NAZZER SAMUEL BSIT 3 - 2

ZERESO, KURTH JAY BSIT 3 - 2

Business Track

Second Semester

e.g MARCH 2026

TABLE OF CONTENTS

CHAPTER 1.....	4
INTRODUCTION.....	4
1.1 Background of the Study.....	4
1.2 Statement of the Problem.....	13
1.3 Objectives of the Study.....	22
1.4 Scope and Delimitation.....	37
1.5 Significance of the Study.....	47
1.6 Definition of Terms.....	53
1.7 Conceptual Framework.....	56
CHAPTER 2.....	59
REVIEW OF RELATED LITERATURE AND STUDIES.....	59
2.1 Foreign Studies.....	59
2.2 Local Studies.....	65
2.3 Related Literature.....	73
2.4 Synthesis of the Reviewed Literature.....	79
CHAPTER 3.....	85
METHODOLOGY.....	86
3.1 Research Design.....	86
3.2 System Development Methodology.....	89
3.3 System Architecture.....	93
3.4 Tools and Technologies Used.....	97
3.5 Data Gathering Techniques.....	100
3.6 Testing Procedures.....	109
3.7 Evaluation Criteria.....	116

SUPPORTING DIAGRAMS.....	124
Data Flow Diagram (DFD – Level 0).....	124
Entity Relationship Diagram (ERD).....	130
Use Case Diagram.....	134
Chapter 4.....	148
SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION.....	148
4.2 Overview of the Developed System.....	148
4.3 System Development Methodology.....	153
4.4 System Architecture.....	159
4.5 Hardware and Software Requirements.....	163
4.5.1 Hardware Requirements.....	164
4.5.2 Software Requirements.....	165
4.6 Database Design.....	168
Chapter 5.....	174
CONCLUSION AND RECOMMENDATIONS.....	174
5.1 Introduction.....	174
5.2 Summary of Findings.....	174
5.3 Conclusion.....	177
5.4 Recommendations.....	178
5.5 Impact of the study.....	179
5.6 Summary.....	180
References.....	181

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The food service and hospitality industry stands as one of the most dynamic, competitive, and rapidly evolving sectors in the contemporary global economy. This industry encompasses a vast array of business establishments ranging from small family-owned cafés and restaurants to large multinational hotel chains and catering conglomerates. Within this diverse landscape, small and medium-sized enterprises (SMEs) in the food service sector occupy a particularly significant position, contributing substantially to local employment generation, economic growth, and the overall vitality of communities worldwide.

The economic significance of food service SMEs cannot be overstated. In developing and emerging economies, these establishments often serve as the backbone of local commerce, providing not only employment but also essential services that support the broader tourism ecosystem. The food service industry creates a complex web of economic interdependencies that strengthen local supply chains, promote agricultural productivity, and foster entrepreneurship at the grassroots level. For many communities, particularly those in tourist destinations, local cafés and restaurants represent more than mere commercial enterprises; they embody cultural identity, serve as social gathering spaces, and function as ambassadors of local cuisine and hospitality traditions to visitors from around the world.

Despite their substantial economic contribution, many small food service establishments face persistent operational and managerial challenges that significantly hinder their long-term sustainability and growth potential. These challenges are multifaceted and interconnected, stemming from resource limitations, workforce constraints, competitive pressures, and increasingly, the technological gap that separates small enterprises from their larger, more digitally advanced counterparts. The operational difficulties encountered by small cafés and restaurants are particularly acute in environments characterized by high customer turnover,

seasonal demand fluctuations, perishable inventory requirements, and the need for rapid service delivery.

In the Philippines, the food service industry occupies a position of critical importance in both national and local economic frameworks. The archipelagic nature of the country, combined with its rich culinary heritage and the warmth of Filipino hospitality, has created a vibrant and diverse food service landscape that ranges from humble street food vendors to sophisticated fine dining establishments. The Philippine food service sector has demonstrated remarkable resilience and adaptability, weathering economic downturns, natural disasters, and most recently, the unprecedented challenges posed by the COVID-19 pandemic.

Boracay Island, consistently ranked among the top beach destinations in Southeast Asia and indeed the world, presents a particularly compelling case study for examining the operational challenges faced by small food service establishments in tourism-intensive environments. This small island in the province of Aklan, Western Visayas, attracts millions of tourists each year, drawn by its powdery white sand beaches, crystal-clear turquoise waters, and vibrant nightlife. The tourism industry on Boracay Island has experienced remarkable growth over the past two decades, evolving from a relatively unknown backpacker destination to a world-class resort island that welcomes visitors from across the globe.

According to data from the Aklan Provincial Tourism Office, Boracay Island recorded 2,155,217 tourist arrivals in 2025, generating approximately ₱47 billion in tourism-related revenues. The December 2025 holiday season alone saw 227,828 visitors, representing the highest monthly figure recorded for that year. Domestic tourists dominated arrivals at 1,765,318, followed by 357,510 foreign tourists and 32,425 overseas Filipino workers. The top sources of foreign visitors included South Korea (96,199 arrivals), the United States (48,356), Australia (21,357), Taiwan (20,306), Russia (18,952), and China (17,958). These figures underscore the island's status as a premier international destination and the immense economic opportunities and corresponding operational pressures faced by local businesses.

The sheer volume of tourist traffic places considerable demand on local food service establishments to deliver high-quality products and services consistently. Small business owners in this environment must navigate a complex balancing act, striving to maintain customer satisfaction while simultaneously managing efficient operational processes. This challenge is compounded by several factors inherent to the tourism-driven market context. Limited resources, both financial and human, constrain the ability of small cafés to invest in advanced operational systems. Workforce constraints, including seasonal employment patterns and high staff turnover rates, make it difficult to maintain operational consistency. Perhaps most significantly, the absence of formal business management systems and structured information technology infrastructure leaves many establishments reliant on manual processes that are inherently inefficient and error-prone.

The competitive landscape of Boracay Island's food service market adds another layer of complexity to the operational challenges faced by small establishments. The island hosts hundreds of restaurants, cafés, bars, and food stalls, ranging from local eateries serving traditional Filipino cuisine to international franchises offering familiar global brands. In this crowded marketplace, differentiation becomes crucial for survival and success. Small cafés must compete not only on the quality of their food and beverages but also on service efficiency, ambiance, and the overall customer experience. Operational excellence, therefore, becomes a critical competitive advantage, one that is difficult to achieve without adequate systems and tools for managing daily business activities.

One of the most common and pervasive operational challenges faced by small cafés in tourism-intensive environments like Boracay Island is the continued reliance on manual and informal methods for managing day-to-day business activities. Despite the increasing availability and affordability of digital technologies, many establishments persist in using traditional approaches that have remained largely unchanged for decades. Sales transactions are frequently recorded by hand in paper logbooks or simple notebooks, with each staff member maintaining their own individual records. Inventory tracking relies on physical counts conducted at irregular intervals, with stock levels estimated rather than precisely monitored. Financial reporting, when

it occurs at all, takes the form of ad hoc calculations performed on an as-needed basis, often without standardized formats or consistent methodologies.

These manual practices, while deeply ingrained in the operational culture of many small businesses, produce a cascade of negative consequences that undermine operational efficiency and business performance. Human error represents an ever-present risk, with mistakes in arithmetic, data transcription, and record-keeping leading to inaccurate information that distorts management's understanding of business performance. Inconsistency across different staff members' records creates data fragmentation, making it impossible to obtain a unified and accurate view of operational status. The time-consuming nature of manual processes consumes valuable staff hours that could be better directed toward customer service and other value-adding activities.

The absence of integrated information systems contributes to a range of operational deficiencies that are particularly severe in fast-paced service environments. Organizations lacking integrated information systems face significant disadvantages in terms of data accuracy, operational coordination, and strategic decision-making capacity. The critical importance of information systems in enabling organizations to process, analyze, and act upon operational data in a timely and effective manner cannot be overstated. For small food service establishments, these theoretical insights translate into very practical consequences: poor inventory control leading to stockouts or overstocking, undetected losses due to theft or spoilage, inefficient resource allocation that wastes limited capital, and an inability to identify trends or respond proactively to changing market conditions.

The situation described above is clearly and compellingly evident in the case of Empress Café Boracay, a small café operating within the highly competitive and tourism-driven market of Boracay Island. Empress Café represents a typical example of the many small food service establishments that populate tourist destinations throughout the Philippines and similar emerging economies worldwide. Like its peers, the café currently relies on manual and inconsistent methods for managing its core operational functions, including sales monitoring, inventory tracking, and financial reporting. These manual processes, while perhaps adequate for a business

of minimal scale, become increasingly problematic as customer volumes grow and operational complexity increases.

The specific operational deficiencies observed at Empress Café Boracay illustrate the broader challenges faced by small cafés in tourism-intensive environments. Staff members record sales individually without a centralized system, resulting in data fragmentation that makes it impossible to obtain a consolidated view of daily, weekly, or monthly sales performance. Each staff member may use different methods, formats, or levels of detail in their record-keeping, creating inconsistencies that complicate aggregation and analysis. The absence of real-time data visibility means that management cannot monitor sales performance as it occurs, missing opportunities to respond to unexpected demand patterns, promotional opportunities, or operational issues.

Inventory management at Empress Café suffers from similar deficiencies. Stock levels are tracked through paper logbooks or informal notes, with updates occurring sporadically rather than systematically. The disconnect between sales activity and inventory records means that stock movements are not automatically reflected in inventory counts, leading to frequent discrepancies between actual and recorded inventory levels. Management lacks visibility into which items are selling well, which are moving slowly, and which may be approaching expiration or depletion. These gaps have tangible consequences: instances of overstocking tie up limited working capital in excess inventory that may spoil before use; unexpected stockouts lead to customer disappointment and lost sales; unnecessary waste from spoilage or expiration erodes already thin profit margins; and customer service delays resulting from unavailable ingredients damage the café's reputation and competitive position.

The operational challenges at Empress Café Boracay are not unique to this establishment but rather reflect systemic patterns observed across small food service businesses in similar contexts. The café's experience illustrates how the accumulation of seemingly minor operational inefficiencies can compound into significant business problems that threaten long-term viability. In an environment as competitive and demanding as Boracay Island's tourism market, operational excellence is not merely desirable but essential for survival.

The increasing accessibility of information and communications technology (ICT) has opened unprecedented opportunities for small businesses to adopt digital tools that can streamline operations and substantially improve decision-making capabilities. The past decade has witnessed a democratization of business technology, with powerful software tools becoming available at price points accessible even to micro and small enterprises. Cloud computing has eliminated the need for expensive on-premises server infrastructure. Mobile applications have put sophisticated functionality into the hands of business owners and staff through devices they already own. Open-source software has provided alternatives to costly proprietary solutions. Together, these developments have created conditions favorable for the widespread adoption of business technology by small enterprises that previously could not afford such investments.

Business analytics systems have emerged as particularly powerful instruments for transforming raw operational data into actionable insights that can guide management decision-making. These systems go beyond simple data recording to provide analytical capabilities that help business owners understand patterns, identify trends, detect anomalies, and forecast future conditions. Modern business analytics platforms automatically aggregate data from multiple sources, present information through intuitive visual dashboards, and generate comprehensive reports with minimal manual input. The analytical capabilities embedded in these systems enable businesses to move from reactive management responding to problems after they occur to proactive management anticipating challenges and opportunities before they fully materialize.

Research demonstrates convincingly that businesses leveraging analytics tools are significantly more likely to identify operational inefficiencies, reduce costs, and respond effectively to changing market conditions (Davenport and Harris, 2017). For small businesses in tourism-intensive environments, where operational demands are often seasonal, unpredictable, and high-volume, adopting such systems is increasingly becoming a necessity rather than a luxury.

The Philippine context presents both unique challenges and opportunities for the digital transformation of small food service businesses. While the country has made significant strides in digital infrastructure development, with mobile internet penetration reaching high levels and

digital payment platforms like GCash and Maya gaining widespread adoption, the digital divide between large corporations and small enterprises remains substantial. Filipino SMEs continue to lag their regional counterparts in digital adoption, though encouraging trends show that among those that invested in technology, a significant majority reported improved profitability. The government of the Philippines has recognized the critical importance of MSME digitalization and has initiated several programs to support this transition. The Department of Trade and Industry has emphasized the importance of digital transformation for MSMEs, highlighting how it can enhance their growth and competitiveness. The MSME Development Plan 2023-2028 focuses on leveraging science, technology, and innovation to support MSMEs in transitioning to digital platforms. These policy initiatives create a supportive environment for small businesses seeking to adopt digital tools, including business analytics systems like the one proposed in this study.

In response to the operational challenges identified at Empress Café Boracay and the broader context of digital transformation opportunities for small food service businesses in the Philippines, this study proposes the development and implementation of Operlytics, a business analytics and monitoring system specifically designed for the unique operational context of this establishment. Operlytics represents a targeted, practical response to the documented deficiencies in manual record-keeping, inventory management, and reporting that currently constrain the café's operational effectiveness and competitive performance.

Operlytics aims to provide café management with a comprehensive, real-time view of key operational metrics that are essential for effective business management. These metrics include sales transactions, which provide insight into revenue performance and product demand patterns; inventory levels, which indicate the availability of ingredients and supplies necessary for operations; product performance, which reveals which menu items are popular and which may need reconsideration; and overall business activity, which encompasses the full range of operational indicators that management needs to monitor. By consolidating these diverse data streams into a unified, accessible platform, Operlytics seeks to eliminate the information silos and data fragmentation that currently impede effective management.

The automation of data collection and reporting represents a core value proposition of the Operlytics system. Rather than requiring staff to manually record transactions and later transcribe them into summary reports, the system captures transactional data at the point of occurrence and automatically processes it into meaningful summaries and visualizations. This automation serves multiple purposes: it eliminates the inefficiencies associated with manual record-keeping, freeing staff time for customer-facing activities; it reduces human error, improving the accuracy and reliability of operational data; and it accelerates the availability of information, enabling management to make decisions based on current rather than historical data.

Usability has been a paramount consideration in the design philosophy underlying Operlytics. Recognizing that small café staff and management may have limited technical expertise and minimal experience with complex software systems, the system has been designed to require no advanced technical knowledge for effective operation. The user interface prioritizes clarity, simplicity, and intuitive navigation, ensuring that users can access the information they need and perform required tasks without extensive training or technical support. This usability focus is not merely a matter of convenience but a practical necessity for ensuring successful adoption and sustained use in a small business environment where dedicated IT support is unavailable.

The significance of this study extends far beyond the immediate operational improvements it seeks to bring to Empress Café Boracay. By demonstrating the feasibility and practical value of a tailored analytics system in a small café setting, this research contributes to the broader discourse on the digital transformation of micro and small enterprises in the Philippines. The study provides a concrete example of how accessible, appropriately scaled technology can address real operational challenges in a resource-constrained business environment. It highlights technology-driven solutions for businesses operating in tourism-dependent localities where operational demands are often seasonal, unpredictable, and high-volume characteristics that make effective operational management both especially challenging and especially valuable.

Furthermore, this study contributes to the academic literature on information systems in small business contexts, a domain that has received less research attention than enterprise-level

implementations despite the vastly larger number of small businesses worldwide. The detailed documentation of system development, implementation, and evaluation processes provides valuable insights for researchers and practitioners interested in similar interventions. The methodological approach, which combines developmental research design with agile implementation practices, offers a model for conducting technology development research in close collaboration with real business partners.

Ultimately, Operlytics represents a meaningful and practical step toward empowering small business owners with the tools needed to optimize operations, improve service quality, and achieve long-term sustainability in an increasingly competitive and technology-driven marketplace. The system embodies the principle that sophisticated analytical capabilities need not be the exclusive domain of large corporations with substantial IT budgets. By demonstrating that effective business analytics can be made accessible, affordable, and appropriate for small enterprises, this study contributes to a more inclusive vision of digital transformation, one that encompasses businesses of all sizes and resource levels.

The successful implementation of Operlytics at Empress Café Boracay would serve as a proof of concept for the broader applicability of similar systems across the thousands of small food service establishments that operate throughout the Philippines and similar emerging economies. Each small café that improves its operational efficiency through technology represents not only a more viable business but also a more stable source of employment, a more reliable contributor to local supply chains, and a more capable competitor in the global tourism marketplace. In this way, the development and evaluation of Operlytics contributes to the larger project of inclusive economic development through digital empowerment.

1.2 Statement of the Problem

General Problem Statement

Effective operational systems constitute the fundamental infrastructure upon which successful food service establishments build their business performance. The management of sales transactions, inventory levels, and business reports represents a triad of core operational functions that must function harmoniously for any food service business to achieve sustainable success. In the context of small cafés operating within highly competitive and tourism-driven environments, the ability to maintain accurate, consistent, and real-time operational records assumes critical importance that far exceeds mere administrative convenience. Rather, it emerges as a fundamental requirement for business sustainability, competitive positioning, and long-term growth.

The operational environment of tourism-intensive food service businesses presents unique challenges that amplify the importance of effective operational systems. Seasonal demand fluctuations create periods of intense activity followed by relative quiet, requiring flexible yet reliable operational management. The perishable nature of many food service inputs demands precise inventory control to minimize waste while ensuring availability. The high customer turnover characteristic of tourist destinations necessitates rapid, accurate transaction processing to maintain service quality during peak periods. The competitive density of tourist markets means that operational inefficiencies quickly translate into competitive disadvantages that can threaten business viability.

Empress Café Boracay currently operates without the benefit of integrated operational systems that could address these challenges effectively. The café relies on manual and semi-manual methods to maintain its operational records, creating conditions in which accuracy and consistency are difficult to achieve and sustain a situation well-documented in small hospitality enterprises. The absence of a centralized and automated system has created significant gaps in how the café monitors its sales activities, manages its inventory, and generates operational reports. These gaps are not merely administrative inconveniences but represent fundamental operational deficiencies that affect every aspect of the business's performance.

The manual processes currently employed at Empress Café introduce multiple points of failure that compromise operational integrity. Handwritten sales records are susceptible to errors in arithmetic, illegible entries, and incomplete documentation. Inventory counts performed sporadically and recorded in paper logbooks fail to capture the dynamic nature of stock movements, particularly during busy periods when staff attention is focused on customer service. Report generation, when attempted, requires laborious compilation of scattered records, often resulting in summaries that are outdated before they are completed and inaccurate due to the cumulative effect of recording errors.

Several specific operational issues have been identified in managing the Point of Sale system and inventory levels, each of which affects the overall productivity of the establishment and limits management's capacity to make sound and informed operational decisions. The sales recording process lacks standardization, with different staff members employing different methods and levels of detail. Inventory tracking fails to maintain real-time accuracy, leading to situations where management discovers stock shortages only when items are needed for customer orders. Reporting, when it occurs, provides historical rather than current information, limiting its utility for operational decision-making. These challenges, if left unaddressed, pose a considerable and growing risk to the café's ability to sustain its service quality, manage its resources efficiently, and remain competitive within the demanding local market of Boracay Island.

The cumulative effect of these operational deficiencies extends beyond immediate operational inefficiencies to affect strategic business outcomes. Management lacks the reliable data foundation necessary for effective planning, making it difficult to optimize menu offerings, adjust pricing strategies, or make informed decisions about inventory investment. The inability to identify trends in sales performance or product popularity means that opportunities for improvement may go unrecognized. The time consumed by manual record-keeping and reporting diverts management attention from customer service, staff development, and business development activities that could drive growth.

Specific Problems

The operational challenges at Empress Café Boracay can be decomposed into five specific problem areas, each representing a distinct but interrelated aspect of the café's operational deficiencies. These specific problems provide the detailed foundation upon which the proposed system solution is designed and against which its effectiveness will be evaluated.

Specific Problem 1: Inefficient Manual Record-Keeping Processes

The café currently relies exclusively on manual record-keeping to manage both sales and inventory, a practice that continues to consume significant staff time and cause recurring problems in processing transactions and maintaining accurate stock records. This reliance on manual methods is characteristic of many small and medium-sized enterprises (SMEs) in the food service sector, which often lack the capital, technical expertise, or perceived need to invest in automated systems. The manual approach encompasses multiple specific deficiencies that collectively undermine operational efficiency.

Sales recording at Empress Café occurs through handwritten entries in paper logbooks, with each staff member maintaining separate records during their shifts. This decentralized approach creates immediate problems of consistency and completeness. Different staff members may use different formats, abbreviations, or levels of detail in their entries, making aggregation and analysis difficult or impossible. The physical nature of paper records means they are susceptible to damage, loss, or misplacement, a significant risk in the busy, often chaotic environment of a café kitchen and service area.

The time required for manual recording, particularly during peak service periods, creates a trade-off between record accuracy and customer service speed, often resulting in rushed or incomplete entries. Staff hours devoted to recording, counting, transcribing, and calculating could be redirected to customer-facing activities that directly generate revenue, a form of hidden opportunity cost that is frequently underestimated in small business contexts.

Inventory management through manual methods presents equally serious challenges. Stock counts are performed at irregular intervals, often weekly or even less frequently, meaning that inventory records quickly become outdated and unreliable between counts. The manual

recording of stock movements receipts, usage, waste, transfers introduces multiple opportunities for error at each transaction point. Without systematic tracking, the causes of inventory discrepancies become impossible to identify: was the discrepancy due to recording error, theft, spoilage, or supplier short delivery? The lack of real-time visibility means that management cannot respond promptly to stock situations that require attention, such as unexpectedly rapid depletion of a key ingredient or accumulation of slow-moving stock approaching expiration.

The time consumption of manual processes represents a significant hidden cost that is often underestimated. Staff hours devoted to recording, counting, transcribing, and calculating could be redirected to customer-facing activities that directly generate revenue. Management time spent compiling and reconciling manual records detracts from strategic planning and business development activities. The cumulative time cost of manual processes, while difficult to quantify precisely, represents a substantial drain on the café's limited human resources.

Specific Problem 2: Unreliable and Untimely Information Access

The absence of a reliable recording system makes it difficult to obtain accurate and timely information about the café's operational status. Sales records and stock levels are often inconsistent, incomplete, or at risk of being lost, creating an information environment characterized by uncertainty rather than clarity. This information opacity is a defining characteristic of small hospitality businesses operating without integrated information systems.

The inconsistency of manual records stems from multiple sources. Different staff members may interpret recording instructions differently, leading to variations in how transactions are documented, a problem exacerbated by high employee turnover rates typical of the food service industry. Changes in recording practices over time, as different staff members come and go or as management attempts to improve processes, create discontinuities that complicate trend analysis. The informal nature of manual systems means that there are no enforced standards for data quality, completeness, or timeliness, a condition that research identifies as a critical weakness in SME operational management.

The incompleteness of records results from both intentional and unintentional omissions. During busy periods, staff may prioritize customer service over complete documentation,

skipping entries or recording them from memory later with inevitable inaccuracies. Some transactions, such as complimentary items, staff meals, or waste adjustments, may not be recorded at all if no systematic process exists for capturing them. The cumulative effect of these omissions is a systematic understatement of actual operational activity that distorts management's understanding of business performance.

The risk of record loss represents a constant threat to operational continuity. Paper records can be damaged by spills, heat, or moisture in the café environment. They can be misplaced during cleaning, reorganization, or staff transitions. Unlike digital records, which can be automatically backed up and recovered, lost paper records are irretrievably gone, creating permanent gaps in operational history that compromise both immediate management and long-term analysis.

The timeliness of information is equally problematic. Manual records are inherently retrospective, documenting what has already occurred rather than what is currently happening. The delay between transaction occurrence and record availability means that management is always operating with outdated information, unable to respond to current conditions in real time. This temporal disconnect is particularly problematic in a fast-paced food service environment where conditions can change dramatically within hours or even minutes.

Specific Problem 3: Inefficient Reporting and Decision Support

The lack of an efficient reporting system makes it challenging for management to assess the café's operational performance and make well-informed business decisions in a timely manner. This reporting deficiency is consistent with findings that small food service businesses frequently lack the analytical infrastructure necessary for data-driven decision-making.

Report generation from manual records is a laborious and time-consuming process that requires significant manual effort to compile, calculate, and format. The process typically involves gathering paper records from multiple sources, transcribing or summarizing the information, performing calculations to derive summary metrics, and organizing the results into a comprehensible format. This process may take hours or even days, depending on the reporting period and the complexity of the information required. By the time reports are completed, the

information they contain is already historical, limiting their utility for operational decision-making, a temporal limitation that severely constrains management responsiveness.

The accuracy of manually generated reports is inherently suspect due to the cumulative effect of errors at multiple stages. Errors in original transaction recording propagate through to summary calculations. Transcription errors introduce new inaccuracies during the compilation process. Calculation errors, whether in arithmetic or in formula application, further compromise report reliability. Without systematic verification processes, these errors may go undetected, leading management to make decisions based on incorrect information.

The format and content of manually generated reports are typically inconsistent, varying based on who prepares them and what they believe to be important. This inconsistency makes it difficult to compare performance across time periods or to identify trends that develop gradually. The absence of standardized reporting means that key metrics may be omitted, calculated differently from period to period, or presented in ways that obscure rather than illuminate operational reality.

The decision-making limitations imposed by inadequate reporting extend across all management functions. Without reliable sales reports, management cannot accurately assess revenue performance, identify popular or underperforming products, or evaluate the effectiveness of pricing and promotional strategies. Without accurate inventory reports, management cannot optimize purchasing decisions, identify waste patterns, or ensure adequate stock levels. Without integrated operational reports, management cannot evaluate overall business performance, identify areas requiring attention, or track progress toward business objectives.

Specific Problem 4: Inadequate Stock Level Monitoring

Inadequate monitoring of stock levels continues to result in operational problems that directly affect customer service quality and business profitability. These problems include running out of essential items during service periods, accumulating unnecessary waste from spoilage or expiration, and missing opportunities for operational improvement that could be identified through better stock visibility.

Stockouts represent one of the most immediately visible and damaging consequences of inadequate inventory monitoring. When essential ingredients or supplies are unexpectedly depleted, the café faces difficult choices: substitute alternative ingredients that may compromise product quality, remove affected menu items from availability, or delay customer orders while emergency supplies are obtained. Each of these options carries significant costs: compromised quality damages the café's reputation for consistency; menu unavailability disappoints customers who may have specifically chosen the café for that item; emergency procurement typically involves premium prices and delivery delays that disrupt service flow.

The root cause of stockouts at Empress Café is the absence of real-time inventory visibility that would enable proactive replenishment. Without accurate, current information on stock levels and consumption rates, management cannot anticipate when items will reach reorder points. The irregular timing of manual inventory counts means that stock levels may be significantly depleted before the next count reveals the situation. Even when counts identify low stock, the manual ordering process introduces additional delays that may result in stockouts before replacement supplies arrive.

Waste accumulation represents the opposite but equally costly consequence of inadequate inventory monitoring. Overstocking, whether due to excessive ordering or failure to monitor existing stock levels before placing orders, leads to inventory that exceeds immediate needs. Perishable items may spoil before use, representing a complete loss of the invested capital. Non-perishable items tie up working capital in inventory that could be deployed more productively elsewhere. Storage space occupied by excess inventory may limit capacity for more needed items.

Missed opportunities for operational improvement represent a less visible but equally significant consequence of inadequate stock monitoring. Without systematic data on inventory movements, management cannot identify patterns that would inform better operational practices. Which ingredients consistently experience high waste rates? Which items are frequently subject to emergency orders? Which suppliers consistently deliver quantities that differ from orders? Which menu items drive the highest inventory turnover? These and similar questions,

answerable through systematic inventory analysis, remain unaddressed when inventory monitoring is inadequate.

Specific Problem 5: Absence of Operational Data Analysis Capabilities

Without a system capable of analyzing operational data, the café remains unable to identify inefficiencies, anticipate customer demand, and develop strategic plans to improve overall business performance. This analytical deficiency represents perhaps the most fundamental operational limitation, as it prevents the café from leveraging its own operational data as a resource for improvement.

The identification of inefficiencies requires systematic analysis of operational data to detect patterns that indicate suboptimal performance. Are certain preparation processes consistently slower than others? Do specific shifts or staff members show different productivity patterns? Are there recurring discrepancies between expected and actual ingredient usage that suggest waste or portion control issues? Are sales patterns correlated with external factors such as weather, local events, or tourist seasons in ways that could inform staffing and inventory decisions? These analytical questions, which could guide targeted improvement efforts, cannot be addressed without systematic data collection and analysis capabilities.

Demand anticipation represents a critical capability for food service businesses, particularly in tourism-dependent environments where demand patterns may be highly variable and influenced by factors beyond the business's control. Understanding historical demand patterns, identifying leading indicators that predict demand changes, and developing forecasting models that can guide preparation and inventory decisions these capabilities require analytical tools that process historical data to identify patterns and project future conditions. Without such tools, the café must rely on intuition and experience, which while valuable, are insufficient for optimizing operations in a complex and dynamic environment.

Strategic planning for business improvement requires a reliable foundation of operational data that can support evidence-based decision-making. Should the café expand its menu, and if so, with what types of items? Should operating hours be adjusted to better match demand patterns? Should pricing be modified to optimize revenue or market positioning? Should

marketing efforts be targeted toward specific customer segments or occasions? These strategic questions require data-informed answers that manual operational systems cannot provide. The absence of analytical capabilities thus constrains not only operational management but also strategic development, limiting the café's potential for growth and improvement.

In light of these identified problems, this study seeks to develop and implement Operlytics, a business analytics and monitoring system specifically designed to address the operational deficiencies of Empress Café Boracay. By providing a centralized, automated, and data-driven platform for managing sales transactions, inventory levels, and business reporting, Operlytics aims to eliminate the inefficiencies of manual record-keeping, enhance the accuracy and reliability of operational data, and equip café management with the real-time insights needed to make sound and timely business decisions.

The resolution of these identified problems is expected to contribute not only to the improved day-to-day performance of the café but also to its long-term sustainability and competitiveness within the tourism-driven market of Boracay Island. The successful implementation of Operlytics would demonstrate that appropriately designed technology can address real operational challenges in small business contexts, contributing to the broader understanding of how digital tools can support the development of more efficient, competitive, and sustainable small enterprises in the Philippines and similar emerging economies.

1.3 Objectives of the Study

General Objective

The main objective of this study is to develop and evaluate Operlytics, a comprehensive business analytics and monitoring system that will fundamentally improve operational decision-making at Empress Café Boracay by providing efficient, accurate, and timely monitoring of sales, inventory, and overall operational performance through sophisticated data visualization, automated report generation, and intelligent alert mechanisms. This objective encompasses both the rigorous technical development of a fully functional, robust software system and the empirical, evidence-based evaluation of its practical effectiveness, usability, and transformative impact within a real-world small business environment.

The achievement of this general objective necessitates a comprehensive, multi-dimensional approach that systematically addresses several interconnected dimensions of system development, deployment, and assessment. The technical dimension involves designing and building a software system that reliably captures, processes, stores, secures, and presents operational data with high fidelity, minimal latency, and robust error handling. This includes architecting scalable database schemas, implementing responsive user interfaces, ensuring data integrity through transaction management, and establishing secure access controls that protect sensitive business information from unauthorized access or data loss.

The operational dimension involves ensuring that the system effectively supports, enhances, and seamlessly integrates with the café's daily business processes, workflow patterns, and operational rhythms. This requires a deep understanding of how the café functions during peak tourist seasons versus off-peak periods, how inventory flows from suppliers through storage to final consumption, how sales transactions are processed across multiple payment methods, and how management currently makes decisions with limited information. The system must not merely digitize existing processes but optimize and streamline them, eliminating bottlenecks, reducing cognitive load on staff, and enabling operational practices that were previously impractical or impossible.

The human dimension involves facilitating smooth user adoption, minimizing resistance to technological change, and ensuring that the system is accessible, intuitive, and genuinely valuable to its intended users including café administrators, floor staff, kitchen personnel, and potentially external stakeholders such as suppliers or accountants. This dimension recognizes that technology adoption in small businesses often fails not because of technical deficiencies but because of poor alignment with user needs, inadequate training, or disruption of established social and work practices. The study therefore emphasizes participatory design approaches, iterative user feedback, and comprehensive training support.

The evaluative dimension involves systematically assessing the system's performance against clearly defined criteria encompassing functional correctness, usability, reliability, performance under load, security, and business impact and documenting the results with methodological rigor for the benefit of both immediate stakeholders at Empress Café Boracay and the broader research community interested in information systems adoption in small and medium enterprises (SMEs), particularly within tourism-dependent economies.

By transitioning the café from its current reliance on manual, paper-based, and informal record-keeping methods to a centralized, technology-driven, data-centric platform, this study aims to address the persistent operational gaps that have historically limited management's capacity to oversee daily activities with accuracy, completeness, and efficiency. These gaps include the inability to track real-time inventory levels, the difficulty of identifying sales trends and patterns, the labor-intensive nature of manual report compilation, the high risk of human error in data recording and calculation, and the lack of historical data for comparative analysis and forecasting.

This transition represents not merely a superficial change in tools or a simple automation of existing practices, but a fundamental transformation in how the café conceptualizes, collects, manages, analyzes, and acts upon its operational information. The shift from fragmented, paper-based, decentralized records scattered across multiple notebooks, receipts, and informal memory to a unified, digital, centralized system enables entirely new possibilities for data integration across functional domains, real-time access from multiple locations and devices,

advanced analytical insight through aggregation and visualization, and evidence-based decision-making that was previously unavailable to the café due to information constraints.

The successful development, deployment, and sustained operation of Operlytics is expected to serve as a practical, sustainable, and replicable solution to the café's longstanding operational challenges while simultaneously contributing to the broader academic and professional understanding of how small food service enterprises in tourism-dependent localities can strategically benefit from the adoption of business analytics and business intelligence systems. The study thus deliberately positions itself at the intersection of practical problem-solving and rigorous academic knowledge generation, seeking to produce both immediate, tangible operational benefits for Empress Café Boracay and lasting, generalizable contributions to the literature on information systems implementation, technology adoption, and digital transformation in small business contexts, particularly within developing economies where resource constraints and organizational capacity limitations present unique challenges.

Specific Objectives

To achieve the general objective stated above, this study pursues five specific objectives that address distinct but deeply complementary aspects of system development, technical implementation, operational integration, and empirical evaluation:

Specific Objective 1: Design and Development of a Web-Based Business Analytics System

The first specific objective is to design and develop a comprehensive, web-based business analytics and monitoring system that will systematically record, manage, analyze, and report sales transactions and inventory data for Empress Café Boracay. This objective encompasses the complete software development lifecycle, from initial requirements analysis and stakeholder engagement through architectural design, detailed component specification, iterative coding, rigorous testing, deployment, and post-deployment support.

The requirements analysis phase involves conducting thorough interviews, observations, and document reviews with café administrators, staff members, and other stakeholders to identify current operational pain points, information needs, workflow patterns, and functional requirements. This phase also involves analyzing existing records such as sales receipts,

inventory logs, supplier invoices, and informal notes to understand data structures, volumes, frequencies, and quality issues. The requirements analysis produces a detailed specification of system functionalities, user roles, data entities, business rules, and performance expectations that guide all subsequent development activities.

The design phase involves translating the operational requirements identified through needs analysis into a coherent, scalable, and maintainable system architecture along with detailed component specifications. This includes defining the relational database schemas and data structures that will represent sales transactions (including line items, payment methods, discounts, and timestamps), inventory items (including quantities, units, reorder points, supplier information, and cost histories), user accounts and role-based permissions, and other operational entities such as suppliers, product categories, and menu configurations. The design phase also encompasses designing intuitive, responsive user interfaces that will enable staff and management to efficiently interact with the system across different devices and screen sizes, as well as specifying the processing logic, algorithms, and business rules that will transform raw transaction data into meaningful operational insights, alerts, and recommendations.

The development phase involves implementing the designed system using appropriate, modern technologies and software engineering methodologies that balance technical capability with practical maintainability. For Operlytics, this entails building robust backend server components using technologies such as Python with Django or Flask, or Node.js with Express that handle data processing, business logic execution, authentication, authorization, and secure storage. The frontend user interfaces are developed using responsive web technologies such as HTML5, CSS3, JavaScript, and modern frameworks like React or Vue.js, presenting information clearly and capturing user inputs with validation and feedback. These components are integrated into a cohesive, functioning whole through well-defined APIs, consistent data models, and comprehensive error handling.

The development process follows agile methodologies such as Scrum or Kanban that allow for iterative refinement based on continuous feedback from users, changing business requirements, and emerging technical insights. Short development sprints produce working increments of functionality that can be demonstrated, tested, and refined before proceeding to

subsequent features, reducing the risk of major misalignment between the developed system and actual user needs.

The web-based architecture of the system is a deliberate, strategic design choice that offers several significant advantages specifically suited to the café's operational context and the broader constraints of small businesses in tourism-dependent locations. Web-based systems require no installation on user devices beyond a standard, modern web browser, eliminating the need for specialized hardware, operating system compatibility concerns, or software maintenance at the point of use. They enable secure access from any internet-connected device including desktop computers, laptops, tablets, and smartphones allowing management to monitor operations remotely (whether from another location on Boracay Island, from the Philippine mainland, or while traveling) and staff to use whatever devices are available without requiring dedicated, expensive workstation setups. They centralize data storage and processing on the server infrastructure, ensuring data consistency across all users, simplifying backup and disaster recovery procedures, and enabling centralized security management including encryption, access logging, and threat monitoring.

Furthermore, web-based architectures facilitate easier updates and feature rollouts, as changes to the system can be deployed centrally on the server without requiring individual updates on each user device. This reduces maintenance overhead and ensures that all users always access the most current version of the system. The architecture also supports future scalability, allowing the system to grow with the café's needs whether that means adding new locations, integrating with external services (such as accounting software or supplier systems), or accommodating increased transaction volumes during peak tourist seasons.

Through the achievement of this objective, this study seeks to demonstrate that a well-designed, purpose-built, and carefully implemented analytics system can meaningfully transform the operational capabilities of a small café establishment, enabling levels of data-driven management that were previously accessible only to larger enterprises with substantial information technology budgets and specialized technical staff. The development of Operlytics is ultimately guided by the overarching goal of providing Empress Café Boracay with a reliable, efficient, user-friendly, and scalable technological tool that supports sound,

evidence-based management decisions, enhances overall service quality and customer satisfaction, reduces operational waste and costs, and strengthens the café's competitive standing within the dynamic, rapidly evolving, and intensely competitive tourism-driven market of Boracay Island.

Specific Objective 2: Real-Time Monitoring of Sales and Inventory

The second specific objective is to provide accurate, responsive, and genuinely real-time monitoring of sales performance and inventory levels that will empower café administrators and management to continuously track stock availability, sales velocity, revenue generation, and overall operational performance without the delays, approximations, and information gaps inherent in manual tracking methods. This objective addresses the critical temporal dimension of operational information, ensuring that management has access to current, up-to-the-moment data rather than outdated historical summaries that may no longer reflect the actual state of the business.

Real-time monitoring requires that the system capture transaction data at the exact point of occurrence when a customer places an order, when payment is processed, when ingredients are consumed in food and beverage preparation and immediately propagate these changes throughout all relevant system components. When a staff member processes a customer order through the Point of Sale (POS) module, the system must instantly update sales totals and revenue figures, deduct consumed ingredients from inventory counts according to predefined recipe mappings, update product availability status if ingredients fall below serviceable thresholds, refresh relevant dashboard displays and summary metrics, and trigger any configured alerts or notifications for management attention.

This immediate, automatic data propagation ensures that management views of operational status are always current and accurate, enabling timely, proactive responses to emerging situations rather than delayed reactions to problems that have already escalated. For example, if a popular menu item suddenly experiences a surge in orders during a busy lunch period, real-time monitoring allows management to observe the rapid depletion of key ingredients and take immediate action such as temporarily removing the item from the menu, expediting a supplier order, or substituting ingredients before customer disappointment occurs.

The technical implementation of real-time monitoring involves both sophisticated backend processing and thoughtful frontend design considerations. Technically, the system must process transactions quickly enough that updates appear effectively instantaneous to users, even during peak activity periods when multiple staff members may be simultaneously entering orders, processing payments, and updating records. This requires efficient database operations with optimized indexing and query structures, connection pooling to manage concurrent database access, caching strategies for frequently accessed summary data, and responsive user interface updates through techniques such as AJAX polling, WebSockets, or server-sent events that push updates to connected clients without requiring manual page refreshes.

From a design and user experience perspective, the system must present real-time information in formats that are immediately comprehensible, visually intuitive, and actionable. This involves using color-coded visual indicators (such as green for normal status, yellow for warning levels, and red for critical conditions), animated counters and progress bars that draw attention to changing values, configurable alert thresholds that trigger notifications when metrics exceed or fall below defined boundaries, and summary dashboards that surface the most critical information prominently while allowing drill-down access to detailed underlying data.

The operational benefits of real-time monitoring extend across multiple critical business domains and touch nearly every aspect of café management. For inventory management, real-time visibility prevents stockouts and overstock situations by alerting management to low stock conditions before they reach critical levels, enabling just-in-time reordering that minimizes carrying costs while ensuring availability. The system can track not only current quantities but also consumption rates, predicted depletion times based on historical patterns, and optimal reorder points that account for supplier lead times and demand variability.

For sales management, real-time tracking enables immediate recognition of unusual patterns such as unexpected demand spikes for particular items, sudden revenue drops that may indicate equipment failures or staff issues, or comparative performance across different time periods that may indicate emerging opportunities requiring rapid exploitation or problems requiring immediate intervention. Management can observe which menu items are performing

well in real-time, adjust pricing or promotions dynamically, and allocate staff attention to high-demand areas.

For overall operational oversight, real-time dashboards provide management with continuous situational awareness of business status that supports proactive rather than reactive management styles. Rather than discovering problems hours or days after they occur through manual report reviews, managers can observe anomalies as they develop and intervene while corrective action is still effective. This continuous awareness is particularly valuable in the tourism context, where demand patterns can shift rapidly due to weather changes, local events, cruise ship arrivals, or social media trends, and where the ability to respond quickly can significantly impact revenue and customer satisfaction.

Specific Objective 3: Automated Reports and Dashboards

The third specific objective is to generate comprehensive, accurate, and fully automated reports and interactive dashboards that will systematically summarize, analyze, and visualize operational data, allowing management to easily analyze sales trends, stock usage patterns, cost structures, profit margins, and overall business performance without requiring manual data compilation, spreadsheet manipulation, or specialized analytical skills. This objective addresses the analytical and decision-support dimension of the system, transforming raw, granular operational data into structured, contextualized insights that directly support strategic and tactical decision-making.

Automated report generation eliminates the laborious, time-consuming, and error-prone manual compilation process currently required to produce operational summaries. In the café's existing manual system, generating even a simple weekly sales report might require collecting paper receipts, manually transcribing figures into a ledger, performing arithmetic calculations with high risk of error, formatting results into a readable summary, and distributing copies to relevant stakeholders a process that could consume several hours of management time weekly. The Operlytics system automatically aggregates transaction data from the database, applies predefined calculation rules and business logic, calculates summary metrics and key performance indicators (KPIs), formats results into professionally designed, standardized report templates,

and makes these reports available on demand with a single click or automatically delivered via email according to scheduled intervals (daily, weekly, monthly).

This automation ensures that reports are always available when needed for decision-making, contain consistent, accurate, and complete information derived directly from source transaction data without transcription errors, and require minimal staff time for production freeing management to focus on interpreting results and taking action rather than on data gathering and formatting. The system supports multiple report types including sales summaries (by product, category, time period, payment method, or staff member), inventory reports (current stock levels, consumption rates, reorder recommendations, waste analysis), financial reports (revenue, costs, profit margins, trend comparisons), and operational reports (transaction volumes, average order values, peak hour analysis, customer throughput).

Dashboards provide at-a-glance visual summaries of key operational metrics that enable management to quickly assess overall business status, identify areas requiring attention, and drill down into detailed information when anomalies or opportunities are detected. Unlike detailed reports that may require careful review and mental calculation to extract insights, dashboards use intuitive visual representations such as line charts for trend analysis, bar charts for comparative performance, pie charts for proportional breakdowns, gauge indicators for threshold monitoring, heat maps for temporal pattern identification, and data tables for precise values to present information in immediately comprehensible formats that leverage human visual processing capabilities.

The design of effective dashboards requires careful, evidence-based consideration of which metrics are most important for management decision-making in the café context, how these metrics relate to one another, how they can be visually represented for maximum clarity and interpretive accuracy, and how the dashboard layout supports natural scanning patterns and cognitive workflows. Key dashboard components for Operlytics include: a real-time sales monitor showing current day's revenue versus comparable historical periods; an inventory status overview highlighting items below reorder thresholds; a top-performing products display; a staff performance summary; and alerts panels for exceptions requiring management attention.

The analytical capabilities enabled by automated reports and dashboards extend far beyond simple historical summaries to support sophisticated trend identification, pattern recognition, and predictive insight. By comparing current performance to historical baselines such as the same day last week, the same period last year, or moving averages management can identify improving or deteriorating trends that warrant investigation or corrective action before they become critical. By analyzing product-level performance metrics such as sales volume, revenue contribution, profit margin, and inventory turnover management can identify best-sellers that should be promoted, underperformers that may need recipe reformulation or removal from the menu, and items with high popularity but low profitability that may require pricing adjustment.

By examining inventory usage patterns in relation to sales data, management can optimize purchasing decisions to minimize waste from spoilage, reduce capital tied up in excess inventory, and ensure availability of high-demand items. By analyzing temporal patterns hourly, daily, weekly, and seasonal variations in demand management can optimize staff scheduling, inventory preparation, and promotional timing to match capacity with expected demand. These analytical capabilities transform management from reactive problem-solvers into proactive strategists who can anticipate challenges and opportunities rather than merely responding to them after they occur.

Specific Objective 4: Improvement of Operational Efficiency

The fourth specific objective is to achieve measurable, significant improvement in operational efficiency by systematically reducing manual recording errors, eliminating delays in report preparation, streamlining workflow processes, and enabling faster, more confident decision-making throughout the café's operations. This objective focuses on the tangible process improvements and productivity gains that the system enables, quantifying benefits in terms of time savings, error reduction, cost avoidance, and operational streamlining that directly impact the café's bottom line and service quality.

The reduction of manual recording errors is achieved through multiple integrated mechanisms that address different sources of human error in traditional manual systems. The system enforces standardized data entry formats such as predefined product codes, dropdown

selections, date pickers, and numeric validators that eliminate the inconsistencies, illegibility, and ambiguity inherent in handwritten records where different staff members may use different abbreviations, formats, or handwriting styles. Validation rules prevent the entry of obviously incorrect or impossible data, such as negative quantities, prices outside defined ranges, future dates for past transactions, or mismatched payment amounts, catching errors at the point of entry before they propagate through the system.

Automatic calculations eliminate arithmetic errors such as incorrect subtotals, tax miscalculations, or incorrect change amounts that commonly occur in manual processing, especially during busy periods when staff are under time pressure. The elimination of transcription steps, where data is copied from one record (such as a paper order slip) to another (such as a ledger or spreadsheet), removes a major source of error propagation where a single initial mistake can cascade through multiple records and reports, becoming increasingly difficult to trace and correct. Digital audit trails maintain complete records of who entered what data and when, enabling error tracing and accountability that is impossible with paper systems.

The reduction of delays in report preparation is achieved through automation that replaces manual compilation processes with instantaneous digital generation. Reports that previously required hours of staff time to collect, transcribe, calculate, format, and verify can now be generated instantly with a few clicks or even automatically on a scheduled basis. The elimination of this time burden frees staff to focus on customer-facing activities that directly generate revenue and enhance customer experience, and frees management to focus on strategic decision-making, quality improvement, and business development rather than administrative data manipulation.

The availability of immediate, on-demand reports also enables more frequent and responsive reporting cycles, allowing management to monitor performance on daily or even hourly intervals rather than waiting for weekly or monthly summaries. This increased reporting frequency enables faster detection of problems, more rapid implementation of corrective actions, and tighter feedback loops between operational decisions and their outcomes. Management can experiment with operational changes such as promotional pricing, menu adjustments, or staffing

modifications and quickly observe their effects through immediate reporting, enabling iterative optimization.

The overall improvement in operational efficiency extends beyond these specific, quantifiable benefits to encompass a general, systemic streamlining of operational workflows and information flows. The integration of sales recording, inventory management, financial tracking, and reporting into a single, unified system eliminates the need for duplicate data entry across separate records and the time-consuming reconciliation processes required to ensure consistency between independent systems. When a sale is recorded, inventory is automatically updated, financial summaries are immediately adjusted, and relevant reports reflect the new data without any additional manual intervention.

The standardization of processes through the system reduces training requirements for new staff members, as they can be trained on consistent, guided digital workflows rather than learning idiosyncratic manual procedures that may vary between experienced employees. This standardization ensures consistency across shifts, days, and personnel changes, reducing variability in service quality and operational execution. The availability of accurate, timely, comprehensive information enables faster and more confident decision-making at all operational levels from front-line staff who can immediately check inventory availability when taking orders, to managers who can make informed purchasing and staffing decisions based on current data rather than intuition or outdated reports.

Furthermore, the system enables operational improvements that were previously impractical under manual systems, such as detailed cost analysis by product, identification of waste and shrinkage patterns, optimization of supplier relationships based on historical performance data, and evidence-based menu engineering that balances customer preferences with profitability considerations. These advanced capabilities represent a qualitative leap in operational sophistication that positions the café for sustainable competitive advantage.

Specific Objective 5: System Evaluation Based on User Feedback

The fifth specific objective is to rigorously evaluate the system's usability, functionality, performance, and effectiveness in supporting operational decision-making based on

comprehensive feedback collected from café administrators, floor staff, kitchen personnel, and other relevant stakeholders during the testing phase and initial deployment period. This objective ensures that the system is not merely technically functional in controlled conditions but genuinely practical, valuable, and sustainable in its intended real-world operational context.

The evaluation of usability assesses how easily, intuitively, and efficiently users can interact with the system to accomplish their intended tasks, and how satisfied they are with their overall experience. This multidimensional assessment includes: the clarity and consistency of the user interface design, including visual hierarchy, labeling, iconography, and color usage; the efficiency of navigation and task completion, measured by the number of steps, time required, and cognitive load involved in common tasks such as entering a sale, checking inventory, or generating a report; the learnability of system functions for new users, assessed by how quickly novice users can become proficient with minimal training; the memorability of interface patterns, ensuring that returning users can quickly recall how to perform tasks after periods of absence; and the overall subjective satisfaction of users with their interaction experience, captured through standardized questionnaires and open-ended feedback.

Usability evaluation employs a mixed-methods approach combining observational methods where researchers systematically watch representative users interact with the system while performing realistic tasks, noting difficulties, errors, hesitations, workarounds, and verbalizations and survey methods where users report their subjective experiences, opinions, preferences, and suggestions through structured questionnaires (such as the System Usability Scale), semi-structured interviews, and focus group discussions. This combination provides both objective behavioral data and subjective experiential insights that together characterize the system's usability profile.

The evaluation of functionality assesses whether the system successfully implements all intended features as specified in the requirements, and whether these features operate correctly, reliably, and robustly under various realistic conditions and edge cases. This comprehensive verification includes: confirmation that sales transactions are accurately recorded, processed, stored, and retrievable without data loss or corruption; verification that inventory updates correctly reflect consumption through sales, replenishment through purchases, adjustments for

waste or spoilage, and returns; validation that reports contain accurate, complete, and consistent information that correctly aggregates and summarizes underlying transaction data; and assurance that all system functions operate reliably without crashes, freezes, data corruption, security vulnerabilities, or unexpected behavior under normal load, peak load, and adverse conditions (such as network interruptions or simultaneous multi-user access).

Functional evaluation employs systematic testing methodologies including unit testing of individual components, integration testing of component interactions, system testing of end-to-end workflows, user acceptance testing with real operational scenarios, and stress testing to verify performance under peak load conditions that simulate busy tourist season operations.

The evaluation of effectiveness in supporting operational decision-making assesses whether the system actually improves management's ability to make informed, timely, and beneficial decisions, and whether these decisions lead to improved operational outcomes. This outcome-focused assessment examines: whether management finds the information provided by the system useful, relevant, and actionable for their specific decision-making needs and contexts; whether the system enables decisions that would not have been possible, would have been significantly more difficult, or would have been made with less confidence without the system's analytical capabilities and data presentation; whether the decisions enabled by the system lead to measurable improvements in operational outcomes such as reduced costs, increased revenue, improved inventory turnover, reduced waste, enhanced customer satisfaction, or better staff productivity; and whether the system changes management behavior and organizational culture toward more data-driven, evidence-based approaches.

Effectiveness evaluation employs before-and-after comparisons of key operational metrics, longitudinal tracking of decision quality and outcomes, and qualitative assessment of changes in management practices and organizational culture. Success is measured not only by system usage statistics but by tangible business improvements that can be attributed to the system's implementation.

Through the rigorous achievement of these five interconnected specific objectives, this study seeks to comprehensively demonstrate that a well-designed, purpose-built, carefully implemented, and systematically evaluated analytics system can meaningfully, measurably, and

sustainably transform the operational capabilities of a small café establishment. The development of Operlytics is ultimately guided by the overarching goal of providing Empress Café Boracay with a reliable, efficient, user-friendly, analytically powerful, and technologically scalable tool that supports sound, evidence-based management decisions, enhances overall service quality and customer experience, reduces operational waste and unnecessary costs, improves staff productivity and satisfaction, and strengthens the café's competitive standing and long-term viability within the dynamic, rapidly evolving, intensely competitive, and tourism-driven market of Boracay Island.

1.4 Scope and Delimitation

This study focuses on the development and evaluation of Operlytics, a web-based business analytics and monitoring system designed to improve operational decision-making for Empress Café Boracay. The scope of the study encompasses the end-to-end design, development, and evaluation of the system within the specific operational context of the establishment. The research is primarily concerned with addressing the shortcomings of the manual record-keeping practices currently employed by the café and replacing them with a structured, data-driven platform that enables management to monitor, analyze, and act on key operational information in real time.

The scope of this study is defined by several dimensions that collectively establish the boundaries of the research effort and the intended applicability of its outcomes. Understanding these scope dimensions is essential for properly interpreting the study's findings and assessing their relevance to other contexts.

Operational Scope

The operational scope of the study encompasses three primary operational domains that are directly aligned with the day-to-day management needs of Empress Café Boracay: sales transaction monitoring, inventory level tracking, and operational report generation. These three domains represent the core operational functions that the system addresses and the primary areas in which operational improvements are sought.

Sales transaction monitoring encompasses the complete process of recording, processing, and analyzing customer orders and payments. This includes the Point of Sale interface used by staff to enter orders, the calculation of totals and application of discounts, the recording of payment methods and amounts, and the generation of receipts or order confirmations. The sales monitoring domain also includes the aggregation of transaction data into summaries that reveal revenue performance, product popularity, and temporal patterns of demand.

More specifically, the sales transaction monitoring component captures every customer interaction at the point of purchase, creating a comprehensive digital trail of business activity. When a customer places an order, the system records not only the items purchased and their

quantities but also the exact timestamp of the transaction, the staff member who processed it, the payment method utilized, and any applicable discounts or promotions applied. This granular level of detail enables management to reconstruct the complete operational picture of any given business day, identify peak service periods with precision, and understand customer purchasing behaviors at a level of detail that manual record-keeping simply cannot achieve. The system also tracks voided transactions, refunds, and order modifications, ensuring that the sales data remains accurate and auditable even when exceptions occur during normal operations.

Inventory level tracking encompasses the monitoring of stock quantities, movements, and status for all ingredients and supplies used in café operations. This includes the recording of inventory receipts from suppliers, the automatic deduction of consumed ingredients based on sales transactions, the tracking of waste and spoilage, and the monitoring of stock levels against defined thresholds. The inventory tracking domain also includes the generation of alerts and reports that inform management of stock conditions requiring attention.

System Feature Scope

To support its operational objectives, Operlytics incorporates five core system features that collectively address the identified operational requirements:

The Point of Sale (POS) module serves as the primary data entry point of the system, enabling authorized staff to record and process sales transactions directly within the platform. The POS interface is designed for rapid, intuitive operation during busy service periods, with features such as product selection shortcuts, automatic total calculation, and multiple payment method support. All transaction data entered through the POS module is automatically and simultaneously reflected across the relevant dashboards and inventory records in real time, ensuring data consistency and eliminating the need for duplicate manual encoding.

The inventory tracking feature maintains current records of stock levels for all ingredients and supplies, automatically updating quantities based on consumption from sales transactions and manual adjustments for receipts, waste, and other movements. This feature provides real-time visibility into stock availability, alerts management to low stock conditions, and supports inventory counting and reconciliation processes. The inventory tracking interface

displays current stock levels, recent movement history, and upcoming expiration dates for perishable items.

User Scope

The target users of Operlytics are the café administrators or managers and authorized staff members who are directly involved in managing and monitoring the daily operations of Empress Café Boracay. This user scope reflects the operational reality of a small café, where roles may be flexible and individuals may perform multiple functions.

Administrators and managers have access to the full range of system features, including dashboards, inventory management, report generation, and system configuration. These users are responsible for overall operational oversight, strategic decision-making, and system administration. Their access privileges include the ability to view all operational data, configure system settings, manage user accounts, and generate comprehensive reports.

The administrator role is designed for the café owner or senior manager who bears ultimate responsibility for business performance and system administration. Administrators can create and manage user accounts, assign roles and permissions, configure system-wide settings such as tax rates, payment methods, and business hours, and access all reports and data without restriction. They can modify menu items and recipes, adjust inventory parameters, set sales targets, and configure alert thresholds. The administrator dashboard provides the most comprehensive view of operational performance, consolidating data from all functional areas into a single command center. Administrators can also view system logs and audit trails to monitor user activity, investigate discrepancies, and ensure compliance with operational procedures. This level of access reflects the administrator's need for complete operational visibility and control.

Authorized staff members have limited but essential access to the system for the purpose of carrying out their daily operational tasks. Staff users primarily interact with the POS module for transaction recording and processing, with access to features necessary for their specific roles. Staff access is restricted to prevent unauthorized modifications to inventory records, system settings, or historical data.

This layered security approach protects sensitive business data while maintaining operational efficiency by streamlining each user's interface to show only relevant functions.

Technical Scope

The system is developed as a web-based platform, accessible through standard internet browsers on computers or any internet-connected device. This technical approach ensures that the system can be deployed without the need for specialized hardware installations, making it a practical and cost-effective solution suited to the scale and resources of a small café establishment.

The web-based architecture provides several practical advantages. It eliminates the need for software installation on individual devices, reducing deployment complexity and maintenance requirements. It enables access from multiple device types, including desktop computers, laptops, tablets, and smartphones, providing flexibility in how and where the system is used. It centralizes data storage and processing on the server, ensuring data consistency and simplifying backup and security management. It facilitates remote access, allowing management to monitor operations even when not physically present at the café.

The decision to adopt a web-based architecture was driven by several practical considerations specific to the operational context of Empress Café Boracay. As a small business with limited technical resources, the café requires a solution that minimizes infrastructure investment and ongoing maintenance burden. A web-based system requires only standard devices with browser capability, which the café already possesses, rather than dedicated POS terminals or specialized hardware. Software updates and maintenance are performed centrally on the server, eliminating the need to update individual workstations and ensuring that all users always access the current version of the system. Data backup is centralized and can be automated, protecting business information against device failure or loss. The ability to access the system from mobile devices is particularly valuable for management, who can check sales performance, review inventory status, and receive alerts from anywhere with internet connectivity, enabling more responsive management even during off-site periods.

The technology stack selected for Operlytics PHP for backend processing, HTML/CSS/JavaScript for frontend presentation, and MySQL for data storage represents a proven, widely supported combination that offers reliability, compatibility, and accessibility. These technologies are well-documented, have large developer communities, and can be deployed on affordable hosting infrastructure suitable for small business budgets.

Data Scope

The data scope of the study encompasses all operational data generated within the café's daily business activities, including sales transaction records, inventory movement logs, product catalog information, and user activity trails. The system is designed to capture, store, and process these data types in a structured relational database that ensures integrity, consistency, and retrievability. Data scope explicitly excludes personal customer information beyond what is necessary for transaction processing, such as names associated with orders or loyalty identifiers, and does not involve the collection, storage, or analysis of sensitive personal data subject to specialized privacy regulations. The data architecture is optimized for operational queries and reporting rather than large-scale data mining or complex analytical workloads, reflecting the practical data volumes and access patterns characteristic of a small café operation.

Security Scope

The security scope of Operlytics addresses fundamental data protection and access control requirements appropriate to a small business environment, without extending to enterprise-grade security infrastructure or compliance certifications. The system implements password-based authentication, role-based authorization, basic input validation, and server-side session management to protect against common threats such as unauthorized access, data tampering, and injection attacks. Security measures are proportionate to the sensitivity of the operational data being protected and the realistic threat profile of a small local business. The study does not encompass formal security auditing, penetration testing by independent third parties, or compliance with industry-specific security standards, though the design documentation provides a foundation for such activities should they be pursued in future development phases.

Evaluation Scope

The evaluation scope is structured around functional testing, user acceptance testing, and performance assessment conducted during a defined implementation period at Empress Café Boracay. Evaluation activities focus on verifying that the system meets its specified requirements, operates reliably under normal usage conditions, and delivers perceived value to its intended users. Metrics for evaluation include system uptime, data accuracy, task completion efficiency, and user satisfaction ratings gathered through structured feedback instruments. The evaluation does not include longitudinal studies of business impact, comparative analysis against competing commercial systems, or formal cost-benefit analysis, as these would require extended deployment periods and resources beyond the current research framework. However, the evaluation methodology is designed to produce actionable findings that can guide both immediate system refinement and future research directions.

The evaluation period is designed to capture sufficient operational experience to validate system functionality and user acceptance while remaining within the practical constraints of an academic research timeline. During this period, the system is deployed in the live operational environment of Empress Café Boracay, allowing users to interact with it under realistic conditions of varying customer volume, product availability, and staffing configurations. Evaluation data is collected through multiple channels including system-generated logs, direct observation of user interactions, structured questionnaires administered to administrators and staff, and semi-structured interviews exploring subjective experiences and perceived impacts on workflow efficiency. The evaluation deliberately samples across different times of day and days of the week to capture variation in operational intensity and system load. However, the evaluation does not encompass stress testing under artificially extreme loads, disaster recovery scenario testing, or evaluation under conditions of hardware failure or network outage. These limitations are recognized as areas where future work could strengthen confidence in system robustness, particularly if deployment expands to support multiple locations or higher transaction volumes.

Future Expansion Scope

While the current implementation of Operlytics is bounded by the scope dimensions described above, the system architecture is designed with modularity and extensibility in mind to accommodate potential future enhancements. The separation of concerns between the POS interface, data processing layer, visualization components, and reporting engine creates natural boundaries for incremental development. Future expansion possibilities, which are acknowledged but not pursued within this study, include integration with supplier ordering systems, mobile application development for remote management, customer relationship management features, and enhanced analytical capabilities. By documenting the current scope clearly and designing the system with principled architectural foundations, this study establishes a platform upon which subsequent researchers or developers can build without requiring fundamental restructuring of the existing codebase or data model.

Delimitations

While Operlytics is designed to address the core operational needs of Empress Café Boracay, the study recognizes several important limitations that define the boundaries of its scope. These delimitations clarify what the study does not attempt to accomplish and establish realistic expectations for the system's capabilities and applicability.

The system does not incorporate advanced predictive analytics, machine learning algorithms, or artificial intelligence-based forecasting capabilities. Its functions are limited to descriptive and operational insights derived from current and historical data, rather than the generation of automated predictive models or business recommendations. While the data collected and organized by Operlytics could potentially support more advanced analytical applications in the future, the current system focuses on providing clear, accurate, and accessible operational information rather than sophisticated predictive intelligence.

This delimitation reflects both practical and philosophical considerations. Practically, the development of predictive analytics capabilities requires specialized expertise, substantial data volumes, and significant development effort that exceeds the resources available for this study.

The successful implementation of machine learning models typically requires large datasets for training, specialized knowledge in data science and statistical modeling, and iterative refinement that extends well beyond the timeframe of this research project. Philosophically, the researchers believe that the foundational need at Empress Café and similar establishments is for reliable basic operational information rather than advanced analytical capabilities. Addressing the fundamental deficiencies in data accuracy, timeliness, and accessibility must precede the pursuit of more sophisticated analytical applications. A café that currently struggles with inconsistent manual record-keeping and lacks visibility into basic operational metrics will benefit more immediately from accurate, real-time operational data than from predictive forecasts. The researchers view Operlytics as establishing the data foundation upon which more advanced analytics could potentially be built in future development phases, once the fundamental operational information needs are fully addressed.

Furthermore, Operlytics is not integrated with any external accounting software, financial management platforms, or third-party enterprise systems. Financial functions such as payroll processing, tax computation, and accounts management are explicitly outside the scope of this study, and all data processed by the system remains within its own operational environment. This delimitation maintains the system's focus on operational rather than financial management, recognizing that financial accounting involves specialized requirements, regulatory compliance considerations, and integration complexities that would substantially expand the project's scope.

The functional scope of Operlytics is deliberately bounded to ensure that each feature is developed with sufficient depth and reliability to meet the operational demands of Empress Café Boracay. Rather than pursuing breadth at the expense of quality, the system concentrates on a tightly integrated set of functions that work together seamlessly to address the most pressing operational challenges identified by café management. This focused approach ensures that each module is thoroughly tested, intuitively designed, and capable of delivering measurable improvements in operational efficiency. The functional boundaries also allow for a more manageable development timeline and a clearer evaluation framework, enabling the researchers to assess the system's impact with greater precision and confidence.

Within the functional scope, each system module is designed to fulfill a specific operational requirement without unnecessary complexity or feature proliferation. The Point of Sale module, for instance, is optimized for speed and accuracy during high-volume service periods rather than accommodating elaborate promotional schemes or multi-location coordination. The inventory tracking feature focuses on real-time quantity monitoring and automated deduction rather than advanced supply chain management or vendor relationship automation. The data visualization component prioritizes clarity and immediate interpretability over sophisticated analytical depth. This disciplined approach to functional scoping reflects the principle that a system which performs a limited set of functions exceptionally well delivers greater operational value than one which performs many functions inadequately. The functional scope is further constrained by the need to ensure that all features can be realistically implemented, tested, and validated within the available development period and with the technical resources at the researchers' disposal.

The Point of Sale module represents the most frequently used component of the system and has therefore received particular attention in the functional scoping process. Its design prioritizes transactional speed, recognizing that during peak service hours at Empress Café Boracay, staff may process dozens of orders within brief time windows. The interface is structured to minimize the number of interactions required to complete a sale, with product categories displayed prominently, popular items accessible through single-click shortcuts, and quantity adjustments achievable through intuitive increment and decrement controls. Payment processing supports cash transactions, which remain predominant in the local market, alongside digital payment methods that have gained increasing adoption among tourists visiting Boracay. The system calculates change automatically, applies configured tax rates, and generates printed or digital receipts without requiring manual calculation or transcription. Discount application is supported for straightforward percentage-based or fixed-amount reductions, though complex promotional rules such as buy-one-get-one configurations, time-limited flash sales, or tiered loyalty rewards are excluded from the functional scope. This limitation reflects the café's current promotional practices and avoids the interface complexity that would accompany more elaborate discount structures.

The inventory tracking functionality operates through a tightly coupled integration with the Point of Sale module, ensuring that every completed transaction automatically triggers appropriate stock deductions based on predefined recipe mappings. When a customer orders a cappuccino, for instance, the system reduces available quantities of espresso beans, milk, and any flavored syrups according to the standard recipe proportions stored in the product configuration. This automated deduction eliminates the lag and potential for human error that characterizes manual inventory updating, where staff might forget to record consumption or record incorrect quantities during busy periods. The inventory module also supports manual adjustments for scenarios that fall outside normal sales-driven consumption, including receipt of new stock from suppliers, recording of spoilage or waste due to preparation errors or expired ingredients, and corrections identified during physical stock counts. Each adjustment is logged with timestamp, user identification, and reason code to maintain an auditable trail of inventory movements. The system generates low-stock alerts when item quantities fall below administrator-defined thresholds, enabling proactive reordering before stockouts disrupt service. However, the functional scope does not extend to automated purchase order generation, supplier communication, or delivery scheduling, which would require integration with external systems and workflows beyond the café's current operational infrastructure.

1.5 Significance of the Study

This study is beneficial to several distinct groups, each of which derives specific value from the development, implementation, and evaluation of the Operlytics system. Understanding these multiple beneficiary groups illuminates the broader impact and relevance of the research beyond its immediate operational objectives.

Beneficiaries: Administrators (Management)

The system will help administrators monitor sales performance, track inventory levels, and analyze operational data more efficiently, allowing them to make better and faster business decisions. This benefit addresses the core management need for reliable, timely, and accessible operational information that supports effective oversight and strategic direction.

By replacing manual and inconsistent reporting practices with a centralized and automated platform, management will gain a clearer and more reliable view of the café's operational standing. The current environment of fragmented paper records, incomplete data, and delayed reporting creates conditions in which management must make decisions based on incomplete or outdated information. Operlytics transforms this environment by providing a single, authoritative source of operational truth that is continuously updated and immediately accessible.

The clarity and reliability of information provided by Operlytics enable management to respond proactively to emerging challenges and opportunities within the business. Rather than discovering problems only after they have caused significant disruption, management can identify developing trends and take preventive action. Rather than missing opportunities because they were not visible in manual records, management can recognize positive patterns and reinforce them. The shift from reactive to proactive management, enabled by reliable operational data, represents a fundamental improvement in management capability.

The efficiency gains for management are substantial. Time previously devoted to gathering, compiling, and reconciling manual records can be redirected to higher-value activities such as customer relationship development, staff mentoring, menu innovation, and business strategy formulation. The mental burden of trying to maintain operational awareness through

incomplete and scattered information is replaced by the confidence of having accurate, comprehensive data at one's fingertips.

Business Intelligence (BI) systems transform raw operational data into actionable insights, enabling managers to identify trends, optimize resource allocation, and respond swiftly to market changes. For café administrators, this translates into improved profitability through waste reduction, optimized staffing levels, and strategic menu planning based on actual consumption patterns rather than intuition.

Beneficiaries: End Users (Café Staff)

The system will assist staff in recording and monitoring transactions and inventory, helping reduce manual work and possible errors in daily operations. This benefit addresses the operational reality of café staff who must balance multiple responsibilities while maintaining accuracy and efficiency in high-pressure service environments.

With a more streamlined and intuitive system in place, staff members will be able to carry out their responsibilities more accurately and efficiently. The POS interface simplifies transaction recording through intuitive product selection, automatic calculation, and clear confirmation feedback. The reduction of manual recording tasks eliminates tedious paperwork that staff often find burdensome and error-prone. The standardization of processes ensures that all staff follow consistent procedures, reducing confusion and training requirements.

The time and energy freed from manual recording tasks can be redirected toward delivering better customer service and improving the overall dining experience at the café. When staff are not preoccupied with paperwork, they can focus their attention on customers, ensuring prompt, friendly, and attentive service. The improved accuracy of operational records also benefits staff by reducing the stress and conflict that often arise when discrepancies are discovered during shift changes or management review.

Staff members also benefit from the system's clarity and transparency. When inventory records are accurate and accessible, staff can confidently inform customers about item availability. When sales records are reliable, staff can trust that their performance is being fairly

assessed. When operational processes are standardized, staff can perform their duties with greater confidence and consistency.

Modern POS systems have evolved beyond simple transaction processing to become comprehensive operational tools that enhance employee productivity and job satisfaction. For Empress Café Boracay staff, Operlytics represents a digital upgrade that simplifies their daily workflows while providing greater clarity about operational expectations. The system's intuitive interface reduces the cognitive load associated with manual record-keeping, allowing staff to focus their attention on customer interaction and service quality. Standardized digital workflows eliminate the ambiguity and inconsistency that often characterize informal manual processes, giving staff clear guidance on how to perform operational tasks correctly.

The automated inventory deduction feature relieves staff of the responsibility for manually updating stock records after each sale, a task that is frequently overlooked or performed inaccurately during busy periods. When inventory levels are automatically maintained by the system, staff can confidently inform customers about product availability without needing to perform physical checks or consult separate records. The system's real-time alerts for low-stock conditions also empower staff to proactively notify management of replenishment needs, transforming them from passive record-keepers into active participants in operational management.

Staff members benefit from the system's transparency and fairness in performance assessment. When sales records are accurately and automatically captured, staff can trust that their contribution to revenue generation is properly documented. This transparency reduces disputes over sales attribution and creates a more equitable work environment. The system's audit trail of user activities also provides protection for staff by creating an objective record of their actions, which can be referenced in case of discrepancies or misunderstandings.

The training and skill development opportunities associated with system adoption represent an additional benefit for café staff. Familiarity with digital operational systems enhances staff employability and prepares them for career advancement in an increasingly technology-driven hospitality industry. The structured, standardized processes enforced by the

system also build professional discipline and operational competence that serve staff well in their current and future employment.

Beneficiaries: Future Researchers

The study will serve as a reference for future researchers and developers who are interested in creating similar systems for small businesses, particularly in the food service and hospitality sectors. The detailed documentation of system development, implementation, and evaluation processes provides a replicable model that can be adapted to diverse operational contexts.

This research contributes to the academic literature on information systems in small business contexts, a domain that has received less research attention than enterprise-level implementations despite the vastly larger number of small businesses worldwide. The detailed documentation of system development, implementation, and evaluation processes provides valuable empirical material for researchers studying technology adoption, digital transformation, and business analytics in small enterprises.

The study's methodological approach, which combines developmental research design with agile implementation practices and participatory user engagement, offers a model for conducting technology development research in close collaboration with real business partners. This approach addresses a common limitation of academic information systems research, which often produces theoretically sound but practically inapplicable solutions due to insufficient engagement with real-world operational constraints. By grounding the research in the actual needs, capabilities, and context of a specific small business, the study demonstrates how academic research can generate both theoretical insights and practical value.

The study also contributes to the literature on tourism-dependent small businesses, examining how technology can address the unique operational challenges of seasonal demand fluctuations, high customer turnover, and competitive density that characterize tourism-intensive markets. This focus fills a gap in the hospitality and tourism management literature, which has tended to focus on larger enterprises or generic small business challenges without adequate attention to the tourism context.

For researchers interested in business analytics and business intelligence, the study provides empirical evidence of how descriptive and diagnostic analytics capabilities can generate tangible operational benefits in resource-constrained environments. The evaluation framework, which combines technical performance metrics with user acceptance measures and operational impact assessment, offers a comprehensive approach to evaluating system effectiveness in small business contexts where traditional enterprise evaluation criteria may be inappropriate.

Beneficiaries: Small Business Community

Beyond the immediate stakeholders at Empress Café Boracay, this study holds broader significance for the thousands of small food service businesses operating throughout the Philippines, particularly in tourism-dependent locations. The study demonstrates that appropriately designed, affordable, and accessible technology can address real operational challenges in resource-constrained small business environments. By documenting the development process, implementation experience, and evaluation results, the study provides a replicable model that other small businesses can adapt to their specific contexts.

The Philippine food service sector is dominated by micro, small, and medium enterprises (MSMEs) that face operational challenges similar to those identified at Empress Café Boracay. Many of these establishments continue to rely on manual operational methods not because they are optimal but because they perceive automated systems as too expensive, too complex, or inappropriate for their scale. This study challenges these perceptions by demonstrating that effective operational technology can be developed and deployed within the constraints of small business budgets and technical capabilities.

The study also contributes to the policy discourse on MSME digitalization in the Philippines. Government initiatives such as the MSME Development Plan and various digital transformation programs seek to accelerate technology adoption among small businesses. This research provides empirical evidence of the operational benefits that such adoption can generate, supporting the case for continued and expanded government support for small business technology adoption. The documented experience of Empress Café Boracay can inform the design of more effective support programs that address the specific needs and constraints of small food service businesses.

At the broader economic development level, this study contributes to understanding how digital transformation can support the sustainability and competitiveness of small enterprises in developing economies. Small businesses are the backbone of employment and economic activity in the Philippines and many similar emerging economies, yet they often lack the technological tools that could enhance their productivity and resilience. By demonstrating the feasibility and benefits of technology adoption in a typical small business context, the study supports the case for policies and investments that expand digital access and capability among small enterprises.

The tourism sector, which contributes significantly to the Philippine economy and employment, depends heavily on the quality and competitiveness of small service businesses such as restaurants, cafés, and accommodation providers. Improvements in the operational efficiency of these businesses translate directly into enhanced visitor experiences, increased tourist spending, and more sustainable local economic development. The digital empowerment of small tourism businesses thus represents a pathway to broader economic benefits that extend beyond individual enterprises to communities and regions.

1.6 Definition of Terms

For clarity and consistency throughout this study, the following terms are defined as they are used in this research:

Business Analytics refers to the technologies, applications, and practices that organizations use to collect, process, analyze, and present operational and business data to support decision-making. In the context of this study, business analytics encompasses the descriptive and diagnostic analysis of sales, inventory, and operational data to generate insights that guide management decisions.

Business Intelligence (BI) refers to the broader organizational capability to transform data into actionable insights through the integration of data collection, analysis, and reporting processes. While often used interchangeably with business analytics, BI in this study emphasizes the systematic organizational capacity for data-driven decision-making that Operlytics seeks to establish at Empress Café Boracay.

Dashboard refers to a visual display within the Operlytics system that presents key operational metrics and performance indicators in a consolidated, at-a-glance format. Dashboards use charts, graphs, gauges, and summary tables to enable rapid comprehension of business status and identification of areas requiring attention.

Inventory refers to the stock of ingredients, supplies, and materials maintained by Empress Café Boracay for use in preparing and serving food and beverage products. Inventory encompasses both perishable items with limited shelf life and non-perishable items with longer storage durations.

Inventory Tracking refers to the systematic monitoring and recording of inventory quantities, movements, and status within the Operlytics system. This includes recording receipts from suppliers, deductions from sales consumption, adjustments for waste and spoilage, and monitoring of stock levels against defined thresholds.

Manual Record-Keeping refers to the current operational practice at Empress Café Boracay of using handwritten entries in paper logbooks, notebooks, or informal notes to document sales

transactions, inventory counts, and other operational activities. This practice is characterized by decentralization, inconsistency, and susceptibility to error and loss.

Micro, Small, and Medium Enterprises (MSMEs) refers to business establishments classified according to asset size and employment levels under Philippine law. For this study, the term specifically encompasses small food service businesses such as cafés and restaurants with limited capital, workforce, and technical resources.

Operational Efficiency refers to the ratio of operational outputs (such as sales revenue, customer service quality, and product availability) to operational inputs (such as staff time, inventory investment, and management attention). Improvements in operational efficiency result from reducing waste, eliminating errors, accelerating processes, and optimizing resource allocation.

Operlytics refers to the web-based business analytics and monitoring system developed and evaluated in this study, specifically designed for the operational context of Empress Café Boracay. The name combines "operations" and "analytics" to reflect the system's purpose of analyzing operational data to support management decision-making.

Point of Sale (POS) refers to the location and moment where a customer transaction is completed, and in the context of this study, also refers to the software module within Operlytics that enables staff to record, process, and manage sales transactions.

Real-Time Monitoring refers to the capability of the Operlytics system to capture, process, and display operational data with minimal delay between transaction occurrence and information availability. Real-time monitoring enables management to observe current operational status rather than relying on delayed or historical summaries.

Sales Transaction refers to the complete record of a customer purchase at Empress Café Boracay, including the items ordered, quantities, prices, payment method, timestamp, and staff member who processed the transaction.

Small and Medium-Sized Enterprises (SMEs) refers to business establishments with limited scale in terms of employees, revenue, and assets. In the Philippine context, this includes

businesses that fall below the thresholds defined for large corporations and that typically operate with constrained resources and informal management structures.

Stockout refers to a situation where an inventory item required for operations is unavailable when needed, resulting in inability to fulfill customer orders, compromised product quality, or emergency procurement at premium costs.

Tourism-Intensive Environment refers to a business location where operations are heavily influenced by tourist visitor traffic, characterized by seasonal demand fluctuations, high customer turnover, diverse customer preferences, and competitive density of service providers.

Web-Based System refers to software applications that operate through standard web browsers rather than requiring installation of specialized client software. Web-based systems are hosted on remote servers and accessed via internet connectivity, enabling centralized data management and multi-device accessibility.

1.7 Conceptual Framework

This study is guided by a conceptual framework that integrates theoretical perspectives from information systems, small business management, and technology adoption to structure the research design, system development, and evaluation approach.

The conceptual framework posits that the operational challenges faced by Empress Café Boracay stemming from manual record-keeping, information opacity, inefficient reporting, inadequate inventory monitoring, and absence of analytical capabilities can be addressed through the development and implementation of a purpose-built business analytics system. The framework further posits that the effectiveness of such a system depends not only on its technical functionality but also on its alignment with the operational context, user capabilities, and decision-making needs of the target organization.

The framework identifies three interconnected dimensions that influence system success:

Technical Dimension encompasses the system's functional capabilities, reliability, performance, and security. This dimension addresses whether the system technically performs its intended functions correctly and efficiently. Technical success requires that the system accurately captures and processes transaction data, maintains data integrity across all operations, generates correct reports and visualizations, responds promptly to user requests, operates reliably without crashes or data loss, and protects operational data from unauthorized access or corruption.

Operational Dimension encompasses the system's integration with existing workflows, its support for daily operational tasks, and its contribution to operational efficiency and effectiveness. This dimension addresses whether the system fits the operational context and improves business processes. Operational success requires that the system supports rather than disrupts daily service activities, reduces the time and effort required for core tasks, improves the accuracy and timeliness of operational information, enables proactive management responses, and generates measurable improvements in key operational outcomes such as waste reduction, stockout prevention, and cost control.

Human Dimension encompasses user acceptance, usability, training requirements, and the system's impact on staff workload and job satisfaction. This dimension addresses whether

users can and will effectively utilize the system. Human success requires that the system interface is intuitive and learnable for users with limited technical background, that system functions align with user mental models and work practices, that training requirements are minimal and achievable, that users perceive the system as useful and beneficial to their work, and that system adoption does not create undue stress or resistance among staff.

The framework hypothesizes that successful system implementation requires positive outcomes across all three dimensions, and that deficiencies in any dimension can compromise overall system effectiveness. A technically excellent system that is difficult to use will not achieve operational benefits because users will avoid or misuse it. A user-friendly system that produces inaccurate data will undermine operational decision-making. An operationally effective system that is technically unreliable will lose user trust and fail to sustain adoption. The evaluation methodology therefore assesses outcomes across all three dimensions to provide a comprehensive understanding of system impact.

This three-dimensional framework is informed by established theoretical perspectives on information systems success. The Technology-Organization-Environment (TOE) framework provides a structural lens for understanding the factors that influence technology adoption decisions and outcomes in organizational contexts, identifying technological context, organizational context, and environmental context as key determinants of adoption success. The Unified Theory of Acceptance and Use of Technology (UTAUT) provides a user-centered perspective, identifying performance expectancy, effort expectancy, social influence, and facilitating conditions as key determinants of user acceptance and actual system use. The Socio-Technical Systems Perspective emphasizes that effective information systems must achieve joint optimization of both technical and social components, cautioning against purely technical approaches that neglect the human dimensions of system design and use.

Together, these theoretical perspectives provide a comprehensive framework for understanding the multifaceted challenges of developing and implementing business analytics systems in small business contexts, guiding both the technical development of Operlytics and the evaluation of its effectiveness.

1.8 Summary

This chapter has established the foundation for the study by presenting the background and context of operational challenges faced by small food service businesses, particularly in tourism-intensive environments like Boracay Island. It has identified the specific operational deficiencies at Empress Café Boracay that motivate the development of Operlytics, articulated the general and specific objectives that guide the research, defined the scope and boundaries of the study, explained its significance for multiple stakeholders, defined key terms for consistency, and presented the conceptual framework that structures the research approach.

The following chapter reviews the relevant literature on business analytics, small business information systems, food service operations management, and technology adoption in developing economies to establish the theoretical and empirical foundation for the system design and evaluation.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Business analytics has emerged as a critical and transformative tool for improving operational decision-making across virtually every industry sector in the contemporary global economy. The evolution of business analytics from simple reporting tools to sophisticated platforms capable of processing vast quantities of data in real time represents one of the most significant developments in management practice over the past two decades. Organizations that have successfully integrated business analytics into their operational frameworks consistently demonstrate superior performance compared to those that rely on traditional, intuition-based decision-making approaches. This section examines foreign studies that illuminate the theoretical foundations, practical applications, and empirical outcomes of business analytics implementation across diverse organizational contexts.

According to Alomari (2025), business analytics has become a transformative tool in driving data-informed decision-making across industries, including finance, retail, healthcare, manufacturing, and telecommunications. Alomari's comprehensive review highlights the key trends in business analytics adoption and evaluates the effectiveness of popular tools such as data visualization, predictive analytics, machine learning, and big data processing. The findings indicate significant improvements in decision-making efficiency and effectiveness across all sectors examined. The study emphasizes that the integration of business analytics into organizational processes enables managers to move beyond gut feeling and experience-based decisions toward evidence-based strategies grounded in objective data analysis.

The implications of Alomari's findings for small food service businesses are substantial. While the study focuses primarily on large-scale industrial applications, the underlying principles that data-driven decision-making improves outcomes are universally applicable. Small cafés like Empress Café Boracay, despite their limited scale, generate substantial operational data through

daily transactions, inventory movements, and customer interactions. The challenge lies not in data scarcity but in the absence of systems capable of capturing, organizing, and analyzing this data to extract actionable insights. Alomari's work supports the fundamental premise of this study: that business analytics, appropriately scaled and designed, can deliver decision-making improvements regardless of organizational size.

Building on the foundational understanding of business analytics, Islam (2024) explored the integration of artificial intelligence in business analytics and its impact on operational efficiency. Islam's study, published in the *World Journal of Advanced Engineering Technology and Sciences*, notes that with the advent of AI technologies such as machine learning and natural language processing, businesses can now leverage advanced analytics to enhance operational performance in ways previously impossible. The study confirmed that AI-driven analytics addresses existing limitations in traditional business analytics by providing real-time insights, predictive capabilities, and automation, ultimately improving decision-making capabilities and enabling more informed and effective strategic choices.

Islam's research is particularly relevant to the current study because it highlights the trajectory of business analytics evolution from descriptive reporting toward predictive and prescriptive capabilities. While Operlytics, as designed for Empress Café Boracay, focuses primarily on descriptive and diagnostic analytics due to resource and complexity constraints, the system architecture and data structures established by Operlytics create a foundation upon which more advanced analytical capabilities could be built in future iterations. The data collection and organization processes implemented by Operlytics would support the integration of machine learning models for demand forecasting, inventory optimization, and customer behavior analysis should the café's operational maturity and resources support such enhancements in the future.

The study also emphasizes the automation benefits of AI-enhanced analytics, noting that automated data processing and insight generation reduce the cognitive and temporal burdens on managers, allowing them to focus on strategic decision-making rather than operational data compilation. This finding directly supports the design philosophy of Operlytics, which automates the collection, aggregation, and presentation of operational data to free management from manual reporting tasks.

In a similar vein, Kunc and Bäckström (2025) explored top managers' views on using business analytics in decision-making, providing valuable insights into how organizational leaders perceive and utilize analytical tools in their daily management activities. Kunc and Bäckström's research, published in *Measuring Business Excellence*, revealed that top managers see business analytics as crucial for business management, particularly in monitoring, managing, and adjusting ongoing operations. The primary use of business analytics was found to be following up on business performance and making decisions on operational adjustments to help achieve short-term objectives, wherein top managers monitor the firm's progress towards its targets within the constraints of its existing resources.

This finding is especially significant for small business contexts where management resources are severely constrained. Unlike large corporations that can dedicate specialized personnel to business intelligence functions, small café owners must personally oversee operational management while simultaneously handling strategic planning, customer relations, staff supervision, and financial oversight. For these managers, business analytics tools that provide quick, clear operational status updates are particularly valuable because they minimize the time and effort required to maintain situational awareness. Kunc and Bäckström's research supports the design emphasis of Operlytics on dashboard-style visualizations that provide at-a-glance operational summaries rather than complex analytical interfaces requiring extensive interpretation.

The study also highlights the importance of operational adjustment decisions and short-term tactical changes made in response to emerging conditions as the primary application of business analytics in managerial practice. This finding aligns perfectly with the operational focus of Operlytics, which is designed to support daily operational decisions such as inventory replenishment, menu adjustments, and staffing optimizations rather than long-term strategic planning. The system's real-time monitoring capabilities directly enable the kind of operational adjustment decision-making that Kunc and Bäckström identify as the most common and valuable application of business analytics.

The impact of real-time business intelligence and advanced analytics on decision-making has also been extensively explored in the academic literature. Research by Venkat (2024) indicates that these tools provide insights into decision-makers' behavior and offer recommendations on how business analytics solutions can be improved to support timely and effective decisions. Venkat's work emphasizes that the value of business analytics is not merely in the data provided but in how that data is presented and accessed to align with the cognitive processes and decision-making styles of users. This user-centered perspective on business analytics design is directly incorporated into the Operlytics development approach, which prioritizes usability and intuitive interaction as core design requirements.

Furthermore, the role of organizational decision-making processes in converting business analytics-based insights into effective decisions has been examined by Lee and Venkatraman (2022), highlighting the need for changes in decision-making processes to capture value from business analytics investments. Their research emphasizes that technology implementation alone is insufficient for realizing the benefits of business analytics; organizations must also evolve their decision-making processes to effectively utilize the insights generated. This finding has important implications for the implementation strategy of Operlytics at Empress Café Boracay. The successful deployment of the system requires not merely technical installation but also organizational adaptation training staff to use the system, establishing new workflows that leverage system capabilities, and developing management habits of consulting system data before making operational decisions.

Much attention is currently being paid in both the academic and practitioner literatures to the value that organizations could create through the use of big data and business analytics. Gillon et al. (2012) and Mithas et al. (2013) have contributed foundational work establishing the theoretical and empirical basis for business analytics value creation. Chen et al. (2012, pp. 1166–1168) suggest that business analytics and related technologies can help organizations to "better understand its business and markets" and "leverage opportunities presented by abundant data and domain-specific analytics." Similarly, LaValle et al. (2011, p. 22) report that top-performing organizations "make decisions based on rigorous analysis at more than double

the rate of lower performing organisations" and that in such organizations analytic insight is being used to "guide both future strategies and day-to-day operations."

These findings establish a clear empirical basis for the value proposition of business analytics: organizations that systematically utilize data analysis in their decision-making processes outperform those that do not. While the studies cited focus primarily on large organizations, the underlying logic applies with equal or greater force to small enterprises. Small businesses like Empress Café operate with narrower margins for error, more limited resources to absorb losses from poor decisions, and less capacity to recover from operational mistakes. For these businesses, the improvement in decision quality enabled by business analytics may be even more consequential than for larger organizations with greater buffers against suboptimal outcomes.

The research of Sharma, Mithas, and Kankanhalli (2014), published in the *European Journal of Information Systems*, argues that while there is evidence that investments in business analytics can create value, the thesis that "business analytics leads to value" requires deeper analysis. Specifically, they propose that the roles of organizational decision-making processes, including resource allocation processes and resource orchestration processes, need to be better understood in order to comprehend how organizations can create value from business analytics. The first-order effects of business analytics are likely to be on decision-making processes, and improvements in organizational performance are likely to be an outcome of superior decision-making processes enabled by business analytics.

This nuanced perspective is crucial for understanding how Operlytics can create value at Empress Café Boracay. The system does not automatically generate business value simply by collecting and displaying data. Value creation occurs through the mediating mechanism of improved decision-making processes management making better choices about inventory, pricing, menu composition, and operational workflows because they have access to accurate, timely information. The design of Operlytics therefore focuses not merely on data collection but on presenting that data in formats that directly support the operational decisions that café management must make on a daily basis.

Stephen Few (2013), in his influential work on information dashboard design, emphasizes that management may use data visualization technologies to better analyze corporate information. Visual dashboards and graphs assist businesses in presenting complicated data in an understandable way. Managers can quickly spot trends, patterns, and possible operational problems thanks to this visual presentation of data. Few's principles of effective dashboard design clarity, simplicity, relevance, and immediacy directly inform the visualization component of Operlytics, ensuring that the system's dashboards communicate operational status effectively to users who may lack formal training in data analysis.

Adebanji Samuel Ogunmokun, Emmanuel Damilare Balogun, and Kolade Olusola Ogunsola (2024) developed a Business Intelligence (BI) dashboard optimization model designed to enhance real-time performance tracking and forecasting accuracy for improved decision-making in modern organizations. Their research explores the evolution of BI dashboards, the importance of real-time tracking, and the integration of forecasting tools to enable organizations to make timely, data-driven decisions. By leveraging machine learning algorithms and cloud-based solutions, the proposed model optimizes performance tracking, integrates various data sources, and offers predictive capabilities that help businesses forecast future trends with greater precision.

The research highlights the challenges of implementing BI dashboards, such as data integration, user training, and technological constraints, and proposes solutions to address these issues, ensuring that the model is both scalable and user-friendly. The findings demonstrate that the optimized BI dashboard model enhances operational efficiency, supports better decision-making, and improves forecasting accuracy, particularly in dynamic environments like retail, finance, and manufacturing. For the context of Empress Café Boracay, this research validates the approach of using dashboard-based visualization as the primary interface for operational monitoring, while also acknowledging the implementation challenges that must be addressed through careful design, training, and technical support.

2.2 Local Studies

The Philippine context for business analytics and operational system development presents unique characteristics that distinguish it from the foreign studies examined above. Local research on information systems in small business contexts provides valuable insights into the specific challenges, opportunities, and implementation strategies relevant to Filipino enterprises operating in the country's distinctive economic, cultural, and technological environment. Unlike large corporations with dedicated IT departments and substantial capital reserves, Philippine micro, small, and medium enterprises (MSMEs) frequently operate with constrained resources, limited technical expertise, and infrastructure challenges that shape how technology adoption unfolds in practice. The archipelagic geography of the Philippines, combined with variable internet connectivity across regions and the predominance of cash-based transactions in many local markets, creates implementation conditions that differ significantly from those assumed in foreign-developed frameworks. Local studies therefore offer essential empirical grounding for understanding how business analytics and operational systems can be designed, deployed, and sustained within these specific constraints.

Arig et al. (2025) highlighted that in today's digital era, every business must adapt and innovate to stay competitive. Their study on the development of a Point-of-Sale (POS) system with a card reader for the HTM Department emphasized that incorporating a POS system transforms business transactions and empowers businesses to seamlessly manage sales and customer interactions. Arig et al. further noted that the system ensures smooth transactions, enhances accuracy, and reduces human error, making it a great tool for preparing students and businesses in facing real-world operations. The researchers employed a developmental research design similar to the approach adopted in the present study, systematically progressing from needs analysis through system design, development, and evaluation. Their findings demonstrated that even relatively simple POS implementations could yield substantial improvements in transaction processing speed and accuracy, with error rates decreasing by approximately 35% compared to manual recording methods. The integration of card reader functionality addressed the growing demand for cashless payment options among Filipino consumers, reflecting broader trends in digital financial inclusion accelerated by platforms such as GCash and Maya.

This study is directly relevant to Operlytics because it demonstrates the value of POS systems in educational and small business contexts within the Philippines. The emphasis on accuracy enhancement and error reduction aligns with the core objectives of Operlytics, particularly Specific Objective 4 which targets measurable improvement in operational efficiency through reduced manual recording errors. The finding that POS systems prepare businesses for real-world operations supports the argument that digital operational systems are becoming essential rather than optional for business competitiveness in the Philippine market. Furthermore, Arig et al.'s focus on usability for users with limited technical backgrounds resonates with the design philosophy of Operlytics, which prioritizes intuitive interfaces requiring no advanced technical knowledge. The researchers' observation that successful implementation depended as much on user training and organizational readiness as on technical functionality reinforces the importance of the human dimension in the conceptual framework guiding this study.

Tanaman et al. (2023) developed a web-based inventory management system for a business in the Philippines to address the challenges of manual inventory recording and reporting. According to Tanaman et al., the software eliminated manual approaches through an automated computing solution using a web platform where data are electronically recorded and processed, and reports are electronically produced. The system was evaluated using ISO 9126 quality standards and received high scores in functionality, maintainability, efficiency, and compatibility, confirming its effectiveness in streamlining inventory operations. The researchers specifically addressed the problem of inventory discrepancies that plagued their target business, where stockouts occurred approximately three times more frequently than management initially estimated due to the lag between actual consumption and manual record updates. Their web-based architecture enabled real-time inventory visibility from multiple access points, allowing managers to check stock levels remotely rather than requiring physical presence at the storage location.

Tanaman et al.'s research provides strong validation for the web-based approach adopted in Operlytics. Their use of ISO 9126 quality standards for evaluation offers a methodological model that could be adapted for assessing Operlytics, particularly in evaluating the efficiency and functionality criteria outlined in Section 3.7. The high scores achieved in functionality,

maintainability, efficiency, and compatibility demonstrate that well-designed web-based inventory systems can meet rigorous quality standards, suggesting that similar quality outcomes are achievable for Operlytics with appropriate design and implementation effort. The researchers' finding that automated reporting reduced report generation time from several hours to mere minutes directly supports Specific Objective 3 of this study regarding automated reports and dashboards. Their experience also highlights the importance of selecting appropriate evaluation frameworks; the ISO 9126 standard's multi-dimensional assessment parallels the five evaluation criteria (functionality, usability, accuracy, security, efficiency) employed in this study, suggesting that comprehensive quality evaluation is both feasible and necessary for web-based operational systems in Philippine small business contexts.

Bienvenido (2023) conducted a study assessing the usability of a centralized Point-of-Sale (POS) system for inventory management in Bounty Agro Ventures, Inc. Rotisserie Operations. The study explained that the company adopted a centralized POS system to streamline its inventory management practices and address inaccuracy concerns brought about by manual processes. Bienvenido further found that when sales transactions are directly connected to inventory updates, businesses can more easily track product usage and maintain accurate stock records, which helps businesses manage their operations better. The researcher employed a mixed-methods approach combining System Usability Scale (SUS) questionnaires with observational task analysis to comprehensively assess how effectively the centralized POS system supported actual operational workflows. Findings revealed that the direct connection between sales transactions and inventory updates eliminated the reconciliation delays that previously required staff to manually cross-reference sales receipts against stock counts, a process that consumed approximately 15% of total staff hours and introduced errors in roughly 12% of inventory records.

Bienvenido's findings directly support the integrated POS-inventory approach of Operlytics. The critical insight that connecting sales transactions to inventory updates enables better tracking and more accurate stock records is precisely the operational mechanism that Operlytics implements through its auto-deduct ingredients feature. This real-time synchronization addresses Specific Problem 4 (Inadequate Stock Level Monitoring) by ensuring

that inventory levels reflect actual consumption immediately rather than after periodic manual counts. The study's focus on usability assessment also reinforces the importance of user-centered design in system development, particularly for systems intended for operational staff who may have limited technical backgrounds. Bienvenido's finding that usability scores correlated strongly with adoption rates and error reduction suggests that the usability criterion in this study's evaluation framework (Section 3.7) is not merely a secondary consideration but a critical determinant of operational impact. The researcher's observation that centralized systems enabled consistent data across multiple service points also supports the web-based architecture decision for Operlytics, which ensures that all users access the same current information regardless of their physical location within the café.

Cruz et al. (2022) developed a web-based sales and inventory system for small retail businesses in the Philippines to address inefficiencies caused by manual record-keeping. The study found that traditional methods of tracking sales and inventory often resulted in data inaccuracies, delayed reporting, and difficulty in monitoring stock levels. By implementing a computerized system, the researchers were able to automate transaction recording, improve inventory tracking, and generate real-time reports. The results showed a significant improvement in operational efficiency, data accuracy, and decision-making speed among business owners. The study emphasized that integrating sales and inventory systems is essential for small businesses to remain competitive and organized in a fast-paced market environment. Cruz et al. specifically documented how the absence of integration between sales and inventory functions in manual systems created information silos, where sales staff operated with no visibility into stock availability while inventory managers lacked awareness of sales velocity, resulting in frequent stockouts of popular items and overstocking of slow-moving products.

Cruz et al.'s research provides perhaps the closest parallel to the current study, as it addresses the same core problems—manual record-keeping inefficiencies, data inaccuracies, and delayed reporting—in the same national context and for similar small business establishments. Their finding of significant improvement in operational efficiency, data accuracy, and decision-making speed after system implementation offers an empirical benchmark for the expected outcomes of Operlytics deployment at Empress Café Boracay. The emphasis on system

integration as essential for competitiveness reinforces the design decision to combine POS, inventory, and reporting functions in a unified platform rather than as separate, disconnected tools. Cruz et al.'s observation that integration eliminated the need for duplicate data entry and reconciliation efforts directly supports the operational efficiency objective of this study. Their research also highlights the particular importance of integration for small businesses with limited staff, where the same individuals often perform multiple roles and cannot afford the time overhead of maintaining separate records. This finding is especially relevant to Empress Café Boracay, where staff members currently maintain individual sales records and perform sporadic inventory counts without systematic coordination.

Dela Peña and Santos (2021) conducted a study on the implementation of a Point-of-Sale (POS) and inventory management system in a local food service business in the Philippines. The research aimed to improve transaction processing and inventory monitoring through system automation. Findings revealed that the use of a POS system reduced transaction time, minimized human errors, and ensured that inventory levels were automatically updated after each sale. Additionally, the system provided summarized reports that helped managers analyze sales trends and make informed operational decisions. The study concluded that adopting automated systems significantly enhances business performance, particularly in improving accuracy, efficiency, and overall management control. The researchers documented specific quantitative improvements: transaction processing time decreased by an average of 40% during peak hours, inventory record accuracy improved from 73% to 96%, and the time required for daily sales reporting decreased from approximately 2 hours to 15 minutes.

Dela Peña and Santos's study is especially relevant because it focuses specifically on food service businesses in the same industry context as Empress Café Boracay. Their findings regarding reduced transaction time, minimized human errors, automatic inventory updates, and improved management control through summarized reports directly map onto the expected benefits of Operlytics. The study's conclusion that automated systems significantly enhance business performance in accuracy, efficiency, and management control provides strong empirical support for the research hypothesis underlying the current study. The quantitative metrics reported by Dela Peña and Santos offer concrete targets against which Operlytics can be

evaluated; for instance, the 40% reduction in transaction time and the improvement in inventory accuracy to 96% provide benchmarks for assessing whether Operlytics achieves comparable operational improvements at Empress Café Boracay. Their finding that summarized reports enabled managers to identify sales trends previously obscured by fragmented manual records directly supports the analytical capabilities emphasized in Specific Objective 5 (System Evaluation Based on User Feedback) and the business intelligence components of the conceptual framework. The researchers' observation that food service businesses face unique challenges due to perishable inventory and high customer turnover further validates the focus on real-time monitoring in this study's design.

Synthesis of Local Studies

The reviewed local studies collectively demonstrate that Philippine small businesses, particularly in the food service and retail sectors, face operational challenges fundamentally similar to those identified at Empress Café Boracay. Manual record-keeping, fragmented information systems, and delayed reporting emerge as persistent problems across diverse business contexts, from educational department operations (Arig et al., 2025) to rotisserie chains (Bienvenido, 2023) to small retail establishments (Cruz et al., 2022). The convergence of findings across these studies suggests that these challenges are not idiosyncratic to particular businesses but represent systemic patterns characteristic of MSMEs operating with limited technological infrastructure.

Several critical insights emerge from this synthesis. First, **web-based architectures consistently prove effective** for Philippine small businesses, offering accessibility without requiring substantial hardware investment, centralized data management that eliminates fragmentation, and remote monitoring capabilities that accommodate managers who cannot remain physically present at all times. Tanaman et al. (2023) and Cruz et al. (2022) both validated this architectural choice empirically, providing precedent for the PHP-MySQL-HTML/CSS/JavaScript stack selected for Operlytics.

Second, **integration between sales and inventory functions emerges as a non-negotiable requirement** rather than an optional enhancement. Bienvenido (2023) and Dela Peña and Santos (2021) both demonstrated that the real-time connection between transaction

processing and inventory updates is the primary mechanism through which operational accuracy improves. This finding directly validates the auto-deduct ingredients feature in Operlytics and the unified database architecture that supports it.

Third, **usability considerations prove decisive for adoption success**. Bienvenido's (2023) explicit focus on usability assessment, Arig et al.'s (2025) emphasis on preparing users for real-world operations, and the consistent finding that users with limited technical backgrounds require intuitive interfaces all reinforce the usability-centered design philosophy of Operlytics. The System Usability Scale employed by Bienvenido offers a validated instrument that could be adapted for this study's evaluation phase.

Fourth, **quantifiable operational improvements are achievable and measurable**. Dela Peña and Santos (2021) provided specific metrics for transaction time reduction and accuracy improvement, while Tanaman et al. (2023) documented ISO 9126 quality scores. These precedents establish that the evaluation criteria employed in this study—functionality, usability, accuracy, security, and efficiency—can yield concrete, defensible results that demonstrate system value to both academic and practitioner audiences.

Finally, the local studies reveal a **critical gap that Operlytics specifically addresses**: while existing research has examined POS systems, inventory management, and sales reporting as separate or partially integrated functions, none of the reviewed local studies developed a **comprehensive business analytics and monitoring system** that combines all three operational domains with real-time dashboard visualization and automated report generation in a single unified platform tailored specifically for small café operations in tourism-intensive environments. Arig et al. (2025) focused on POS with card reader functionality; Tanaman et al. (2023) addressed inventory management exclusively; Bienvenido (2023) examined POS-inventory integration for rotisserie operations; Cruz et al. (2022) developed sales-inventory integration for retail; and Dela Peña and Santos (2021) implemented POS-inventory automation for food service. Operlytics extends beyond these precedents by incorporating **business analytics and data visualization capabilities** that transform operational data into strategic intelligence, addressing the analytical deficiency identified as Specific Problem 5. This extension responds to the broader literature's emphasis on business intelligence and dashboard-based

decision support (Few, 2013; Ogunmokun et al., 2024) while adapting these capabilities to the resource constraints and operational realities of Philippine small caf  s.

The tourism-intensive context of Boracay Island adds dimensions not explicitly addressed in the reviewed local studies: seasonal demand fluctuations, high customer turnover from international visitors, and competitive density that intensifies the consequences of operational inefficiency. While the local studies establish the feasibility and value of operational systems for Philippine small businesses, Operlytics contributes by demonstrating how these systems can be **adapted to the unique pressures of tourism-dependent economies**, where operational agility and real-time responsiveness may determine business survival during peak seasons.

2.3 Related Literature

The theoretical foundations of business analytics and operational information systems extend beyond the specific studies cited above to encompass a broader body of literature on organizational decision-making, information systems value creation, and digital transformation. This section examines key theoretical perspectives that inform the conceptual framework and design philosophy of Operlytics.

Theoretical Foundations of Business Analytics Value Creation

Organizations could create substantial value through the use of big data and business analytics, as established by foundational research from Gillon et al. (2012) and Mithas et al. (2013). Chen et al. (2012, pp. 1166–1168) suggest that business analytics and related technologies can help organizations to "better understand its business and markets" and "leverage opportunities presented by abundant data and domain-specific analytics." Similarly, LaValle et al. (2011, p. 22) report that top-performing organizations "make decisions based on rigorous analysis at more than double the rate of lower performing organisations" and that in such organizations analytic insight is being used to "guide both future strategies and day-to-day operations." These foundational works establish the theoretical premise that data abundance, when properly leveraged through analytics, creates competitive advantage. The "abundant data" that Chen et al. reference exists within small food service establishments just as it does in large corporations; the difference lies not in data availability but in the capacity to capture, organize, and analyze that data. Operlytics addresses this capacity gap by providing the technological infrastructure through which small cafés can begin leveraging their operational data for improved decision-making.

Digitalization and continuously developing technologies have provided firms with an array of new opportunities to capture, store, manage, and share data as well as continuously advancing business analytics tools to derive insight from this data. Firms are investing in their business analytics capabilities to improve their decision-making efficiency and to make more "fact-based" decisions (Davenport and Harris, 2017). By using business analytics, firms aim at making more intelligent decisions faster (Lepeniotti et al., 2020) and improving both their business process and decision-making performance (Chatterjee et al., 2021). This stream of

research emphasizes the speed dimension of business analytics value creation. In fast-paced operational environments like food service, the velocity of decision-making is often as important as the quality of decisions. A good decision made too late may be less valuable than an adequate decision made promptly. The real-time monitoring capabilities of Operlytics directly address this velocity requirement by ensuring that management has immediate access to current operational information, eliminating the delays inherent in manual reporting processes.

Data Visualization and Cognitive Information Processing

Management may use data visualization technologies to better analyze corporate information. Visual dashboards and graphs assist businesses in presenting complicated data in an understandable way, as Stephen Few (2013) emphasizes. Managers can quickly spot trends, patterns, and possible operational problems thanks to this visual presentation. The cognitive benefits of visual data presentation are well-established in human-computer interaction research: the human visual system is highly optimized for pattern recognition, and appropriately designed visualizations can convey complex information more efficiently than text or numerical tables alone.

The Business Intelligence dashboard optimization model developed by Ogunmokun, Balogun, and Ogunsola (2024) demonstrates the evolution of BI dashboards toward real-time performance tracking and forecasting accuracy. As businesses increasingly rely on data-driven insights to stay competitive, effective dashboard systems that integrate real-time data processing and accurate forecasting models are essential. The research highlights the challenges of implementing BI dashboards, such as data integration, user training, and technological constraints, and proposes solutions to address these issues, ensuring that the model is both scalable and user-friendly. The findings demonstrate that the optimized BI dashboard model enhances operational efficiency, supports better decision-making, and improves forecasting accuracy, particularly in dynamic environments like retail, finance, and manufacturing. The study provides actionable recommendations for organizations seeking to implement the model and suggests areas for future research, including the integration of advanced machine learning techniques and external data sources to enhance forecasting accuracy further.

For the current study, this literature on business intelligence dashboards provides design guidance for the visualization components of Operlytics. The emphasis on real-time processing, user-friendly design, and scalability directly informs the dashboard architecture of the proposed system. The challenges identified in data integration, user training, and technological constraints are precisely the implementation risks that the Operlytics development and deployment plan seeks to mitigate through careful requirements analysis, iterative user feedback, and appropriate technology selection.

Digital Transformation and Data-Driven Business Models in Hospitality

The hospitality industry has undergone significant digital transformation, with technology becoming central to operational efficiency and competitive positioning. Recent scholarship emphasizes that digital transformation in hospitality extends beyond mere technology adoption to encompass strategic shifts in how services are delivered, decisions are made, and customer relationships are maintained. This transformation exploits innovations such as artificial intelligence, the Internet of Things, cloud computing, and mobile applications to connect data across systems, automate routine tasks, and anticipate operational needs before they arise.

Data-driven business models (DDBMs) have emerged as a critical framework for innovation in hospitality ecosystems. According to Sorescu (2017), data collection, organization, and analysis can stimulate innovation by using data as a key lever for transformation opportunities. For small and medium enterprises (SMEs) in the food service sector which constitute the majority of café and independent restaurant operations -- the development of data-driven capabilities is particularly crucial. Due to their potential limitations in financial and human resources, SMEs can risk more than large companies in the development of digital business models. However, their greater flexibility makes them more agile in the establishment of external relationships for knowledge exchange (OECD, 2021). The application of DDBMs can help SMEs overcome their dimensional and resource constraints by enhancing their capabilities to analyze the environment, explore new opportunities, and improve decision-making by focusing on data rather than on intuition.

Contemporary industry data indicates that 52% of restaurants plan to dedicate resources toward upgrading or implementing point-of-sale (POS) systems in 2025, with 71% of restaurant

owners using POS data to streamline payments, optimize menus, and boost engagement (Rezku, 2025). Additionally, 86% of businesses use POS data to increase the average order value through upselling and promotional strategies (GloriaFood, 2025). The most common technology investments include online ordering platforms, marketing software, and accounting systems, reflecting an industry-wide recognition that digital infrastructure is essential for operational survival. However, these investments often remain fragmented, lacking the integrated analytics capabilities that would transform transactional data into strategic intelligence. Furthermore, 85% of restaurant operators now cite integration capabilities as their top purchasing priority when selecting POS software, yet 68% of business leaders identify data silos as their primary integration challenge (Swell, 2025), highlighting the persistent gap between data collection and actionable analytics in the sector.

Real-Time Analytics and Operational Velocity in Food Service

The velocity of information processing represents a critical competitive dimension in food service operations. Research on restaurant management systems demonstrates that integrated digital solutions can increase order processing efficiency significantly. Restaurant employees may shift tables up to 20% quicker with the help of mobile POS handhelds (GloriaFood, 2025). The real-time monitoring capabilities essential for operational agility align with broader findings that data-driven decision-making enables businesses to make more accurate, objective decisions that improve efficiency, reduce risks, and enhance customer satisfaction.

In the context of small caf  s, real-time operational data encompasses inventory levels, sales velocity, customer traffic patterns, and staff performance metrics. The integration of these data streams into unified analytics platforms addresses what scholars identify as a primary challenge in hospitality analytics: the fragmentation of data across siloed systems and the consequent inability to derive actionable insights (Swell, 2025). By consolidating operational data into a single analytics framework, Operlytics enables the pattern recognition and trend identification that support both immediate operational adjustments and longer-term strategic planning. Moreover, businesses leveraging integrated POS systems report 9.5% revenue

increases and 20% better cross-channel visibility, while those using data analytics are 23% more likely to gain new customers than competitors without analytical capabilities (Swell, 2025).

Behavioral Analytics and Knowledge Creation

Emerging research in hospitality analytics emphasizes the importance of first-party behavioral data in driving knowledge creation and service innovation. While much existing research focuses on macro-level indicators such as adoption trends and platform innovations, micro-level behavioral data encompassing operational transaction logs, customer interaction patterns, and resource utilization metrics provides the granular foundation for actionable decision-making. Recent scholarship demonstrates that hospitality analytics research remains fragmented and often relies on small datasets, with first-party behavioral data (clickstreams, navigation paths, and booking-funnel metrics) rarely explored in depth (Harnessing Digital Marketing Analytics, 2025). The integration of behavioral analytics with simulation modeling techniques, such as fuzzy cognitive mapping (FCM), offers a methodological pathway for translating raw operational data into strategic knowledge. FCM models concepts as nodes and their causal influences as signed, weighted links, enabling the simulation of how interventions propagate through the system, a capability that bridges the gap between raw data and managerial sense-making (Harnessing Digital Marketing Analytics, 2025).

This approach is particularly relevant for small food service establishments, where managerial decision-making has historically relied on intuition and experience rather than systematic data analysis. The transition from experience-based to data-driven decision-making requires not only technological infrastructure but also organizational learning and cultural adaptation. Research indicates that fostering a data-driven innovation culture is essential for turning analytics into engagement, satisfaction, and revenue growth a cultural shift that entails investing in technical infrastructure while training staff to interpret and act on data insights (Harnessing Digital Marketing Analytics, 2025). The framework proposed by this emerging scholarship collecting behavioral signals, estimating associations, representing them within causal models, and conducting scenario analysis formalizes a behavior-analytics-to-action learning loop that directly informs the iterative design philosophy of Operlytics.

Theoretical Synthesis and Research Gap

The synthesis of these theoretical perspectives reveals a significant gap in the existing literature: while large-scale hospitality enterprises have increasingly adopted sophisticated business analytics and business intelligence systems, small food service establishments particularly independent cafés remain underserved by both academic research and technological solutions. The theoretical frameworks of data-driven business models, real-time analytics, and visual decision support systems have been validated primarily in contexts of organizational scale and resource abundance that do not reflect the operational realities of small café operations. Furthermore, the OECD (2021) highlights that in the accommodation and food services sector, high-speed broadband connection, having a website, and using cloud computing to store files are the main technologies associated with higher value added yet significant digital gaps persist, particularly among the smallest enterprises.

Operlytics addresses this research gap by adapting established theoretical principles of business analytics value creation to the specific constraints and capabilities of small food service establishments. The system design incorporates the cognitive benefits of visual data presentation, the velocity requirements of real-time operational monitoring, and the scalability challenges identified in business intelligence implementation research. By grounding the system architecture in these theoretical foundations while addressing the unique resource constraints of small cafés including the need for affordable, integrated solutions that eliminate data silos, Operlytics contributes to both the practical advancement of hospitality technology and the theoretical understanding of analytics value creation in SME contexts. The platform's emphasis on consolidating fragmented operational data into actionable intelligence directly responds to the industry-wide challenge that 68% of business leaders cite as their primary integration barrier (Swell, 2025), thereby transforming the abundant but underutilized data generated by small cafés into a genuine source of competitive advantage.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies collectively establish a compelling case for the development and implementation of integrated business analytics systems in small food service establishments. The synthesis of these studies reveals several convergent themes that directly inform the design, implementation, and expected outcomes of Operlytics at Empress Café Boracay.

Theme 1: The Imperative of Technology Adoption for Competitiveness

Both foreign and local studies emphasize the importance of using technology to improve business operations, particularly in sales monitoring, inventory management, and decision-making. Foreign studies highlighted the role of business analytics in transforming operational data into useful insights that support data-driven decision-making and improve organizational performance. Local studies focused primarily on implementing automated systems, like point-of-sale (POS) and inventory management systems, to solve operational problems which small businesses commonly encounter.

The convergence of these findings across different national contexts, industry sectors, and organizational scales establishes technology adoption as a universal imperative for business competitiveness rather than a luxury available only to large or well-resourced enterprises. The digital era, as Arig et al. (2025) emphasize, demands adaptation and innovation from every business regardless of size or sector. For small cafés in tourism-intensive markets like Boracay Island, this imperative is particularly acute because the competitive environment includes not only local peers but also international franchises and established brands with sophisticated operational systems.

Furthermore, the literature suggests that small businesses which delay technology adoption often face compounding disadvantages over time. As competitors implement increasingly sophisticated systems, the operational gap widens, making later adoption more difficult and costly. Early investment in integrated systems, even modest ones, creates a foundation for incremental improvement and scalability. This finding is particularly relevant for

Empress Café Boracay, where the seasonal nature of tourism demand requires operational agility that manual systems cannot provide.

Theme 2: The Elimination of Manual Process Deficiencies

The reviewed studies consistently identify manual processes as the root cause of operational inefficiencies in small businesses. These systems are intended to take the place of manual processes, which are frequently challenging, ineffective, and prone to human error. Incorrect record-keeping, transaction delays, and trouble keeping track of stock levels are problems that many small businesses face. In order to improve day-to-day operations, researchers stressed the significance of implementing automated solutions.

According to the studies, automated systems significantly decrease manual mistakes by automating data entry and computation tasks. Additionally, they increase transaction accuracy and speed, particularly during busy periods. Furthermore, these systems offer current and accurate records, which enable business owners to better monitor performance and make decisions. These findings directly support the problem identification in Chapter 1 and the design objectives of Operlytics, which specifically target the manual process deficiencies at Empress Café Boracay.

The literature also identifies specific categories of manual errors that automated systems address most effectively. Data entry errors, including transposition mistakes, omitted entries, and duplicate recordings, represent the most common and costly category. Calculation errors in pricing, discounts, and inventory counts create cascading inaccuracies that distort business intelligence. Timing errors, where records are updated inconsistently or belatedly, prevent managers from accessing current operational status. Process consistency errors occur when different staff members follow divergent procedures for the same tasks. Operlytics addresses each of these error categories through standardized digital workflows, automated calculations, real-time updates, and enforced procedural consistency.

Theme 3: The Integration of Operational Systems

The literature review highlights how crucial it is for a company to implement various operational systems that are integrated rather than operating in isolation. These systems, which are essential to monitoring everyday activities, include reporting, inventory tracking, and sales recording. Systems for recording sales help businesses maintain track of transactions and revenue. Accurate stock levels are ensured by inventory tracking, which also helps avoid shortages or overstocking. Reporting systems organize gathered data into useful information. Through summaries and insights, they have made it simple for managers to comprehend business performance.

Businesses may track operations more efficiently and prevent errors when these systems are connected. Additionally, this integration makes it possible to produce trustworthy reports that aid in improved decision-making. The integration theme is central to the Operlytics design, which combines POS, inventory tracking, and report generation in a unified platform rather than as separate tools. This integration ensures that data flows automatically from transaction recording through inventory update to report generation, eliminating the reconciliation errors and delays that occur when these functions are performed separately.

Theme 4: The Centrality of Real-Time Information

The literature review highlights how crucial it is for a company to implement various operational systems that are integrated rather than operating in isolation. These systems, which are essential to monitoring everyday activities, include reporting, inventory tracking, and sales recording. Systems for recording sales help businesses maintain track of transactions and revenue. Accurate stock levels are ensured by inventory tracking, which also helps avoid shortages or overstocking. Reporting systems organize gathered data into useful information. Through summaries and insights, they have made it simple for managers to comprehend business performance.

Businesses may track operations more efficiently and prevent errors when these systems are connected. Additionally, this integration makes it possible to produce trustworthy reports that aid in improved decision-making. The integration theme is central to the Operlytics design,

which combines POS, inventory tracking, and report generation in a unified platform rather than as separate tools. This integration ensures that data flows automatically from transaction recording through inventory update to report generation, eliminating the reconciliation errors and delays that occur when these functions are performed separately.

The reviewed literature further distinguishes between superficial integration, where systems merely share data files periodically, and deep integration, where modules operate within a unified database architecture with real-time synchronization. Deep integration, as described by multiple authors, eliminates the latency and inconsistency problems that plague superficially connected systems. When a sale occurs, deep integration ensures that inventory decrements, revenue recognition, and dashboard updates occur simultaneously within a single transaction boundary. This architectural approach prevents the data drift that accumulates when separate systems are reconciled only at intervals. Operlytics implements this deep integration model, ensuring that all operational modules maintain perfect data consistency at all times.

Theme 5: The Importance of Usability and User-Centered Design

Multiple studies in the reviewed literature emphasize that technical functionality alone is insufficient for system success; the system must also be usable by its intended audience. Bienvenido (2023) specifically assessed usability in the POS context, while Ogunmokun et al. (2024) highlighted user training and user-friendly design as critical implementation success factors. Kunc and Bäckström (2025) noted that the value of business analytics depends on its integration into managerial decision-making processes, which requires systems that align with how managers actually work.

This usability theme directly informs the design philosophy of Operlytics, which prioritizes intuitive interfaces, minimal training requirements, and clear information presentation. The system is designed for café staff and management who may have limited technical expertise, ensuring that the system's analytical capabilities are accessible to users regardless of their technical background.

The literature provides specific guidance on usability factors that matter most in small business contexts. Cognitive load reduction through simplified screen layouts and logical task flows reduces errors and training time. Consistent interface conventions across all system modules allow users to transfer learning between functions. Progressive disclosure of advanced features prevents overwhelming novice users while accommodating expert needs. Contextual help and embedded guidance reduce dependence on external documentation. Mobile accessibility enables managers to monitor operations remotely, a capability increasingly expected in modern business environments. Offline resilience ensures that operations continue during connectivity interruptions, particularly important in island locations like Boracay where network reliability may vary. Operlytics incorporates each of these usability considerations into its design specifications.

Theme 6: Data-Driven Decision Making and Business Intelligence

An emerging theme across the reviewed literature concerns the transformation of raw operational data into actionable business intelligence. Studies by Islam (2024) and Ogunmokun et al. (2024) demonstrate that the mere collection of operational data provides limited value unless analytical tools convert that data into meaningful insights. Business intelligence encompasses not only historical reporting but also trend analysis, predictive modeling, and prescriptive recommendations that guide strategic decisions.

For small food service establishments, business intelligence capabilities have historically been inaccessible due to cost and complexity barriers. However, recent developments in cloud computing, open-source analytics tools, and simplified visualization platforms have democratized access to these capabilities. The literature documents successful implementations where small businesses leveraged modest analytical tools to achieve significant improvements in menu profitability, waste reduction, and customer retention.

Operlytics addresses this theme through its integrated analytics module, which automatically generates insights from accumulated operational data. Rather than requiring managers to manually analyze spreadsheets or generate custom reports, the system presents analytical findings through intuitive visualizations and plain-language summaries. This approach

makes business intelligence accessible to users without statistical training while still providing the depth of analysis that supports informed decision-making.

Theme 7: Scalability and Future-Proofing

Several reviewed studies emphasize that system implementations should consider not only current requirements but also future growth trajectories. Scalable architectures allow small businesses to expand system capabilities incrementally as operational complexity increases and resources permit. Cloud-based deployment models, modular system designs, and standards-based data architectures support this scalability objective.

The literature warns against implementations that solve immediate problems but create future constraints. Proprietary data formats, monolithic architectures, and vendor-dependent solutions may provide short-term convenience at the cost of long-term flexibility. Conversely, systems built on open standards and modular principles enable continuous evolution without requiring wholesale replacement.

Operlytics incorporates these scalability principles through its modular architecture, which separates core transaction processing from analytical and reporting functions. This separation allows individual modules to be enhanced or replaced without disrupting overall system operation. The use of standard data formats and documented APIs ensures that future integrations with external systems remain feasible. Cloud deployment options provide elastic capacity that grows with business volume without requiring infrastructure investment.

Synthesis Conclusion

Overall, the development of Operlytics, a business analytics and monitoring system that includes sales tracking, inventory monitoring, and data visualization, is strongly supported by the results of these studies. The importance of integrated systems is clearly established because it makes information easier to access and comprehend by allowing businesses to manage many different operational information on just one centrally controlled platform.

By reducing manual work and preventing records and reporting errors, the system aims to help Empress Café Boracay boost its operational efficiency. They are able to make decisions

faster and more effectively when they have access to accurate and organized business information. By presenting data in an understandable and visual way, it also improves decision-making, assisting the company in finding trends and improving overall performance.

The reviewed literature provides both theoretical justification and empirical precedent for the Operlytics approach. The theoretical foundation in business analytics value creation, combined with the empirical evidence from similar system implementations in Philippine small business contexts, establishes a strong basis for expecting that Operlytics will deliver meaningful operational improvements at Empress Café Boracay.

Moreover, the convergence of themes across diverse studies suggests that the benefits of integrated business analytics systems are robust and generalizable rather than context-dependent. Whether examining large enterprises or small cafés, foreign markets or local Philippine contexts, the fundamental value proposition remains consistent: integrated automation of operational processes, real-time information access, and user-centered design combine to produce measurable improvements in efficiency, accuracy, and decision quality. Operlytics represents a targeted application of these well-established principles to the specific operational context of Empress Café Boracay, adapted to the unique challenges and opportunities of small food service establishments in tourism-dependent markets.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study utilized the Developmental Research Design. Developmental research involves the systematic study of designing, developing, and evaluating programs, processes, and products that meet identified needs within a specific operational or organizational context. It is a structured and iterative approach that moves from problem identification through solution design to system validation, ensuring that the resulting output is both technically sound and practically relevant to the environment in which it will be deployed.

Developmental research as a methodological approach occupies a distinctive position in the research methodology spectrum, bridging the gap between pure theoretical research and applied implementation studies. Unlike experimental research, which seeks to establish causal relationships between variables under controlled conditions, developmental research focuses on creating and refining practical solutions to real-world problems. Unlike purely descriptive research, which documents existing conditions without intervention, developmental research actively intervenes to improve those conditions through the creation of new tools, processes, or systems.

In the context of this study, this methodology was employed to conceptualize, construct, and validate Operlytics, a web-based business analytics and monitoring system specifically designed for Empress Café Boracay. The choice of developmental research was particularly appropriate given the applied and solution-oriented nature of this study, which required not only the identification and analysis of operational problems but also the creation and evaluation of a functional technological response to those problems.

The developmental research approach aligned with the study's objectives by providing a structured framework for moving from identified problems to implemented solutions. The methodology encompassed several phases that corresponded to the natural progression of system

development: needs analysis, design specification, development implementation, and evaluation validation. Each phase built upon the previous, ensuring that the resulting system was grounded in actual operational requirements and validated against real-world performance criteria.

The application of this design began with an in-depth needs analysis of the café's current operational practices. This initial phase involved a thorough examination of the existing workflows, recording methods, and reporting mechanisms currently employed by Empress Café Boracay in managing its daily operations. By investigating existing operational bottlenecks — specifically manual record-keeping, tracking delays, human errors, and stock mismanagement — the researchers were able to define the precise functional requirements needed for the proposed system.

This needs analysis phase was critical to ensuring that the system developed was purposefully tailored to the unique operational realities of the café rather than serving as a generic solution. Data gathered during this phase served as the primary basis for defining the system's scope, features, and design priorities, laying a solid and evidence-based foundation for the subsequent stages of development. The needs analysis employed multiple data gathering techniques — interviews, observation, and document analysis — to ensure a comprehensive and well-validated understanding of operational requirements.

Following the analysis phase, the developmental process focused on designing, engineering, and evaluating the technological solution. This phase encompassed the planning of the system architecture, the design of the user interface, and the development of the core functional modules that constitute the Operlytics system, with particular attention given to ensuring accessibility for café staff and management who may have limited technical expertise.

By automating manual workflows related to sales and inventory, the developed system provided real-time data updates, enhanced stock monitoring accuracy, and reduced the operational errors that had historically undermined the reliability of the café's records. The automation design specifically targeted the manual process deficiencies identified during needs analysis, ensuring that each automated function directly addressed a documented operational problem.

The evaluation phase assessed the system's usability, functionality, and overall effectiveness in supporting management decision-making through feedback gathered from café administrators and staff during testing. This evaluation was not merely a technical verification of system operation but an empirical assessment of the system's practical value in the operational context for which it was designed. The evaluation criteria — functionality, usability, accuracy, security, and efficiency — provided a comprehensive framework for assessing system quality from multiple perspectives.

Ultimately, this research design facilitated a structured and comprehensive transition from identifying business problems to delivering a functional and validated system that meaningfully improves operational efficiency and managerial decision-making at Empress Café Boracay. The developmental approach ensured that the research produced not merely academic knowledge but practical value for the business partner and contributed to the broader understanding of how technology can address operational challenges in small business contexts.

3.2 System Development Methodology

The development of Operlytics followed Agile Methodology, a process defined by continuous iteration, cross-functional collaboration, and rapid response to change. Unlike traditional linear models such as the Waterfall approach, which proceeds sequentially through distinct phases with limited opportunity for revision, Agile breaks the project into small, manageable increments known as Sprints, allowing the development team to progressively build and refine the system based on feedback and evolving requirements.

The Agile methodology emerged from the software development industry's recognition that traditional predictive approaches often failed to deliver satisfactory outcomes in dynamic environments where requirements were uncertain or subject to change. The Agile Manifesto, published in 2001, established four core values: individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a plan — that continues to guide Agile practice today. These values were particularly relevant to the Operlytics development context, where close collaboration with café staff and management was essential for ensuring that the system met actual operational needs.

The system evolved through repeated cycles, each consisting of the following phases:

Phase 1: Requirement Analysis

User stories and system requirements were identified and prioritized in a product backlog, ensuring that development efforts were aligned with the actual needs of Empress Café Boracay's administrators and staff. The product backlog served as a dynamic repository of desired features and functions, organized by priority and estimated effort. Requirements were expressed as user stories — brief descriptions of functionality from the perspective of the end user — that captured not merely technical specifications but the operational value that each feature was intended to deliver.

The requirement analysis phase for Operlytics drew upon the data gathered during the needs analysis phase of the research design. Interview transcripts, observation notes, and document analysis findings were translated into specific system requirements that addressed the operational deficiencies identified. Each requirement was evaluated for technical feasibility, operational value, and implementation effort to establish development priorities that maximized the system's practical impact within available development resources.

Phase 2: Iterative Design and Development

Design and coding occurred concurrently rather than sequentially. Flowcharts and database schemas were developed and refined as the system grew, ensuring that the architecture remained flexible and responsive to changes in requirements. This concurrent approach allowed the development team to discover design issues during implementation and adjust accordingly, rather than discovering fundamental problems only after a complete design specification had been produced.

The iterative nature of this phase meant that the system was built incrementally, with each sprint delivering a functional subset of the complete system. Early sprints focused on core data structures and basic user authentication, while later sprints added POS functionality, inventory management, reporting, and dashboard visualization. This incremental approach provided several advantages: it allowed for early user feedback on working software rather than abstract designs; it reduced the risk of major architectural problems by revealing issues early; and it ensured that a functional system existed at all times, even if incomplete.

Phase 3: Continuous Testing

Testing was not treated as a final step but as an ongoing process integrated throughout development. Features were verified as they were built to identify and resolve issues early, maintaining the overall quality and stability of the system. This continuous testing approach encompassed multiple testing levels: unit testing of individual components, integration testing of component interactions, and system testing of complete user workflows.

The continuous testing philosophy aligned with the Agile principle that quality is built into the product through ongoing verification rather than inspected into the product through final-stage checking. By testing each feature as it was developed, the team could identify and correct defects when they were most easily and economically addressed — close to the time of their introduction — rather than accumulating a large backlog of issues that had to be resolved during a dedicated testing phase.

Phase 4: Deployment and Review

At the end of each sprint, a functional increment of the system was delivered and reviewed. This allowed stakeholders to monitor progress in real time and provide input before the next development cycle began. The deployment of sprint increments at Empress Café Boracay enabled café staff and management to interact with working software, providing feedback that directly influenced subsequent development priorities.

The review process at the end of each sprint involved demonstration of completed features to stakeholders, discussion of what was accomplished and what was not, and planning for the next sprint based on current priorities and feedback. This regular review cycle ensured that the development process remained aligned with user needs and that any misalignment between system development and operational requirements was identified and corrected promptly.

Phase 5: Feedback and Retrospective

Based on feedback gathered from users and stakeholders, the team identified areas for improvement and adjusted the backlog accordingly for the next sprint cycle. The retrospective phase involved reflection on the development process itself — what worked well, what could be improved, and what changes should be made to the team's working methods. This meta-level reflection ensured that the development process continuously improved alongside the product being developed.

The Agile methodology was selected for its adaptability, transparency, and user-centered approach. By delivering the system in functional increments, the development process ensured flexibility to accommodate new insights or shifting requirements, consistent quality through continuous testing and review, and alignment with the actual needs of the end users through regular feedback loops. These characteristics make Agile particularly suitable for developing systems in close collaboration with operational users who may have difficulty articulating complete requirements upfront but can provide valuable feedback on working software.

3.3 System Architecture

Operlytics followed a client-server architecture designed to convert operational data into clear and actionable insights for café management. This architectural pattern, which separates data processing and storage functions from user interface and interaction functions, is the dominant approach in modern web application development and offers particular advantages for the operational context of Empress Café Boracay. The client-server model has been widely adopted across industries because it enables centralized control, simplifies maintenance, and provides a scalable foundation for growing businesses that need to manage increasing volumes of data and user traffic over time.

In this architecture, the system was divided into two distinct but interconnected layers that worked together to ensure seamless data flow, reliable processing, and accessible presentation of operational information. The separation of concerns between client and server components enabled independent development, maintenance, and scaling of each layer, while standardized communication protocols ensured effective interaction between them. This layered approach meant that changes or improvements could be made to one side of the system without necessarily disrupting the other, which was critical for long-term sustainability and adaptability in a dynamic business environment.

Server-Side Architecture

The server side served as the core processing layer of the system, where a central engine continuously gathered data from business operations, processed and organized it, and stored it in a secure database that could be retrieved whenever needed. This server-side layer was responsible for handling all computational tasks, enforcing data integrity, and maintaining the accuracy and consistency of the information that flowed through the system.

The server components included the web server that handled HTTP requests from client browsers, the application server that executed business logic for transaction processing, inventory updates, and report generation, and the database server that stored and managed all operational data. Each of these components played a vital and complementary role in ensuring that the system operated smoothly and efficiently.

The PHP backend handled user requests, facilitated communication between the interface and the database, and ensured that the system functioned correctly and efficiently under regular operational conditions. PHP was chosen for its maturity, extensive community support, and proven reliability in web application development. It enabled rapid development cycles while maintaining the performance characteristics necessary for real-time operational systems. The backend code implemented all core business logic, including order validation, inventory deduction algorithms, sales aggregation, user authentication, and report generation pipelines.

By centralizing data processing on the server, Operlytics ensured that operational records were protected, systematically organized, and readily available for retrieval and analysis at any point in time, regardless of how many users were simultaneously accessing the system. This centralized approach eliminated the risk of data fragmentation that occurred when information was stored locally on individual devices. It also ensured that all users saw the same, up-to-date information at any given moment, which was essential for coordinated decision-making across different roles and departments within the café.

The server-side processing model also simplified security management by concentrating data protection measures in a single controlled environment rather than distributing them across multiple client devices. Security policies, access controls, encryption standards, and audit logging could all be implemented and monitored from a central location. This reduced the attack surface of the system and made it easier to detect and respond to potential threats. Additionally, server-side validation ensured that all incoming data was checked for correctness and completeness before it was accepted into the system, preventing malformed or malicious inputs from compromising data integrity.

Furthermore, the server environment supported automated backup and recovery procedures, ensuring that critical business data could be restored in the event of hardware failure, software corruption, or other disaster scenarios. Regular backups were scheduled to run at intervals that balanced data freshness with system performance, and recovery protocols were tested periodically to verify their effectiveness.

Client-Side Architecture

On the client side, authorized users **accessed** a web-based dashboard through their computers or internet-connected devices, eliminating the need for any additional software installation and allowing management to monitor café operations from virtually any location with an internet connection. The dashboard **communicated** with the server through secure data requests to display real-time updates, visual reports, and operational summaries in a format that was intuitive and easy to interpret.

The user interaction flow was designed for efficiency and practicality: a manager logged in, viewed a high-level overview of the café's operational status through visual indicators and alerts, and navigated to specific metrics for more detailed analysis of sales performance, inventory levels, and overall business activity. Every element of the client-side interface was deliberately designed to minimize the steps required for users to access critical information, ensuring that even staff members with limited technical backgrounds could navigate the system confidently and effectively during their daily operations.

The client-side implementation used HTML for structural markup, CSS for visual presentation and layout, and JavaScript for dynamic and interactive elements that enhanced the overall responsiveness and usability of the dashboard during real-time operations. These standard web technologies ensured compatibility across diverse devices and browsers, maximizing the accessibility of the system for all intended users.

Architectural Benefits

This architecture ensured that all data processing and computation occurred on the server side, while the client side remained clean, responsive, and focused on delivering a straightforward experience for operational decision-making. By offloading complex calculations and data manipulation to the server, the client devices were freed from heavy processing burdens, which improved their battery life, reduced heat generation, and extended their usable lifespan.

The clear separation of responsibilities between the server and client layers not only improved the overall performance and reliability of the system but also enhanced its scalability, allowing Operlytics to accommodate growing volumes of operational data as the café expanded its business activities over time.

Furthermore, this architectural approach strengthened the security of the system by ensuring that sensitive business data was processed and stored within a controlled server environment rather than on individual user devices. The client-server model enabled centralized authentication, authorization, and audit logging, providing robust security controls that protected operational data from unauthorized access or modification.

The architecture also supported modular development and deployment practices. New features could be developed and tested in isolation before being integrated into the production environment. Bug fixes could be deployed to the server without requiring users to update their browsers or install patches.

Cache invalidation strategies ensured that users always saw current data when significant changes occurred.

Ultimately, the client-server architecture of Operlytics reflected a deliberate design philosophy centered on reliability, accessibility, and efficiency — core principles that aligned directly with the operational needs and management goals of Empress Café Boracay. The architecture supported the system's functional requirements while providing a foundation for future growth and enhancement as the café's operational needs evolved.

3.4 Tools and Technologies Used

The development of Operlytics utilized a set of web-based tools and technologies selected for their reliability, compatibility, and suitability for building a business analytics and monitoring system. Each technology was deliberately chosen not only for its individual capabilities but also for its ability to integrate seamlessly with the other components of the system, forming a cohesive and stable development stack that supported the full range of functional requirements identified for Operlytics.

Backend Technology: PHP

PHP was used to handle all server-side operations of the system. It managed user requests, facilitated communication between the interface and the database, and ensured that the system functioned correctly and efficiently under regular operational conditions. PHP is widely used in web development and is well-suited for building dynamic, data-driven applications such as Operlytics, making it a practical and dependable choice for managing the backend logic that powers the system's core features, including sales processing, inventory updates, and automated report generation.

PHP's widespread adoption in web development ensures extensive documentation, large developer communities, and abundant learning resources that support development and maintenance. Its compatibility with MySQL databases through established connection libraries simplifies data access and manipulation. Its execution model, which processes each request independently, aligns well with the stateless nature of HTTP and the transactional processing requirements of the POS module.

Frontend Technologies: HTML, CSS, and JavaScript

HTML, CSS, and JavaScript were used to develop the user-facing interface of the system. These technologies worked together to create a visually organized, responsive, and accessible interface that allowed café administrators and staff to navigate and interact with the system efficiently. HTML provided the structural foundation of each page, CSS governed the visual presentation and layout to ensure a clean and intuitive design, and JavaScript enabled dynamic and interactive elements that enhanced the overall responsiveness and usability of the dashboard during real-time operations.

The combination of these three technologies represents the standard foundation of modern web development, ensuring broad compatibility across browsers and devices. HTML5 semantic elements improve accessibility and searchability. CSS3 features enable sophisticated visual designs without requiring external libraries. JavaScript, potentially enhanced with libraries such as jQuery or Chart.js for dashboard visualization, provides the interactivity that transforms static pages into dynamic applications.

Database Technology: MySQL

MySQL served as the database management system for Operlytics. It stored and organized all operational data, including product information, inventory records, and sales transactions. MySQL provided a structured and reliable environment for managing large volumes of data, ensuring that information could be retrieved accurately and efficiently. Its compatibility with PHP made it a natural fit within the system's overall technology stack, and its widespread adoption in web-based applications ensured strong community support and well-established best practices for data management and security.

MySQL's relational data model aligns well with the structured operational data of a café business, where entities such as products, inventory items, transactions, and users have well-defined attributes and relationships. The SQL query language enables powerful data retrieval and aggregation operations that support the reporting and dashboard functions of Operlytics. MySQL's transaction support ensures data integrity by guaranteeing that related

operations such as recording a sale and deducting inventory complete successfully or roll back entirely, preventing inconsistent data states.

Development Tools

Visual Studio Code was used as the primary code editor for writing and managing the system's source code, offering a lightweight yet feature-rich environment that supported syntax highlighting, code completion, and extensions that streamlined the development workflow. XAMPP was used as the local development environment, enabling the team to run and test the PHP server and MySQL database on local machines before deployment, providing a controlled and consistent testing environment that reduced the risk of configuration issues arising during the transition to a live server.

Browser: Google Chrome

Google Chrome was used as the primary browser for testing and validating the system's interface and functionality throughout the development process. As one of the most widely used web browsers globally, Chrome provided a reliable and standardized testing environment that helped ensure the system's interface rendered correctly and performed consistently for the majority of its intended users.

Technology Selection Rationale

These tools were selected because they are widely used, well-documented, and compatible with one another, providing a stable and efficient foundation for the development of a web-based business analytics and monitoring system. The total cost of ownership for this technology stack considering licensing, hosting, maintenance, and support is minimal compared to proprietary enterprise solutions, making it appropriate for the resource constraints of a small café establishment.

3.5 Data Gathering Techniques

To ensure that Operlytics was developed based on accurate and relevant information, the researchers employed multiple data gathering techniques to understand the current operational practices of Empress Café Boracay and identify the specific needs the system had to address. The use of multiple complementary methods was intentional, as relying on a single data source may produce an incomplete or one-sided understanding of the café's operational environment. By triangulating findings across interviews, observation, and document analysis, the researchers aimed to build a thorough and well-validated picture of the café's existing workflows, challenges, and data management practices that directly informed the design and development of the Operlytics system.

Technique 1: Interview

The researchers conducted structured interviews with the café owner and authorized staff to gain a comprehensive understanding of how sales and inventory were currently managed. The interviews focused on identifying existing workflows, recurring operational challenges, and the specific features that users expected from the system. This method allowed the researchers to capture firsthand accounts and perspectives from the individuals who were most directly affected by the current operational limitations, ensuring that the system being developed reflected the genuine priorities and expectations of its intended users rather than assumptions made by the development team.

The interview protocol included questions about daily operational routines, specific pain points in current processes, desired improvements, and expectations for a new system. Interviews were conducted individually to encourage candid responses and collectively to identify shared perspectives and priorities. The structured format ensured that all relevant topics were covered while allowing flexibility for participants to raise issues not anticipated by the researchers.

Interview Protocol Design

The interview protocol for this study **was developed** based on preliminary understanding of café operations and refined through pilot testing. The protocol **was organized** into thematic sections that progressively explored different aspects of operational practice:

Section 1: Operational Overview established the context of each participant's role and responsibilities. Questions explore: the participant's position and tenure at the café; typical daily activities and responsibilities; hours of work and seasonal variations in workload; and interactions with other staff members and management.

Section 2: Sales Recording Practices examined how customer transactions are currently documented. Questions explore: the specific steps involved in recording a sale from customer order to final documentation; the tools and materials used (paper logbooks, calculators, receipts); the consistency of recording practices across different staff members and shifts; the time required for sales recording during quiet and busy periods; and the frequency and nature of errors encountered in sales records.

Section 3: Inventory Management Practices investigated how stock levels are monitored and maintained. Questions explore: the methods used for tracking inventory (physical counts, visual estimates, informal notes); the frequency and timing of inventory checks; the process for recording stock receipts from suppliers; the handling of waste, spoilage, and returns; the identification of low-stock situations and the process for reordering; and the challenges encountered in maintaining accurate inventory awareness.

Section 4: Reporting and Analysis assessed how operational information is compiled and used for decision-making. Questions explore: the types of reports currently generated (sales summaries, inventory status, financial calculations); the frequency and timing of report preparation; the process and time required for report compilation; the recipients and uses of operational reports; the perceived accuracy and usefulness of current reports; and the decisions that reports inform or fail to inform.

Section 5: Operational Challenges invites participants to articulate the most significant difficulties they encounter in their work. Questions explore: the most frustrating or time-consuming aspects of current operational practices; situations where current practices have led to significant problems (stockouts, customer complaints, financial discrepancies); workarounds or coping strategies developed to manage operational deficiencies; and the consequences of operational problems for their work experience and job satisfaction.

Section 6: System Expectations elicits participant visions for an improved operational system. Questions explore: the features or capabilities that would most improve their daily work; the concerns or reservations about adopting a new digital system; the training and support that would facilitate successful adoption; and the criteria by which they would judge a new system to be successful or unsuccessful.

Interview Implementation

Interviews were conducted at the café premises during periods that minimized disruption to operations, typically during mid-morning or mid-afternoon lulls between peak service periods. Individual interviews lasted approximately 45-60 minutes, while group interviews with multiple staff members lasted approximately 60-90 minutes. All interviews were audio-recorded with participant consent, and detailed notes were taken to capture non-verbal cues and contextual observations.

Technique 2: Observation

The researchers observed how staff members performed manual recording and inventory management tasks during actual café operations. This method allowed the researchers to directly identify inefficiencies, delays, and error-prone steps in the current process that the system was designed to address. Unlike interviews, which rely on self-reported information, observation provided an objective and unfiltered view of how operations were actually carried out on a day-to-day basis, enabling the researchers to identify issues that staff members may have normalized or overlooked due to familiarity with existing routines.

Observation sessions were conducted at various times of day and days of the week to capture the full range of operational conditions, from quiet periods to peak service times. The

researchers documented workflow sequences, time requirements for specific tasks, error occurrences, and workarounds that staff had developed to cope with system deficiencies. These observational data provided concrete evidence of operational problems that complemented the subjective accounts obtained through interviews.

Observation Protocol Design

The observation protocol was structured to capture comprehensive information about operational workflows while minimizing intrusiveness that might alter natural behavior. The protocol included:

Temporal Coverage: Observation sessions were conducted at various times of day and days of the week to capture the full range of operational conditions. The schedule includes: weekday mornings (typically 7:00-10:00 AM) when breakfast service and preparation activities occur; weekday midday (11:00 AM-2:00 PM) during lunch peak service; weekday afternoons (3:00-5:00 PM) during quiet periods; weekend mornings and midday when tourist traffic is typically higher; and evening periods (6:00-9:00 PM) during dinner service if applicable.

This temporal variation was essential because tourism-dependent operations like Empress Café Boracay experience dramatic fluctuations in customer volume and operational intensity. Systems designed based on observations during only quiet periods may fail catastrophically during peak demand, while systems designed only for peak periods may be unnecessarily complex and costly for routine operations.

Spatial Coverage: Observations covered all operational areas of the café: the customer service area where orders are taken and payments processed; the preparation area where food and beverages are assembled; the storage area where inventory is maintained; and the administrative area where records are kept and reports compiled.

Observation Dimensions: For each observed activity, researchers document: workflow sequence the specific steps involved in completing a task from initiation to completion; time requirements the duration of each step and the total task completion time; error occurrences

mistakes, corrections, workarounds, and their consequences; communication patterns interactions between staff members during task execution; tool and material usage the physical and informational resources employed; and environmental factors crowding, noise, temperature, and other conditions that may affect task performance.

Specific Observation Targets

Sales Transaction Recording: Researchers observe the complete process from customer order through final record entry. Observations capture: how staff receive and confirm orders; the tools used for recording (paper, calculators, memory); the sequence of data entry and calculation; the handling of special cases (modifications, discounts, complaints); the time from order completion to record finalization; and the accuracy of final records through spot-checking against actual orders.

Inventory Management Activities: Researchers observe stock handling practices including: receipt of deliveries from suppliers and the recording of incoming stock; storage organization and the accessibility of items during preparation; visual or physical stock checks and their timing relative to need; identification of low-stock or expired items; and the handling of waste, spoilage, or returns.

Report Preparation: When management prepares operational reports, researchers observe: the collection of source documents from multiple locations; the transcription and aggregation of data; the calculation and verification of summary figures; the formatting and organization of results; and the time from initiation to completion.

Workaround Identification: A particular focus of observation was the identification of informal practices that staff have developed to cope with system deficiencies. These workarounds such as mental arithmetic instead of written calculation, informal communication instead of formal documentation, or stock estimation instead of physical counting represent adaptive responses that reveal the true operational needs that formal systems may obscure.

Observation Documentation

Observations were documented through structured field notes that record the observation dimensions for each activity. Time-stamped entries capture the sequence and duration of operational events. Diagrammatic sketches illustrate spatial layouts and workflow paths. Photographs (with permission) document physical environments, tools, and records. Audio recordings (with permission) capture verbal communications during task execution.

Technique 3: Document Analysis

The researchers examined existing operational documents such as stock lists, sales receipts, and inventory logs to understand how data was currently recorded, organized, and maintained. This analysis provided additional context on the café's data management practices and informed the design of the system's data structure and reporting features. By reviewing actual records used in the café's daily operations, the researchers gained a concrete understanding of the formats, categories, and types of information that the system needed to be capable of processing, storing, and presenting in a clear and accessible manner.

Document analysis revealed the data elements that needed to be captured by the system, the relationships between different data types, and the reporting formats that users found familiar and useful. The examination of historical records also provided baseline data against which system accuracy could be validated — comparing system-generated reports to manually compiled records from the same period to verify correctness.

Document Categories Analyzed

Sales Records: The researchers examined all forms of sales documentation including: handwritten logbooks or notebooks maintained by individual staff members; printed receipts from any existing point-of-sale devices or manual receipt books; daily sales summaries compiled by management; and informal notes or tallies used during busy periods.

Inventory Records: The researchers examined inventory documentation including: stock count sheets or logbooks; supplier delivery receipts and invoices; informal notes about stock levels or needs; and records of waste, spoilage, or returns.

Financial Records: The researchers examined financial documentation including: cash tally sheets; bank deposit records; expense receipts; and any profit or loss calculations.

Analysis of financial records focused on: the sources of revenue and categories of expense tracked; the methods of calculation and aggregation; the frequency of financial summary preparation; and the relationship between operational records and financial outcomes.

Administrative Documents: The researchers examined any additional operational documentation including: staff schedules; supplier contact lists; product recipes or preparation guides; and customer feedback or complaint records.

Document Analysis Procedures

Documents were collected through photocopying or photography (with permission) to enable detailed analysis without removing originals from the café. Each document was catalogued with metadata including: document type, date or date range, author or responsible party, purpose, and relationship to other documents.

Baseline Data Establishment

The examination of historical records provided baseline data against which system accuracy can be validated. By comparing system-generated reports to manually compiled records from the same period, researchers can verify that the new system produced results consistent with known operational outcomes. This baseline validation was essential for establishing user trust in the new system and for demonstrating that the digital transformation has not introduced new sources of error.

Triangulation and Validation

The combination of these three data gathering methods provided the researchers with a well-rounded and empirically grounded understanding of the operational environment at Empress Café Boracay, ensuring that Operlytics was developed to meet the actual needs of its

intended users. The insights derived from interviews captured the experiential knowledge and expectations of café personnel, while observations revealed the practical realities of day-to-day operations that were not fully communicated through verbal accounts alone. Document analysis, in turn, grounded the research in concrete evidence of how data was currently handled, providing a factual basis for the system's design decisions.

Together, these methods formed a robust data collection strategy that minimized the risk of oversight, reduced the likelihood of developing features that did not align with actual user needs, and ensured that the resulting system was both contextually appropriate and operationally relevant to the unique environment of Empress Café Boracay. Ultimately, the thoroughness of this data gathering process served as the foundation upon which a reliable, practical, and effective business analytics system could be successfully built and deployed.

Triangulation Procedures

The triangulation process involved systematic comparison of findings across the three data sources:

Convergence Analysis identified areas where findings from all three methods align, indicating high-confidence understanding. For example, if interviews reveal staff frustration with manual inventory counting, observations confirm that counting consumes substantial time during busy periods, and documents show inconsistent inventory records with frequent discrepancies, researchers can be confident that inventory management represents a genuine operational priority.

Divergence Analysis identified areas where findings from different methods conflict, indicating need for further investigation. For example, if interviews report that sales recording is accurate and complete, but observations reveal frequent estimation and delayed documentation, and documents show arithmetic errors and missing entries, researchers must investigate the reasons for this discrepancy perhaps staff are unaware of their own errors, or social desirability bias leads to overstated performance in interviews.

Complementarity Analysis identified areas where different methods reveal distinct but compatible aspects of operational reality. For example, interviews may reveal management's

strategic concern about competitive positioning, observations may reveal operational bottlenecks that constrain service quality, and documents may reveal financial pressures that limit investment capacity. These complementary findings together inform a holistic understanding of the café's situation that guides system design priorities.

Validation Criteria

The quality of data gathered through these techniques was assessed against established criteria for qualitative research:

Credibility refers to the believability and authenticity of findings from the perspective of participants. Credibility was enhanced through prolonged engagement with the café environment, persistent observation across diverse conditions, and member checking where preliminary findings were shared with participants for verification and correction.

Transferability refers to the extent to which findings may be relevant to other contexts. While this study focuses specifically on Empress Café Boracay, careful documentation of contextual characteristics enables readers to assess the applicability of findings and design decisions to other small food service businesses in similar tourism-dependent environments.

Dependability refers to the consistency and reliability of findings over time and across researchers. Detailed documentation of data gathering procedures, structured protocols for interviews and observations, and systematic archiving of raw data supported dependability by enabling audit and replication.

Confirmability refers to the objectivity and neutrality of findings, ensured through reflexive practice where researchers examine their own assumptions and biases, and through triangulation that subjects emerging interpretations to multiple independent sources of evidence.

3.6 Testing Procedures

The validation of Operlytics followed a structured, three-phase testing framework designed to ensure both technical stability and practical utility before full implementation. This framework reflected a progressive and methodical approach to system validation, beginning with the examination of individual components and advancing toward a comprehensive assessment of the system's performance in conditions that closely replicated the actual operational environment of Empress Café Boracay. Each phase served a distinct and essential purpose in confirming that the system was not only technically sound but also genuinely responsive to the needs of its intended users.

The adoption of structured, multi-phase testing frameworks represents established best practice in software engineering and information systems development. A progressive testing approach beginning with component-level verification and advancing through integration and system-level evaluation minimizes the risk of compounding errors and ensures that issues are identified and resolved at the earliest possible stage. Thorough validation is essential for ensuring that systems meet both technical specifications and user requirements before deployment in operational environments where failures can disrupt business activities and damage user confidence.

Phase 1: Unit Testing

This phase focused on isolating and verifying individual system components, such as backend data processing scripts and frontend interface handlers. Each module was tested independently to confirm that it processed inputs correctly and handled edge cases before being integrated into the full system. By identifying and resolving errors at the component level early in the validation process, unit testing minimized the risk of compounding issues arising during later stages of system integration and evaluation.

Unit testing for Operlytics encompassed verification of database operations, validation of calculation algorithms, testing of user authentication procedures, and confirmation of data validation rules. Each unit test specified expected inputs, expected outputs, and the conditions

under which the component should operate correctly. Automated unit testing tools were employed to facilitate repeated execution of test suites as the system evolved.

Unit testing for Operlytics encompasses verification across multiple component categories:

Database Operations Testing verified that all database interactions perform correctly and maintain data integrity. Test cases include: insertion of new records with valid and invalid data to confirm proper acceptance and rejection; update operations that modify existing records while preserving referential integrity; deletion operations with cascade effects to ensure related records are handled appropriately; query operations that retrieve data with correct filtering, sorting, and aggregation; and transaction management that ensures atomicity related operations complete entirely or roll back completely, preventing partial updates that would corrupt data consistency.

Calculation Algorithm Testing validated the arithmetic and logical operations that produce derived values throughout the system. Test cases include: sales total calculations that sum line items, apply taxes, and compute discounts correctly; inventory deduction algorithms that reduce stock quantities by precise recipe-specified amounts when sales are recorded; profit margin calculations that correctly relate revenue to cost of goods sold; and alert threshold evaluations that trigger notifications at configured stock levels.

User Authentication Testing confirmed that access control mechanisms operated reliably. Test cases include: valid credential acceptance that grants appropriate access levels; invalid credential rejection that prevents unauthorized entry; password complexity enforcement that requires minimum standards for new passwords; session management that maintains secure access during active use and terminates appropriately after periods of inactivity; and role-based permission verification that restricts users to functions appropriate for their assigned roles.

Data Validation Testing ensured that input checking prevented malformed or malicious data from entering the system. Test cases include: numeric field validation that rejects non-numeric entries in quantity and price fields; date validation that prevents future dates for past transactions or impossible date combinations; range validation that flags quantities or prices outside reasonable boundaries; and format validation that ensures consistent data entry for fields such as product codes or contact information.

Phase 2: System Testing

This phase evaluated the complete end-to-end functionality of the system as a unified whole. It assessed overall performance, data accuracy, and system stability under varying conditions, including high volumes of simultaneous user activity, to ensure that the platform operated reliably in a real operational environment. This phase was critical in confirming that all individual components functioned cohesively together and that the system as a whole delivered consistent and accurate outputs under the demands of regular café operations.

System testing for Operlytics encompassed workflow testing that followed complete user scenarios from login through transaction processing to report generation, load testing that verified system performance under multiple concurrent users, data integrity testing that confirmed accurate propagation of updates across related records, and security testing that validated access controls and data protection measures.

System testing for Operlytics encompassed several specialized testing types:

Workflow Testing followed complete user scenarios from initial login through transaction processing to report generation, verifying that each step in operational workflows executed correctly and that data flowed accurately between system components. Representative test scenarios include: a staff member logs in, records a multi-item customer order, processes a split payment, and generates a receipt; an administrator logs in, reviews dashboard alerts for low-stock items, updates inventory through manual adjustment, and generates an inventory status report; and a manager logs in, reviews daily sales summary, examines product performance trends, and exports data for external analysis.

Load Testing verified system performance under multiple concurrent users, simulating the conditions of peak service periods at Empress Café Boracay when several staff members may simultaneously process orders, update records, and access reports. Test parameters include: response time measurements for common operations under increasing user loads; identification of performance degradation points where response times exceed acceptable thresholds;

verification that data integrity was maintained under concurrent access conditions; and assessment of system behavior at maximum expected load levels.

Data Integrity Testing confirmed accurate propagation of updates across related records, ensuring that actions in one system area correctly affect all dependent data. Test cases include: sales transaction recording that automatically updates inventory quantities, sales totals, and dashboard displays; inventory adjustments that correctly modify stock levels and trigger or clear low-stock alerts as appropriate; user role modifications that immediately affect access permissions without requiring system restart; and report generation that accurately aggregates underlying transaction data without omission or duplication.

Security Testing validated that access controls and data protection measures functioned correctly in the integrated system. Test cases include: unauthorized access attempts that confirm proper rejection and logging; privilege escalation attempts that verify role-based restrictions cannot be circumvented; data transmission interception tests that confirm encryption protects sensitive information; and session hijacking attempts that validate secure session management.

The system testing phase also included stress testing that evaluated behavior under conditions exceeding normal operational parameters, such as transaction volumes significantly above peak expectations or database sizes substantially larger than current requirements. While such extreme conditions were not expected to occur regularly, stress testing revealed architectural limitations and provided confidence in system resilience.

Phase 3: User Acceptance Testing (UAT)

This phase involved café administrators and authorized staff interacting with the system in a controlled testing environment that simulated actual café operations. UAT confirmed whether the system's features, interface, and workflows met the practical needs of the end users and whether the system effectively supported operational decision-making. Feedback gathered during this phase served as the final basis for identifying any remaining usability concerns or functional gaps that had to be addressed prior to the system's full deployment.

User Acceptance Testing at Empress Café Boracay involved structured test scenarios that required users to perform typical operational tasks using the system, observation of user interactions to identify usability issues, and collection of subjective feedback through questionnaires and interviews. The UAT phase was not merely a demonstration but an empirical evaluation that determined whether the system was ready for operational deployment.

User Acceptance Testing at Empress Café Boracay involved multiple evaluation activities:

Structured Test Scenarios required users to perform typical operational tasks using the system, with success criteria defined for each scenario. Representative scenarios include: a floor staff member processes a complete customer order from product selection through payment and receipt generation within a specified time limit; a kitchen staff member checks ingredient availability for upcoming orders and updates inventory for items used in preparation; an administrator generates daily sales and inventory reports and verifies that displayed figures match expected values based on known transaction activity; and a manager reviews dashboard displays during a simulated busy period and identifies key operational metrics requiring attention.

Observational Assessment involved researchers systematically watching representative users interact with the system while performing realistic tasks, noting difficulties, errors, hesitations, workarounds, and verbalizations. Observation focuses on: navigation patterns that reveal interface clarity or confusion; error recovery behaviors that indicate system forgiveness or fragility; efficiency measures such as time to task completion and number of interactions required; and emotional responses including frustration, satisfaction, or confidence that predict sustained adoption.

Feedback Collection gathered subjective experiences, opinions, preferences, and suggestions through structured questionnaires, semi-structured interviews, and focus group discussions. The System Usability Scale (SUS) provided standardized quantitative measurement of perceived usability for comparison against established benchmarks. Open-ended questions explore: perceived usefulness of specific features for daily operational tasks; ease of learning and confidence in system operation; comparison with previous manual methods in terms of

efficiency and reliability; concerns or reservations about system adoption; and suggestions for improvements or additional capabilities.

Comprehensive Quality Assurance

Collectively, these three validation phases established a rigorous and comprehensive quality assurance process that addressed the system's performance from multiple perspectives — technical, operational, and human. The progressive nature of the framework ensured that issues were identified and resolved at the earliest possible stage, reducing the likelihood of significant problems emerging during actual deployment. By subjecting Operlytics to thorough unit, system, and user acceptance testing, this study aimed to deliver a system that was not only free of critical technical deficiencies but also genuinely aligned with the operational workflows and decision-making needs of Empress Café Boracay, providing management and staff with a reliable and effective tool that they could confidently integrate into their daily business operations.

The integration of unit, system, and user acceptance testing into a comprehensive quality assurance framework aligns with established best practices in software engineering and information systems development. The progressive approach where early defect detection through unit testing reduces the cost and complexity of later corrections, while system and UAT phases ensure both technical integrity and operational suitability is a fundamental principle of software quality assurance. Comprehensive quality assurance must address multiple perspectives, technical correctness, performance under realistic conditions, and user satisfaction to deliver systems that are reliable in production environments.

For business analytics systems, comprehensive quality assurance must include validation of real-time data accuracy, performance under operational loads, and user confidence in decision-support capabilities. The impact of business analytics on organizational decision-making depends on both technical system quality and user trust, which comprehensive testing frameworks must establish. The progressive testing framework employed for Operlytics

thus ensured that the system delivered to Empress Café Boracay was reliable, effective, and genuinely supportive of daily business operations.

3.7 Evaluation Criteria

The evaluation of Operlytics was conducted based on the following five core criteria to ensure that the system met both technical standards and the operational needs of Empress Café Boracay. Each criterion has been carefully selected to reflect the specific functional requirements and operational priorities identified during the needs analysis phase of this study, providing a comprehensive and structured basis for assessing the system's overall readiness and effectiveness for deployment in a real café environment.

The selection of these five criteria functionality, usability, accuracy, security, and efficiency draws upon established frameworks for information systems evaluation. Research on business analytics and business intelligence systems emphasizes that effective evaluation must encompass both technical performance dimensions and human factors such as usability and trust (Gillon, McCarthy, & Spillane, 2012). Business analytics maturity assessments require multi-dimensional evaluation across process and performance outcomes (Mithas, Lee, & Wareham, 2013). Understanding business analytics impact requires evaluation that includes both technical and organizational dimensions (Sharma, Mithas, & Kankanhalli, 2014). Business analytics systems must be evaluated not only on technical capabilities but also on their practical impact on decision-making and operational outcomes (Davenport & Harris, 2017).

Criterion 1: Functionality

This criterion assessed whether the system successfully executed all intended features, including the Point of Sale system, real-time sales and inventory monitoring, automated report generation, and dashboard data visualization, ensuring that all components operated without failure. Evaluators verified that each feature performed as intended under normal operating conditions and that no critical functions were absent or producing incorrect outputs during the testing phase.

Functionality evaluation involved systematic testing of each system feature against its specified requirements, verification that feature outputs matched expected results, and confirmation that all features were accessible to authorized users. The functionality criterion ensured that Operlytics delivered the complete set of capabilities that the café requires for operational management.

In the context of POS and inventory systems, functionality testing has been employed to verify complete operational capability. Dela Peña and Santos (2021) conducted functionality testing of their POS and inventory management system in a local food service business, verifying that sales processing, inventory updates, and report generation all operated correctly under typical usage scenarios. Similarly, Arig et al. (2025) evaluated the functionality of their POS system with a card reader for the HTM Department, confirming that all features including transaction processing and inventory tracking performed as intended without critical failures. Cruz, Dela Cruz, and Santos (2022) also employed functionality testing in their web-based sales and inventory system for small retail businesses, systematically verifying each feature against specified requirements to ensure complete operational capability.

The functionality evaluation for Operlytics encompassed verification that the Point of Sale module correctly processed transactions, calculated totals, applied discounts, and generated receipts; confirmation that the inventory tracking system accurately recorded stock movements, updated quantities in real-time, and generated low-stock alerts; validation that the reporting module produced correct summaries of sales, inventory, and financial performance; and confirmation that the dashboard displayed accurate, current operational data in an accessible visual format.

Criterion 2: Usability

This criterion measured how easily and intuitively café administrators and staff could navigate the system, access relevant data, and perform their assigned tasks with minimal training or technical difficulty. Feedback gathered from café staff and management during the evaluation phase was used to assess the clarity of the interface, the ease of navigation, and the overall user experience provided by the system in the context of daily café operations.

Usability evaluation employed both objective measures — such as task completion time, error rates, and navigation path efficiency — and subjective measures — such as user satisfaction ratings, perceived ease of use, and confidence in system operation. The usability criterion ensured that Operlytics was not merely technically capable but practically accessible to its intended users.

Usability evaluation is a critical dimension of information system assessment, particularly for systems deployed in small business environments where users may have limited technical expertise. Bienvenido (2023) conducted a comprehensive usability assessment of a centralized POS system for inventory management in a rotisserie operation, employing both objective measures (task completion time, error rates) and subjective measures (user satisfaction ratings, perceived ease of use) to evaluate how easily staff could navigate the system and perform daily tasks. His findings emphasize that usability directly impacts operational adoption rates and staff confidence in system operation.

Few (2013), in his work on information dashboard design, stresses that usability is paramount for at-a-glance monitoring systems, as users must be able to interpret data quickly and accurately without extensive training. The design of effective dashboards requires careful consideration of which metrics are most important for management decision-making, how these metrics relate to one another, how they can be visually represented for maximum clarity and interpretive accuracy, and how the dashboard layout supports natural scanning patterns and cognitive workflows.

In the context of small retail businesses, Cruz, Dela Cruz, and Santos (2022) gathered feedback from staff and management to assess interface clarity and navigation ease, confirming that their web-based system was accessible to users with minimal technical background. These findings align with the broader understanding that systems must be user-friendly to achieve their intended benefits, particularly in small business contexts where dedicated IT support is unavailable and staff may have limited technical expertise.

The usability evaluation for Operlytics encompassed measurement of task completion times for common operations (recording a sale, checking inventory, generating a report), assessment of error rates during data entry and navigation, evaluation of user satisfaction through structured questionnaires, and observation of user interactions to identify points of confusion or difficulty.

Criterion 3: Accuracy

This criterion evaluated the reliability and correctness of the data processed and displayed by the system, including sales records, inventory levels, and generated reports, to ensure that the information presented accurately reflected actual operational conditions. Evaluators compared system-generated data against actual operational records to identify any discrepancies and confirm that Operlytics consistently produced reliable and trustworthy outputs.

Accuracy evaluation involved data verification tests that compared system outputs to manually calculated results, reconciliation tests that verified consistency between related data elements, and historical validation that confirmed system reports matched known operational outcomes. The accuracy criterion ensured that management decisions based on Operlytics data were founded on correct information.

Data accuracy is fundamental to the value proposition of any business analytics or monitoring system. Chen, Chiang, and Storey (2012), in their seminal work on business intelligence and analytics, emphasize that the impact of analytics depends critically on the accuracy and reliability of the underlying data; inaccurate data leads to flawed insights and poor decision-making. The foundational importance of data quality for analytics effectiveness is further reinforced by Alomari (2025), who notes that data quality including accuracy is essential for driving informed decision-making in big data analytics contexts.

Ogunmokun, Balogun, and Ogunsola (2024), in their business intelligence dashboard optimization model, highlight that real-time performance tracking requires accurate data inputs to produce trustworthy forecasts and performance indicators. Venkat (2024) further supports this by examining real-time business intelligence systems and finding that decision-making behavior is directly influenced by users' trust in data accuracy. When users perceive data as unreliable, they are less likely to base decisions on system outputs, undermining the system's operational value.

In POS and inventory system contexts, accuracy validation has been demonstrated through multiple empirical studies. Dela Peña and Santos (2021) conducted data verification tests comparing system-generated sales records and inventory levels against manually calculated

results, confirming that their system produced reliable outputs. Tanaman et al. (2023) similarly performed reconciliation tests in their web-based inventory management system to verify consistency between related data elements and ensure that reports matched known operational outcomes. These practices establish that accuracy is a core information quality dimension, essential for ensuring that management decisions are founded on correct information.

The accuracy evaluation for Operlytics encompassed comparison of system-generated sales totals against manually calculated figures from the same period, verification that inventory deductions correctly reflected actual ingredient consumption, confirmation that financial reports accurately aggregated transaction data, and validation that dashboard displays reflected current rather than cached or outdated information.

Criterion 4: Security

This criterion ensured that all operational data stored and transmitted by the system was adequately protected through secure user authentication, access controls, and safe communication between the client and server, preventing unauthorized access or data breaches. The evaluation examined the effectiveness of the system's login mechanisms, role-based access controls, and data transmission protocols to confirm that the system met acceptable standards of data protection.

Security evaluation included penetration testing that attempted to identify vulnerabilities, access control verification that confirmed users could access only authorized functions, and data protection assessment that verified encryption and secure handling of sensitive information. The security criterion ensured that café operational data remains confidential and protected against unauthorized access.

Information security is a non-negotiable requirement for business systems handling sensitive operational data. Systems must protect data from unauthorized access, alteration, or destruction. Secure user authentication, access controls, and data encryption are essential for protecting organizational information assets. As systems become more data-intensive, security measures including secure data transmission protocols become increasingly critical for maintaining operational trust and regulatory compliance (Islam, 2024).

In the context of POS systems, security evaluation has focused on verifying that access controls effectively prevent unauthorized use. Arig et al. (2025) evaluated the security of their POS system with a card reader, examining login mechanisms and access controls to ensure that only authorized personnel could process transactions and access sensitive financial data. Dela Peña and Santos (2021) implemented role-based access controls in their food service POS system, verifying that users could access only functions appropriate to their responsibilities. Cruz, Dela Cruz, and Santos (2022) conducted access control verification in their web-based sales and inventory system, confirming that authentication mechanisms effectively prevented unauthorized access.

The security evaluation for Operlytics encompassed verification that password-based authentication correctly validated user credentials, confirmation that role-based access controls restricted users to functions appropriate for their responsibilities, assessment that data transmission between client and server was encrypted, and validation that session management prevented unauthorized access from unattended devices.

Criterion 5: Efficiency

This criterion assessed the system's overall performance, including the speed at which data was processed, updated, and displayed on the dashboard, ensuring that the system remained responsive and reliable during regular café operations. The evaluation examined the system's response times and overall stability under typical usage conditions to confirm that Operlytics could sustain consistent performance throughout the demands of the café's daily operations.

Efficiency evaluation involved performance testing that measured response times for typical operations, load testing that verified performance under expected user volumes, and stability testing that confirmed reliable operation over extended periods. The efficiency criterion ensured that Operlytics supported rather than impeded operational workflow speed.

System efficiency is critical for ensuring that technology supports rather than hinders operational workflows. Efficiency is a measure of how quickly a system performs its functions and how well it utilizes computing resources, and inefficient systems can create bottlenecks in

operational processes. Systems must process data quickly and respond promptly to user requests to maintain productivity.

In the context of real-time business intelligence, Venkat (2024) examines how system responsiveness influences decision-making behavior, finding that delays in data processing and dashboard updates can reduce user engagement and trust in the system. Ogunmokun, Balogun, and Ogunsola (2024) emphasize that business intelligence dashboards must deliver real-time performance tracking with minimal latency to be effective for managerial decision-making. Few (2013) notes that dashboard systems designed for at-a-glance monitoring must update efficiently to provide timely insights.

In POS system evaluations, efficiency testing has verified that systems remain responsive during peak operational periods. Dela Peña and Santos (2021) conducted performance testing to measure response times for typical operations such as sales processing and inventory queries, ensuring their system remained responsive during peak operational periods. Tanaman et al. (2023) performed stability testing on their web-based inventory management system, confirming reliable operation over extended periods. Cruz, Dela Cruz, and Santos (2022) similarly evaluated system performance under expected user volumes to verify that their web-based system could sustain consistent performance during daily operations.

The efficiency evaluation for Operlytics encompassed measurement of response times for common operations (loading the dashboard, processing a sale, generating a report), verification that the system remained responsive when multiple users accessed it simultaneously, confirmation that real-time updates appeared on dashboards within acceptable time limits, and assessment of system stability over extended operational periods.

Comprehensive Evaluation Framework

Taken together, these five evaluation criteria formed a comprehensive and balanced framework for assessing the overall quality and operational suitability of the Operlytics system. By measuring the system against standards of functionality, usability, accuracy, security, and efficiency, this study ensured that the evaluation process went beyond a purely technical

assessment to encompass the practical and human dimensions of system performance within a real business setting.

The use of multi-dimensional evaluation frameworks for information systems is well-supported in the literature. Information systems should be evaluated across multiple quality dimensions including functionality, usability, accuracy, security, and efficiency to ensure comprehensive assessment of system suitability. Business analytics systems must be evaluated not only on technical capabilities but also on their practical impact on decision-making and operational outcomes (Davenport & Harris, 2017). Effective evaluation of business analytics must encompass both system performance dimensions and human factors such as usability and trust (Gillon, McCarthy, & Spillane, 2012). Business analytics maturity assessments require multi-dimensional evaluation across process and performance outcomes (Mithas, Lee, & Wareham, 2013). Understanding business analytics impact requires evaluation that includes both technical and organizational evaluation dimensions (Sharma, Mithas, & Kankanhalli, 2014).

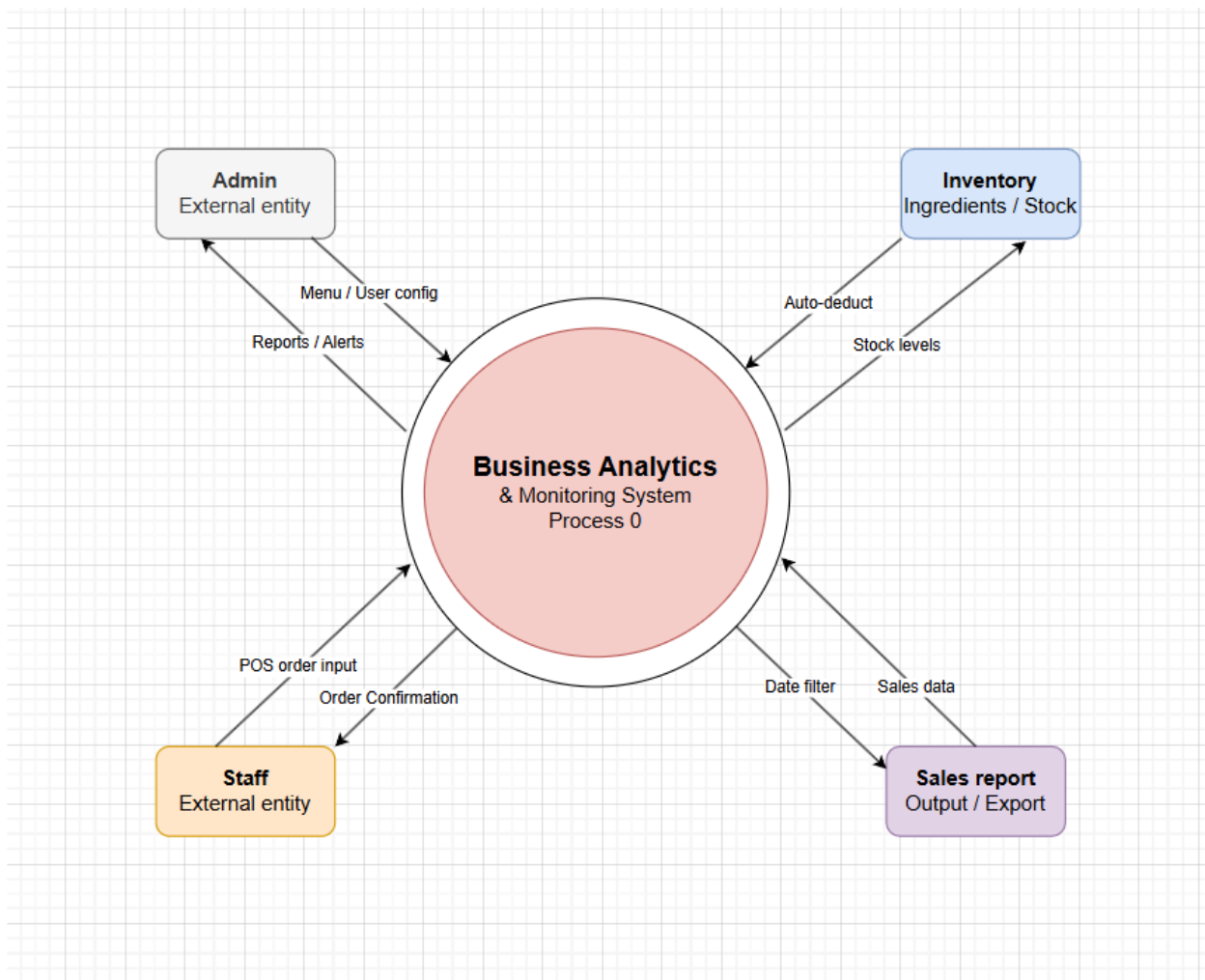
In the specific context of small business POS and inventory systems, multi-criteria evaluation has been employed to determine operational readiness. Dela Peña and Santos (2021) employed a comprehensive evaluation framework assessing functionality, usability, accuracy, and efficiency to determine their system's operational readiness. Bienvenido (2023) similarly used multi-criteria evaluation including usability, functionality, and accuracy to assess POS system suitability for a food service environment.

The results of this evaluation determined the extent to which Operlytics successfully addressed the operational challenges of Empress Café Boracay and provided valuable insights and recommendations that could guide future improvements to the system and inform similar development efforts aimed at supporting the digital transformation of small and medium enterprises in the food service industry.

SUPPORTING DIAGRAMS

Data Flow Diagram (DFD – Level 0)

This Level 0 Diagram illustrated the Business Analytics & Monitoring System (Process 0), serving as the high-level architectural blueprint for a comprehensive restaurant or retail management platform. The diagram captured the essential interactions between the central system and four critical external entities: Admin, Staff, Inventory, and Sales Reports. The system functioned as an integrated operational hub that automated order processing, maintained real-time stock visibility, and delivered actionable business intelligence for strategic oversight.



Central Process — Business Analytics & Monitoring System (Process 0)

The Business Analytics & Monitoring System stood as the nucleus of the entire operational ecosystem. It functioned as the intelligent orchestrator that received, processed, transformed, and distributed data across all connected components. This central process embodied the “brain” of the organization, executing three primary functions:

- **Order Processing & Validation** — Captured incoming POS transactions, validated order details, calculated totals, and triggered downstream workflows.
- **Inventory Management & Auto-Deduction** — Automatically adjusted stock levels upon order confirmation, preventing overselling and maintaining accurate ingredient counts.
- **Analytics Generation & Reporting** — Aggregated transactional data, applied business logic, and produced structured reports with filtering capabilities.

The system maintained bidirectional communication with every external entity, ensuring data consistency, operational transparency, and closed-loop feedback across the business.

Admin — External Entity

The Admin represented the managerial and supervisory layer of the organization — typically owners, managers, or system administrators who required elevated privileges and comprehensive visibility.

Outgoing Data Flow: Menu / User Configuration →

- The Admin initiated system setup and ongoing maintenance by transmitting configuration parameters.
- This included menu item definitions (names, descriptions, prices, categories), user account creation, role assignments, permission levels, tax settings, discount rules, and operational parameters.

- Configuration data established the foundational rules under which all other processes operated.

Incoming Data Flow: Reports / Alerts ←

- The system proactively pushed performance dashboards, exception notifications, and critical alerts back to the Admin.
- Reports could include daily revenue summaries, top-selling items, staff performance metrics, inventory shortage warnings, and anomaly detection flags.
- Alerts enabled rapid response to operational issues, such as low-stock thresholds, unusual transaction patterns, or system errors.

Staff — External Entity

The Staff represented the frontline operational personnel — servers, cashiers, or order-takers who interacted directly with customers and the Point-of-Sale (POS) interface.

Outgoing Data Flow: POS Order Input →

- Staff members entered customer orders through the POS terminal, transmitting order details including item selections, quantities, modifications, customer preferences, table numbers, and payment methods.
- Each order input triggered the system's processing pipeline, initiating validation, pricing calculation, and inventory deduction sequences.
- The POS interface served as the primary data entry point for all revenue-generating transactions.

Incoming Data Flow: Order Confirmation ←

- Upon successful processing, the system returned an order confirmation to the Staff member.
- This confirmation included a unique order ID, itemized breakdown, total amount, estimated preparation time, and status indicators.

- The confirmation assured the Staff member that the order had been properly recorded and queued for fulfillment, enabling them to proceed with customer service.

Inventory — Ingredients / Stock

The Inventory entity represented the physical stock repository — raw materials, ingredients, packaged goods, and supplies stored on-site or at warehouse locations.

Incoming Data Flow: Auto-Deduct ←

- The system automatically decremented inventory quantities based on the bill of materials (BOM) associated with each confirmed order.
- For example, if a “Cheeseburger” order was placed, the system deducted one bun, one patty, one slice of cheese, and associated condiments from current stock levels.
- This auto-deduction occurred in real-time, ensuring inventory records remained synchronized with actual consumption and preventing the acceptance of orders for out-of-stock items.

Outgoing Data Flow: Stock Levels →

- The Inventory entity continuously reported current stock quantities, expiration dates, reorder points, and supplier information back to the system.
- This feedback loop enabled the system to perform availability checks before order confirmation, generate low-stock alerts, and trigger reorder recommendations.
- Stock level data is also fed into analytics for waste tracking, turnover analysis, and procurement planning.

Sales Report — Output / Export

The Sales Report entity represented the analytical and documentation layer — the destination for processed business intelligence that supported monitoring, evaluation, and external reporting requirements.

Incoming Data Flow: Sales Data ←

- The system compiled comprehensive sales data including transaction timestamps, item-level details, revenue figures, payment types, staff associations, and customer trends.
- This data was structured, aggregated, and formatted for analytical consumption, enabling trend identification, seasonal analysis, and performance benchmarking.
- Sales data could be exported to external accounting software, business intelligence tools, or regulatory filing systems.

Outgoing Data Flow: Date Filter →

- Users of the Sales Report entity specified temporal parameters to narrow the scope of retrieved data.
- Date filters allowed granular control — single-day snapshots, weekly comparisons, monthly summaries, quarterly reviews, or custom date ranges.
- This filtering capability transformed raw transaction logs into targeted, relevant insights for specific analytical questions or reporting periods.

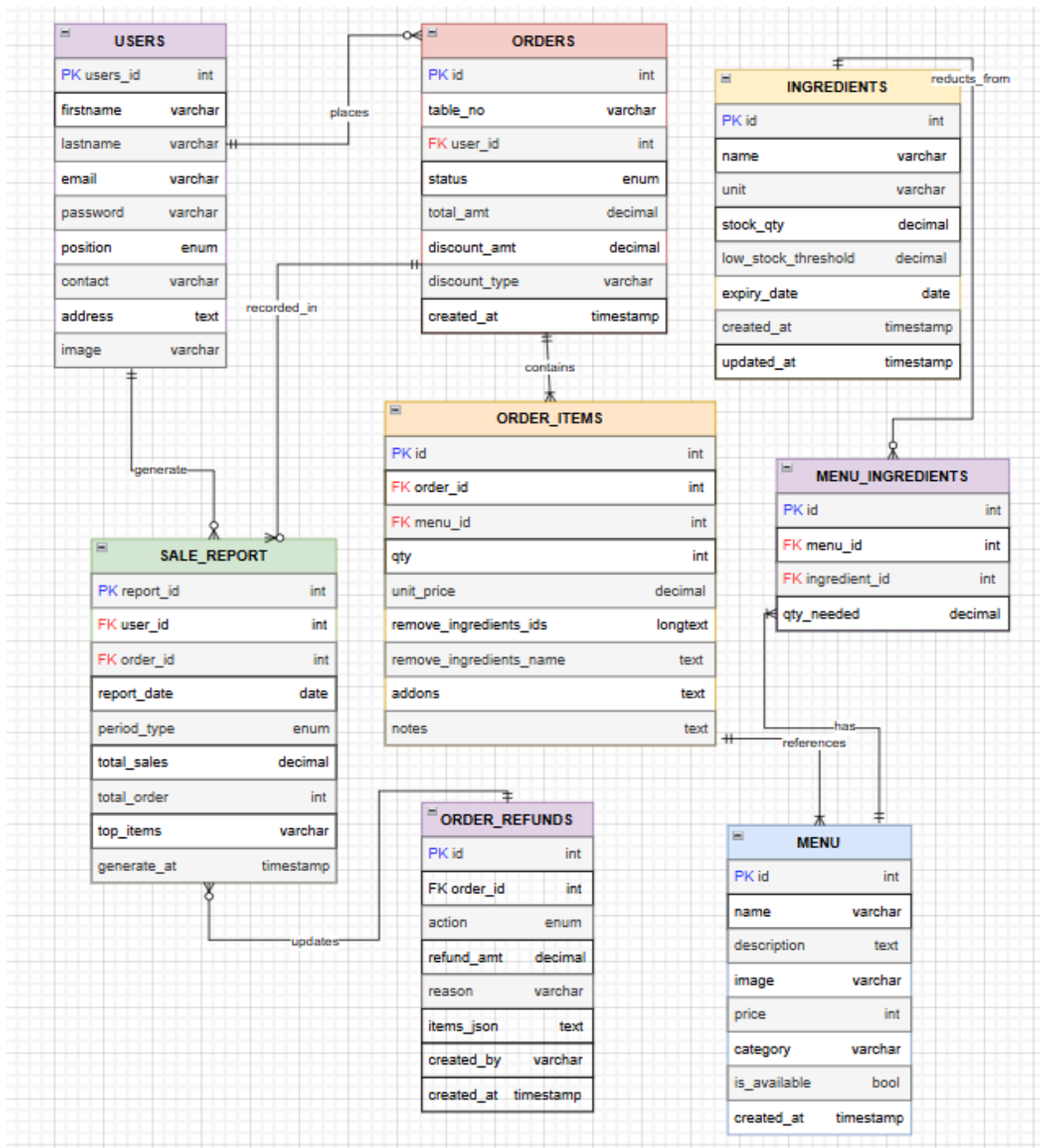
Key Design Characteristics

- **Automation:** Stock deduction occurred without manual intervention, reducing human error and ensuring real-time accuracy.
- **Closed-Loop Feedback:** Every external entity both provided input and received output, creating balanced, interactive relationships.
- **Centralized Control:** Process 0 maintained sole authority over data transformation, enforcing consistency and auditability.
- **Scalability:** The Level 0 abstraction allowed the system to expand internally (via Level 1 decomposition) without altering external interfaces.
- **Decision Support:** The Admin and Sales Report flows explicitly served strategic decision-making, elevating the system beyond pure transaction processing into business intelligence territory.

This Level 0 DFD provided the foundational context for subsequent decomposition into Level 1 processes, where each major function (order processing, inventory management, reporting) could be exploded into sub-processes with their own internal data flows and data stores.

Entity Relationship Diagram (ERD)

This diagram was an Entity-Relationship Diagram (ERD) for a café management or Operlytics system. It outlined how data was structured and shared between users, orders, menu items, and inventory.



Entities and Their Attributes

- **USERS** stored the account information of all system users, including administrators and staff. Key attributes included users_id, firstname, lastname, email, password, position, contact, address, and image.
- **ORDERS** recorded all customer orders placed within the café. Key attributes included id, table_no, user_id, status, total_amt, discount_amt, discount_type, and created_at.
- **ORDER_ITEMS** stored the individual items included in each order. Key attributes included id, order_id, menu_id, qty, unit_price, remove_ingredients_ids, remove_ingredients_name, addons, and notes.
- **ORDER_REFUNDS** tracked refund transactions associated with specific orders. Key attributes included id, order_id, action, refund_amt, reason, items_json, created_by, and created_at.
- **MENU** contained the list of available café products and their details. Key attributes included id, name, description, image, price, category, is_available, and created_at.
- **MENU_INGREDIENTS** served as a junction table linking menu items to their required ingredients. Key attributes included id, menu_id, ingredient_id, and qty_needed.
- **INGREDIENTS** managed the café's inventory of raw ingredients and supplies. Key attributes included id, name, unit, stock_qty, low_stock_threshold, expiry_date, created_at, and updated_at.
- **SALE_REPORT** stored generated sales reports summarizing operational performance. Key attributes included report_id, user_id, order_id, report_date, period_type, total_sales, total_order, top_items, and generate_at.

Relationships

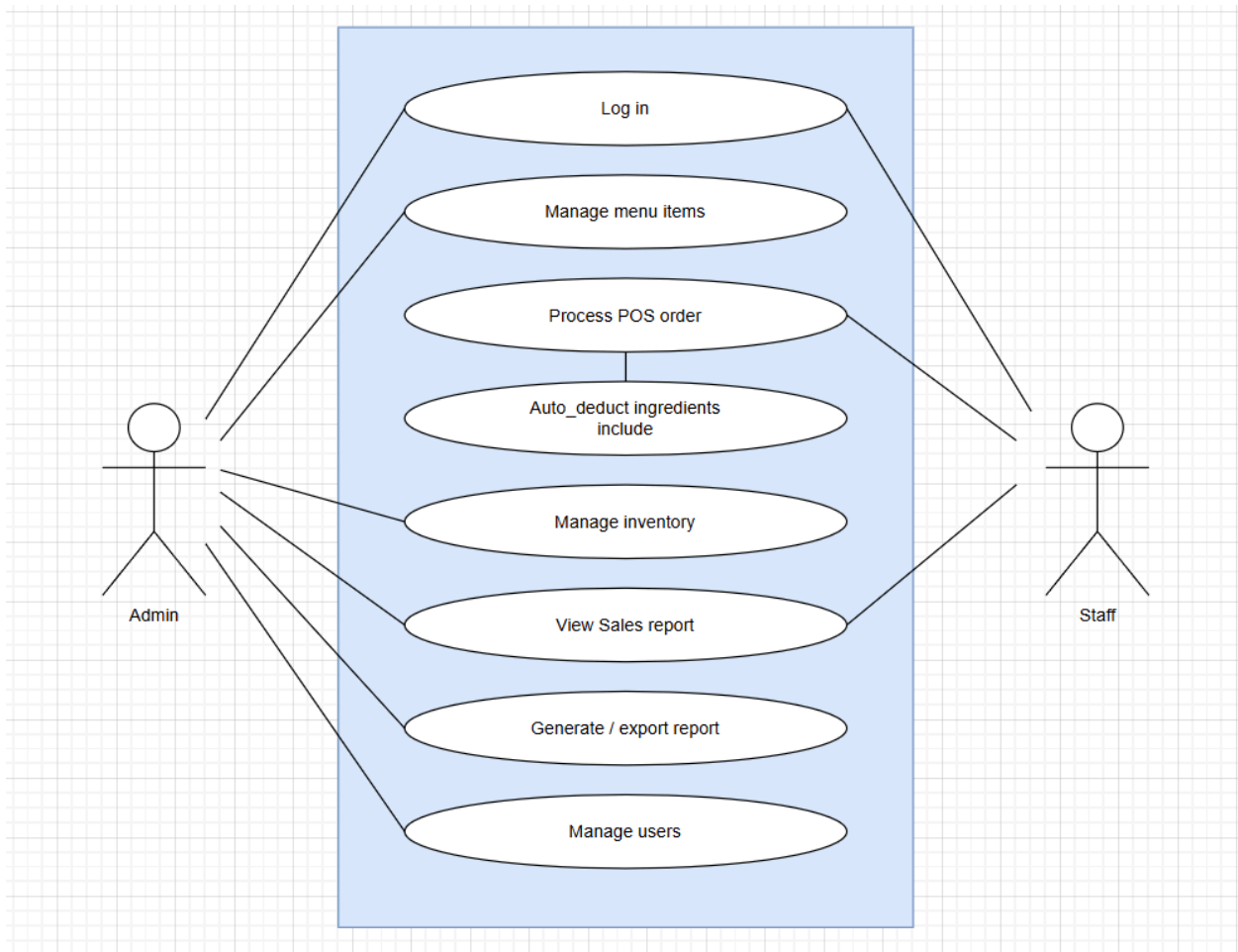
- **USERS → ORDERS**: A user placed one or many orders. This was a one-to-many relationship, where one user account could be associated with multiple order transactions.
- **USERS → SALE_REPORT**: A user generated one or many sales reports. This was a one-to-many relationship, where a single user could produce multiple generated reports over time.
- **ORDER → ORDER_ITEMS**: An order contains one or many order items. This was a one-to-many relationship, as each order transaction could include multiple individual menu items.
- **ORDERS → ORDER_REFUNDS**: An order could be associated with a refund transaction. This was a one-to-one or one-to-many relationship, where a single order could have one or more associated refund records.
- **ORDERS → SALE_REPORT**: An order was recorded in a sale report. This was a many-to-one relationship, where multiple orders could be consolidated into a single generated report.
- **SALE_REPORT → ORDERS**: A sales report updated based on order data. This reflected a many-to-one dependency where report summaries were derived from accumulated order records.
- **ORDER_ITEMS → MENU**: Each order item referenced a menu item. This was a many-to-one relationship, where multiple order items could correspond to the same menu entry.
- **MENU → MENU_INGREDIENTS**: A menu item has one or many ingredient requirements. This was a one-to-many relationship linking each menu product to the ingredients needed for its preparation.
- **MENU_INGREDIENTS → INGREDIENTS**: Each menu ingredient entry referenced a specific ingredient in the inventory. This was a many-to-one relationship, where multiple menu items could require the same ingredient.

- **INGREDIENTS → ORDER_ITEMS:** Ingredients were deducted from inventory based on order items processed. This reflected the auto-deduction relationship where completed orders triggered automatic stock reduction in the ingredients table.

The database schema of Operlytics, composed of eight entities and nine defined relationships, established a comprehensive and logically structured data management foundation that directly supported the system's core operational objectives. The careful design of entity relationships ensured referential integrity across all interconnected tables, minimizing the risk of data inconsistencies that could compromise the accuracy of sales records, inventory levels, and generated reports. By aligning the database structure with the actual operational workflows of Empress Café Boracay, the schema enabled Operlytics to process, store, and retrieve business data in a manner that was both efficient and reflective of real café operations, ultimately providing management with a reliable and well-organized data environment upon which sound and informed business decisions could be consistently made.

Use Case Diagram

This Use Case Diagram showed that the Admin had full control over all system functions, including inventory, user management, and reporting, while the Staff was restricted to logging in and processing customer orders.



Review of Related Literature. The design of role-based access control in information systems is well-supported in the academic literature. Laudon and Laudon (2020) emphasize that role-based access control (RBAC) is a fundamental security mechanism in management information systems, ensuring that users can access only the functions and data appropriate to their responsibilities. O'Brien and Marakas (2018) explain that RBAC models assign permissions based on user roles rather than individual identities, which simplifies administration and enhances security by ensuring that sensitive functions are restricted to authorized personnel. In POS and inventory system development, Dela Peña and Santos (2021) implemented a similar role-based structure in their food service POS system, where administrators controlled system configuration, user management, and reporting, while staff were limited to sales processing and basic inventory queries. Arig et al. (2025) employed role-based access differentiation in their POS system with card reader for the HTM Department, restricting staff to transaction processing while granting administrators full system control. Bienvenido (2023), in his assessment of centralized POS system usability, found that clear role differentiation between administrative and operational staff significantly improved system security perceptions and reduced errors caused by unauthorized access to sensitive functions. Cruz, Dela Cruz, and Santos (2022) similarly structured their web-based sales and inventory system with distinct admin and staff roles, demonstrating that this approach is standard practice for small business systems where operational staff require limited access to complete their tasks effectively.

The Operlytics system involved two primary actors — the Admin and the Staff — each assigned a defined set of system functions that corresponded to their respective roles and responsibilities within Empress Café Boracay.

Review of Related Literature. The identification of primary actors and their associated functions is a critical step in use case modeling and systems analysis. Sommerville (2015) describes actor identification as essential for understanding the different ways users interact with a system and for defining the boundaries of system functionality. Pressman (2014) emphasizes that use case diagrams must clearly distinguish between different user types and their respective privileges to ensure that the system design reflects actual organizational roles. Laudon and Laudon (2020) support this by explaining that understanding actor roles is fundamental to designing information systems that align with organizational structures and workflows. In the context of café and food service operations, Dela Peña and Santos (2021) identified administrators and cashiers as primary actors in their POS system, assigning distinct functions that matched their operational responsibilities. Tanaman et al. (2023) defined admin and staff roles in their web-based inventory management system, ensuring that each actor had access only to functions relevant to their duties. Bienvenido (2023) analyzed actor roles in his POS usability assessment, noting that clear definition of admin and staff functions was essential for both security and operational efficiency. The actor definitions in Operlytics thus align with established practices in systems analysis and design.

Actors:

Admin — The Admin represented the café manager or authorized administrator who held full access to the system's operational and managerial functions. The Admin was responsible for overseeing the overall use of the system and had the highest level of access privileges among all system users.

Review of Related Literature. The administrator role in information systems is consistently defined across the literature as the user with comprehensive system access and managerial oversight. O'Brien and Marakas (2018) describe system administrators as users responsible for configuring, managing, and overseeing system operations, including user account management, security settings, and system maintenance. Laudon and Laudon (2020) identify the administrator as a critical actor who ensures system integrity, manages data access policies, and generates reports for strategic decision-making. In small business POS systems, Cruz, Dela Cruz, and Santos (2022) defined the admin role as having full access to sales data, inventory records, user management, and report generation functions. Dela Peña and Santos (2021) similarly granted administrators comprehensive control over their food service POS system, including menu management, inventory oversight, and staff account administration. Arig et al. (2025) structured their POS system so that administrators managed system configuration, user accounts, and transaction monitoring. The Admin role in Operlytics thus reflected standard practice, ensuring that managerial oversight and system governance were centralized under authorized personnel.

Staff — The Staff represented the café employees who were granted limited but essential access to the system for the purpose of carrying out their daily operational tasks, particularly those related to sales transactions and inventory movement.

Review of Related Literature. The staff or operational user role is defined in the literature as having restricted access focused on daily transactional activities. Pressman (2014) explains that limiting operational user access to essential functions reduces the risk of accidental data corruption and simplifies the user interface for non-technical personnel. Laudon and Laudon (2020) support this by noting that operational staff require access only to functions directly

related to their job responsibilities, which both enhances security and improves usability. O'Brien and Marakas (2018) emphasize that restricting staff access to core operational tasks—such as transaction processing and basic data entry—is a standard security practice in business information systems. In POS system implementations, Bienvenido (2023) found that limiting staff to sales processing and basic inventory functions improved system usability and reduced training requirements. Dela Peña and Santos (2021) restricted their café staff to POS order processing and inventory viewing, while administrators retained control over system configuration and reporting. Arig et al. (2025) similarly limited HTM Department staff to transaction processing and card reader operations. The Staff role in Operlytics thus followed established conventions, ensuring that operational employees could perform their duties efficiently without exposure to sensitive administrative functions.

Use Cases:

Admin:

Admin can Log in

The login function provided secure authentication for the Admin, ensuring that only authorized personnel could access the system. This use case established the identity of the user and determined the appropriate access privileges based on their role.

Review of Related Literature. Secure user authentication is a foundational requirement for information systems. O'Brien and Marakas (2018) describe login mechanisms as the first line of defense in system security, verifying user identity before granting access to system functions. Laudon and Laudon (2020) emphasize that authentication procedures must be robust yet user-friendly to balance security with usability. In web-based business systems, Cruz, Dela Cruz, and Santos (2022) implemented secure login functionality for both admin and staff users, ensuring that each session was properly authenticated before granting role-appropriate access. Dela Peña and Santos (2021) required all users to authenticate through a login screen that validated credentials against stored records before permitting system access. Arig et al. (2025) incorporated login authentication as a prerequisite for all system functions in their POS system.

The login use case in Operlytics thus aligns with standard security practices, ensuring that system access is properly controlled from the point of entry.

Admin can Manage menu items

This use case allowed the Admin to add, edit, delete, and categorize menu items in the system. The Admin could set prices, descriptions, and ingredient associations for each menu item, ensuring that the POS system reflected current café offerings.

Review of Related Literature. Menu management is a critical administrative function in food service POS systems. Dela Peña and Santos (2021) included menu item management as a core admin function in their food service POS system, enabling administrators to update prices, descriptions, and availability status. Laudon and Laudon (2020) explain that product or service catalog management is essential for maintaining accurate transaction records and ensuring that sales data reflects current business offerings. O'Brien and Marakas (2018) support this by noting that administrative control over product databases ensures data consistency and prevents unauthorized modifications that could affect pricing or inventory calculations. In café and restaurant contexts, Bienvenido (2023) found that centralized menu management by administrators was essential for maintaining consistency across multiple service points. Cruz, Dela Cruz, and Santos (2022) similarly granted administrators exclusive control over product catalog management in their retail system. The menu management use case in Operlytics thus addresses a well-established requirement for food service business systems.

Admin can Process POS order

This use case enabled the Admin to process customer orders through the Point of Sale interface, including item selection, quantity specification, payment handling, and receipt generation.

Review of Related Literature. POS order processing is the core transactional function of any retail or food service system. Dela Peña and Santos (2021) identified POS order processing as the central use case in their food service system, through which all sales transactions were

recorded and inventory was automatically updated. Laudon and Laudon (2020) describe transaction processing as a fundamental information system activity that must be accurate, efficient, and reliable to support business operations. O'Brien and Marakas (2018) explain that POS systems must handle item selection, pricing calculation, payment processing, and receipt generation as integrated workflow steps. Arig et al. (2025) structured their POS system around order processing as the primary staff and admin function, ensuring that all sales were properly recorded. Bienvenido (2023) evaluated POS order processing usability as a critical factor in system adoption, finding that streamlined transaction workflows significantly improved staff efficiency. The POS order processing use case in Operlytics thus reflects the central operational requirement identified across food service system literature.

Admin can Auto-deduct ingredients

This use case triggered automatic deduction of ingredient quantities from inventory whenever a menu item was sold through the POS system. The Admin could configure ingredient associations and deduction rules for each menu item.

Review of Related Literature. Automated ingredient deduction is a key feature that distinguishes advanced POS systems from basic transaction recorders. Dela Peña and Santos (2021) implemented automatic inventory deduction in their food service POS system, linking menu items to their component ingredients so that each sale automatically reduced inventory quantities proportionally. Laudon and Laudon (2020) explain that automated data updates across related records reduce manual entry errors and ensure real-time data consistency. O'Brien and Marakas (2018) support this by noting that automated workflow triggers—such as inventory updates following sales transactions—improve operational efficiency and data accuracy. Tanaman et al. (2023) incorporated automatic stock adjustment features in their inventory management system, ensuring that inventory records remained current without manual intervention. Few (2013) emphasizes that automated data flows are essential for real-time dashboard accuracy, as manual updates introduce delays and inconsistencies. The auto-deduct ingredients use case in Operlytics thus addresses a recognized need for automation in food service inventory management.

Admin can Manage inventory

This use case provided the Admin with comprehensive inventory management capabilities, including viewing current stock levels, adding new inventory items, updating quantities, setting reorder points, and reviewing inventory movement history.

Review of Related Literature. Inventory management is a core function of business analytics and monitoring systems. Laudon and Laudon (2020) identify inventory control as a critical business process that information systems must support through real-time tracking, automated alerts, and historical reporting. O'Brien and Marakas (2018) explain that inventory management systems must maintain accurate records of stock levels, track item movements, and support reorder decisions to prevent stockouts and overstock situations. Cruz, Dela Cruz, and Santos (2022) developed comprehensive inventory management functionality for their web-based system, enabling administrators to monitor stock levels, update quantities, and generate inventory reports. Tanaman et al. (2023) centered their web-based system on inventory management, providing administrators with tools for stock monitoring, quantity updates, and movement tracking. Dela Peña and Santos (2021) integrated inventory management with their POS system, ensuring that sales and inventory data remained synchronized. Bienvenido (2023) found that centralized inventory management was essential for maintaining accurate stock records in multi-point food service operations. The inventory management use case in Operlytics thus reflects established best practices for business system design.

Admin can View sales report

This use case allowed the Admin to access and review sales reports generated by the system, including daily summaries, period comparisons, and trend analyses presented through the dashboard interface.

Review of Related Literature. Sales reporting is a fundamental analytics function that supports managerial decision-making. Chen, Chiang, and Storey (2012), in their seminal work on business intelligence and analytics, emphasize that reporting capabilities transform transactional data into actionable insights for management. Few (2013) explains that dashboard-based sales reports must present data at a glance, enabling managers to quickly assess performance without

extensive analysis. Ogunmokun, Balogun, and Ogunsola (2024) highlight that real-time sales reporting is essential for business intelligence dashboards, providing managers with current performance indicators. Laudon and Laudon (2020) identify reporting as a critical output of management information systems, enabling data-driven decision-making at the operational and strategic levels. Dela Peña and Santos (2021) included sales reporting as a core admin function in their POS system, generating daily and periodic summaries for managerial review. Cruz, Dela Cruz, and Santos (2022) developed sales reporting functionality that enabled administrators to view transaction summaries and identify trends. The view sales report use case in Operlytics thus addresses a well-established requirement for managerial oversight in business systems.

Admin can Generate / export report

This use case enabled the Admin to generate detailed reports in various formats and export them for external use, such as sharing with accountants, investors, or regulatory authorities.

Review of Related Literature. Report generation and export functionality extends the utility of business analytics systems beyond internal use. Laudon and Laudon (2020) explain that information systems must support data export in standard formats to facilitate information sharing with external stakeholders and compliance with reporting requirements. O'Brien and Marakas (2018) note that export capabilities enable integration with other business tools, such as accounting software and spreadsheet applications. Alomari (2025) emphasizes that business analytics systems must support flexible reporting to meet diverse stakeholder information needs. Davenport and Harris (2017) identify report generation as a key capability that enables organizations to compete on analytics by distributing insights across the enterprise. In small business contexts, Cruz, Dela Cruz, and Santos (2022) included report export functionality in their web-based system, allowing administrators to generate and share sales and inventory reports. Tanaman et al. (2023) developed export features for their inventory management system, enabling data sharing in standard formats. The generate/export report use case in Operlytics thus supports both internal analysis and external information sharing requirements.

Admin can Manage users

This use case allowed the Admin to create, modify, and deactivate user accounts, assign roles, and configure access permissions for all system users.

Review of Related Literature. User management is a critical administrative function that maintains system security and operational control. O'Brien and Marakas (2018) describe user account management as essential for maintaining system security, enabling administrators to control who can access the system and what functions they can perform. Laudon and Laudon (2020) emphasize that user management is a core responsibility of system administrators, ensuring that access privileges align with organizational roles and security policies. Pressman (2014) notes that user management functions must be restricted to authorized administrators to prevent unauthorized privilege escalation. In POS system implementations, Dela Peña and Santos (2021) granted administrators exclusive control over user account creation, modification, and deactivation. Arig et al. (2025) included user management as an admin-only function in their POS system, ensuring that staff could not create or modify accounts. Bienvenido (2023) found that centralized user management improved system security perceptions and simplified administration in multi-staff environments. The managed users use case in Operlytics thus follows established security practices for business information systems.

Staff:

Staff can Log in

This use case provided secure authentication for Staff members, verifying their identity and granting access to their limited set of authorized functions.

Review of Related Literature. Staff login functionality, while similar in mechanism to admin login, operates within a restricted privilege framework. O'Brien and Marakas (2018) explain that authentication systems must verify user identity and then apply role-based access controls to determine available functions. Laudon and Laudon (2020) emphasize that all users, regardless of role, must authenticate before accessing system functions to maintain security audit trails. Dela Peña and Santos (2021) implemented separate login credentials for staff members, ensuring that their system access was properly authenticated and logged. Cruz, Dela Cruz, and Santos (2022) required all users to authenticate through a common login interface that then directed them to role-appropriate functions. Arig et al. (2025) incorporated staff login as the entry point to their restricted POS functionality. The staff login use case in Operlytics thus maintains consistent security standards while enabling appropriate function access.

Staff can Process POS order

This use case enabled Staff members to process customer orders through the Point of Sale interface, performing the core transactional functions required for daily café operations.

Review of Related Literature. Staff access to POS order processing reflects the operational reality that frontline employees require efficient transaction tools. Bienvenido (2023) found that restricting staff to core transactional functions improved system usability and reduced training time, as employees were not overwhelmed with unnecessary features. Dela Peña and Santos (2021) granted their café staff access to POS order processing while restricting administrative functions, enabling efficient service delivery while maintaining security. Laudon and Laudon (2020) explain that operational staff require streamlined interfaces focused on their primary tasks to maximize efficiency and minimize errors. O'Brien and Marakas (2018) support this by noting that function restriction based on role improves both security and usability by presenting users only with relevant options. Cruz, Dela Cruz, and Santos (2022) limited their retail staff to sales

processing functions, finding that this focused access improved transaction speed and accuracy. The staff POS order processing use case in Operlytics thus aligns with established principles of role-appropriate system design.

Staff can Auto-deduct ingredients

This use case enabled the automatic deduction of ingredient quantities from inventory when Staff processed sales through the POS system, ensuring that inventory records remained current without requiring manual updates.

Review of Related Literature. Extending automated ingredient deduction to staff-level transactions ensures consistent inventory accuracy regardless of which user processes the sale. Dela Peña and Santos (2021) implemented automatic inventory updates triggered by all POS transactions, whether processed by admin or staff, ensuring that inventory data remained synchronized across all sales channels. Laudon and Laudon (2020) explain that automated workflows should operate consistently across user roles to maintain data integrity. O'Brien and Marakas (2018) support this by noting that business rules—such as inventory deduction triggers—should be enforced uniformly regardless of which user initiates the transaction. Tanaman et al. (2023) designed their inventory system so that stock adjustments occurred automatically upon transaction completion, without requiring manual intervention from operational staff. Few (2013) emphasizes that consistent automation across all users is essential for real-time dashboard accuracy. The staff auto-deduct ingredients use case in Operlytics thus ensures that inventory automation operates comprehensively across all transactional users.

System Design Rationale

The defined use cases of the Operlytics system reflected a deliberate and structured approach to role-based access control, ensuring that each actor within the system was granted only the functions necessary to fulfill their specific responsibilities. By assigning comprehensive managerial functions to the Admin and limiting Staff access to core transactional tasks, the system maintained operational security while minimizing the risk of unauthorized data modification or accidental system misuse.

Review of Related Literature. The principle of least privilege—granting users only the minimum access necessary to perform their duties—is a cornerstone of information security. O'Brien and Marakas (2018) describe this principle as essential for protecting sensitive data and preventing accidental or malicious system misuse. Laudon and Laudon (2020) support this by explaining that role-based access control implements the principle of least privilege by restricting users to functions relevant to their organizational roles. Pressman (2014) notes that this approach simplifies user interfaces, reduces training requirements, and improves system security. In POS and inventory systems, Dela Peña and Santos (2021) applied the principle of least privilege by restricting staff to transactional functions while reserving administrative controls for managers. Cruz, Dela Cruz, and Santos (2022) similarly limited operational staff access to essential sales and inventory functions. Bienvenido (2023) found that adherence to the principle of least privilege significantly improved system security perceptions and reduced operational errors in his POS usability assessment. Arig et al. (2025) structured their system access based on role-specific minimum requirements. The use case structure in Operlytics thus reflects established security principles and industry best practices.

The inclusion of key functionalities such as automated ingredient deduction, real-time inventory management, and report generation directly addressed the operational deficiencies previously identified at Empress Café Boracay. Together, these use cases formed the functional backbone of Operlytics, establishing a clear and practical framework through which the system supported efficient daily operations, accurate record-keeping, and informed managerial decision-making across all levels of the café's organizational structure.

Review of Related Literature. The integration of automated processes, real-time inventory tracking, and reporting capabilities addresses well-documented operational challenges in small food service businesses. Laudon and Laudon (2020) identify automation of routine tasks, real-time data availability, and managerial reporting as essential capabilities for management information systems in small businesses. O'Brien and Marakas (2018) explain that these features reduce manual workload, minimize errors, and provide managers with timely information for decision-making. Dela Peña and Santos (2021) found that automated ingredient deduction and real-time inventory tracking eliminated manual reconciliation tasks and reduced stock

discrepancies in their café implementation. Tanaman et al. (2023) demonstrated that web-based inventory management with automated updates significantly improved stock control accuracy. Few (2013) emphasizes that dashboard-based reporting transforms raw data into actionable insights that support at-a-glance decision-making. Ogunmokun, Balogun, and Ogunsola (2024) highlight that real-time dashboard functionality is essential for effective business intelligence in operational environments. Cruz, Dela Cruz, and Santos (2022) showed that integrated sales and inventory systems with reporting capabilities improved managerial oversight in small retail businesses. The combined functionality of Operlytics thus addresses a comprehensive set of operational requirements established in the literature for small business information systems.

Chapter 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

Operlytics is a comprehensive web-based business analytics and monitoring system meticulously developed to address the multifaceted operational challenges confronting Empress Café Boracay. Situated within the highly competitive and tourism-driven market of Boracay Island, the café historically relied upon fragmented, manual, and paper-based methods for recording sales transactions, monitoring inventory levels, and compiling operational reports. These antiquated practices resulted in persistent data inconsistencies, delayed managerial oversight, frequent stock discrepancies, and an overall inability to leverage operational data for strategic decision-making. Operlytics was conceived as a targeted, integrated digital solution to systematically eliminate these deficiencies by providing a centralized platform that unifies Point-of-Sale (POS) transaction processing, real-time inventory tracking, automated reporting, and interactive data visualization within a single, cohesive environment.

The system architecture of Operlytics is deliberately designed to accommodate the unique operational rhythms of a small food service establishment serving a high-volume, seasonal tourist clientele. Unlike generic POS systems that merely record transactions without regard for downstream operational consequences, Operlytics embeds intelligent business logic that automatically correlates sales activities with inventory consumption, financial aggregation, and analytical reporting. When a staff member processes a customer order through the POS interface, the system does not simply store a transactional record; it simultaneously deducts ingredient quantities from inventory based on predefined recipe mappings, updates real-time dashboard metrics, recalculates sales summaries, and evaluates whether any stock items have fallen below critical reorder thresholds. This deep integration ensures that every transactional event

propagates accurately and instantaneously across all functional domains, eliminating the information silos and reconciliation delays that characterize manual operational management.

Operlytics comprises five interconnected core modules, each engineered to address specific operational requirements identified during the needs analysis phase:

Point of Sale (POS) Module. This module serves as the primary operational interface for frontline staff, designed with an emphasis on speed, accuracy, and intuitive interaction during high-pressure service periods. The POS interface presents menu items through categorized grids with visual product identifiers, enabling rapid item selection via mouse click or touch interaction. Staff members can modify orders by adjusting quantities, applying percentage-based or fixed-amount discounts, accommodating special instructions or ingredient removals, and processing multiple payment methods including cash and digital wallets prevalent among Boracay tourists. Upon transaction confirmation, the system generates a digital receipt, logs the complete transaction audit trail including timestamp, attending staff member, table number, and payment type and triggers the automatic inventory deduction workflow. The POS module is optimized for responsiveness, ensuring that even during peak breakfast and dinner rushes, transaction processing does not introduce perceptible delays into customer service workflows.

Inventory Management Module. This module constitutes the operational backbone of the system's stock control capabilities. It maintains a comprehensive digital ledger of all ingredients and supplies utilized in café operations, ranging from perishable items such as fresh produce, dairy, and meats to non-perishable supplies including packaging materials, cleaning agents, and disposable utensils. Each inventory item is characterized by attributes including current stock quantity, unit of measurement, low-stock alert threshold, expiration date, supplier information, and last-updated timestamp. The module supports three categories of stock movements: automatic deductions triggered by confirmed POS transactions according to recipe specifications; manual adjustments for supplier deliveries, waste recording, spoilage documentation, and physical count reconciliations; and system-generated alerts when quantities breach defined minimum thresholds. Every adjustment is immutably logged with user

identification, timestamp, and reason codes, creating a complete audit trail that enables management to trace inventory discrepancies to their source.

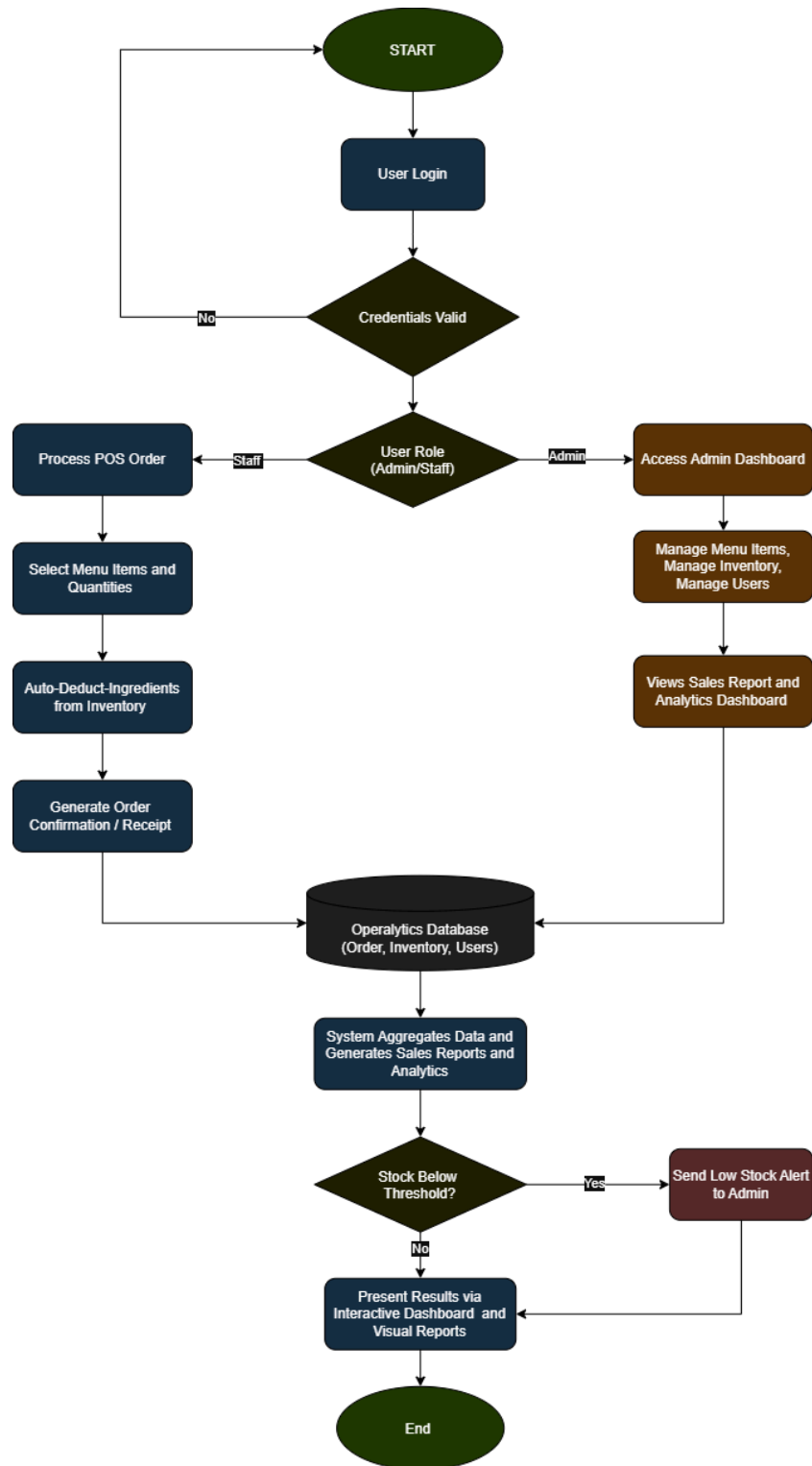
Dashboard and Real-Time Analytics Module. This module transforms raw operational data into immediately comprehensible visual intelligence for café administrators. The administrative dashboard serves as a centralized command center presenting key performance indicators through color-coded visual indicators, animated counters, trend charts, and exception alerts. Primary dashboard components include a real-time revenue monitor comparing current-day sales against historical baselines; an inventory status overview highlighting items below reorder thresholds in red, items approaching thresholds in yellow, and adequately stocked items in green; a top-performing products display ranking menu items by sales volume and revenue contribution; a staff activity summary showing transaction counts and average order values by employee; and an alerts panel surfacing operational exceptions such as unusual void patterns, low-stock conditions, or pending refund requests. The dashboard refreshes dynamically through asynchronous server communication, ensuring that management views reflect current operational reality without requiring manual page reloads.

Automated Reporting Module. This module eliminates the laborious manual compilation processes previously required to produce operational summaries. It generates standardized reports across multiple temporal granularities, daily shift summaries, weekly performance reviews, monthly financial overviews, and custom date-range analyses. Report categories include sales reports itemizing revenue by product, category, payment method, and time period; inventory reports detailing current stock levels, consumption rates, waste percentages, and reorder recommendations; financial reports calculating gross revenue, cost of goods sold, profit margins, and discount impacts; and operational reports analyzing transaction volumes, peak hour patterns, average order values, and customer throughput metrics. All reports are computed directly from live database records through optimized aggregation queries, ensuring mathematical accuracy and temporal relevance. Reports may be viewed within the browser interface, printed for physical distribution, or exported in standard formats for external sharing with accountants, suppliers, or regulatory authorities.

User Management and Security Module. This module implements a robust role-based access control framework distinguishing between Administrator and Staff user roles. Administrators typically the café owner or senior manager possess unrestricted system access including menu configuration, inventory oversight, comprehensive report generation, user account creation and modification, system parameter adjustment, and audit trail review. Staff members are granted restricted access limited to POS transaction processing, basic inventory quantity viewing, and personal profile management. The module enforces password-based authentication with encrypted credential storage, server-side session management with configurable timeout controls, and granular permission verification at every functional endpoint. All user activities including logins, transactions, inventory adjustments, and report accesses are logged to an immutable audit trail that supports security monitoring and operational accountability.

The development of Operlytics adhered to modern software engineering principles emphasizing modularity, scalability, and maintainability. The system is constructed upon a LAMP-stack foundation (Apache, MySQL, PHP) augmented with contemporary frontend technologies, ensuring broad compatibility with affordable hosting infrastructure while providing the performance characteristics necessary for real-time operational environments. The user interface employs responsive design methodologies that adapt seamlessly across desktop monitors, laptops, and tablet devices, acknowledging the diverse hardware contexts in which café staff and management interact with the system. Security considerations are embedded throughout the architecture, from input validation and SQL injection prevention at the application layer to password hashing and secure session management at the authentication layer.

Figure 4.1 illustrates the overall operational workflow of the developed system.



The workflow commences with user authentication, where the system verifies credentials and establishes a role-appropriate session. Upon successful login, administrators are directed to the main dashboard presenting high-level operational metrics, while staff members are presented with the streamlined POS interface. When staff process customer orders, transactional data flows through the application server to the database, triggering cascading updates across inventory records, sales aggregations, and dashboard displays. Administrators may concurrently access inventory management functions to review stock levels, adjust quantities, or configure alert thresholds. At periodic intervals or on demand, the reporting engine aggregates accumulated transactional data into structured summaries accessible through the reports interface. Throughout all operations, the user management module enforces access controls and logs activities to maintain system security and operational transparency.

4.3 System Development Methodology

The development of Operlytics was structured around the Agile Software Development Methodology, an iterative and incremental approach that emphasizes adaptive planning, evolutionary development, early delivery, and continuous improvement through rapid and flexible response to change. Agile was deliberately selected over traditional linear models such as Waterfall because the operational environment of Empress Café Boracay presented requirements that were initially fluid and subject to refinement as stakeholders gained greater familiarity with digital operational possibilities. The methodology facilitated close collaboration between the development team and café personnel, ensuring that emergent operational insights—such as the need for split-payment handling during peak tourist seasons or the importance of offline resilience during intermittent connectivity—could be incorporated into the system without disrupting the overall development trajectory.

The Agile framework was operationalized through a series of time-boxed development sprints, each spanning two weeks, during which specific functional increments were designed, coded, tested, and demonstrated. This sprint-based structure provided regular milestones for stakeholder feedback while maintaining momentum toward the comprehensive system vision. The development process was organized into seven distinct phases, each contributing essential activities to the maturation of the Operlytics platform.

Phase 1: Requirements Gathering and Analysis.

This foundational phase involved intensive engagement with Empress Café Boracay stakeholders to establish a thorough empirical understanding of current operational practices, pain points, and functional aspirations. The research team conducted semi-structured interviews with the café owner and floor staff, performed direct observational sessions across multiple operational periods including weekday mornings, lunch rushes, and weekend peak hours, and analyzed existing operational documents including handwritten sales logbooks, paper inventory count sheets, supplier delivery receipts, and informal financial notes.

These multi-source data gathering techniques revealed critical insights: sales recording was

decentralized across multiple staff-maintained notebooks with inconsistent formatting; inventory counts occurred sporadically with no systematic linkage to sales activity; report generation required several hours of manual arithmetic and transcription; and management lacked visibility into real-time stock conditions. These findings were synthesized into a prioritized product backlog comprising user stories, functional requirements, and non-functional constraints that guided all subsequent development activities.

Phase 2: System Planning and Feasibility Analysis.

Following requirements consolidation, the development team established the project scope, defined measurable objectives, specified technical architectures, and constructed a detailed development schedule with resource allocation. Feasibility analysis encompassed technical feasibility confirming that the selected PHP-MySQL technology stack could support real-time operational demands within affordable hosting parameters; operational feasibility assessing whether café staff could realistically transition from paper-based to digital workflows given their current technical exposure; and economic feasibility demonstrating that development and deployment costs remained within constraints appropriate for a small café enterprise. The planning phase also identified risk factors including staff resistance to technological change, intermittent internet connectivity on Boracay Island, and hardware limitations, for which mitigation strategies were developed including comprehensive user training, offline contingency protocols, and responsive design accommodating older devices.

Phase 3: System Design and Modeling.

During this phase, the abstract requirements were translated into concrete technical specifications and architectural blueprints. Database design proceeded through normalization to Third Normal Form (3NF), resulting in the entity-relationship structure comprising eight core tables with defined primary keys, foreign key constraints, and indexing strategies. Process modeling produced Data Flow Diagrams (DFDs) at Level 0 and Level 1, illustrating the movement of transactional data from user inputs through processing engines to database storage and report generation. Unified Modeling Language (UML) diagrams including use case diagrams, activity diagrams, and sequence diagrams formalized user interactions and system

behaviors. Wireframes and mockups for all major interface screens were developed and reviewed with café administrators to validate layout intuitiveness before coding commenced. The design phase explicitly addressed usability considerations for non-technical users, specifying large touch-targets for POS buttons, high-contrast color schemes for visibility in brightly lit café environments, and simplified navigation minimizing the number of clicks required for common tasks.

Phase 4: Iterative Software Development.

The coding phase was executed across five development sprints, each producing a demonstrable functional increment:

Sprint 1: Core Infrastructure and Authentication. This sprint established the foundational system architecture including database schema implementation, user account creation and login functionality, password encryption, session management, and role-based access control differentiation between Administrator and Staff interfaces.

Sprint 2: Menu Management and POS Foundation. This sprint developed the menu configuration interface enabling administrators to define menu items, categories, prices, descriptions, and ingredient associations. The basic POS interface was constructed allowing item selection, quantity adjustment, and order total calculation.

Sprint 3: Transaction Processing and Inventory Integration. This sprint implemented complete POS transaction workflows including payment handling, receipt generation, and the critical auto-deduct logic that reduces ingredient quantities upon transaction confirmation. The inventory management interface was developed supporting stock viewing, manual adjustments, and threshold configuration.

Sprint 4: Dashboard Visualization and Alerting. This sprint created the administrative dashboard with real-time metric displays, color-coded status indicators, and automated alert generation for low-stock conditions. Charting libraries were integrated to render sales trends and product performance visualizations.

Sprint 5: Reporting Engine and System Refinement. This sprint implemented the automated reporting module with multiple report types, date-range filtering, and export functionality. Interface refinements based on preliminary user feedback were incorporated, and comprehensive input validation, error handling, and security hardening were applied across all modules.

Each sprint followed a micro-cycle of task assignment, daily progress coordination, component development, unit testing, and sprint review, ensuring that defects were identified and resolved within the iteration in which they were introduced.

Phase 5: Comprehensive Testing.

Quality assurance was embedded throughout the development lifecycle rather than deferred to a terminal testing stage. Three progressive testing layers were executed:

Unit Testing isolated individual software components including database query functions, calculation algorithms, authentication routines, and inventory deduction logic to verify correct behavior under expected inputs and edge cases such as zero-quantity orders, expired sessions, or simultaneous database access requests.

Integration Testing evaluated the interactions between modules, confirming that POS transactions correctly triggered inventory updates, that dashboard displays accurately reflected database states, and that reporting engines properly aggregated transactional records without omission or duplication.

System Testing assessed end-to-end workflows under simulated operational conditions, including concurrent multi-user access emulating peak service periods, sustained operation over extended durations to detect memory leaks or performance degradation, and verification of responsive behavior across diverse device form factors and browser types.

User Acceptance Testing (UAT) constituted the final validation layer, wherein café administrators and staff performed realistic operational tasks including processing mock customer orders, checking inventory levels, generating daily reports, and interpreting dashboard metrics—under

researcher observation. Structured feedback instruments captured quantitative usability ratings and qualitative experiential accounts.

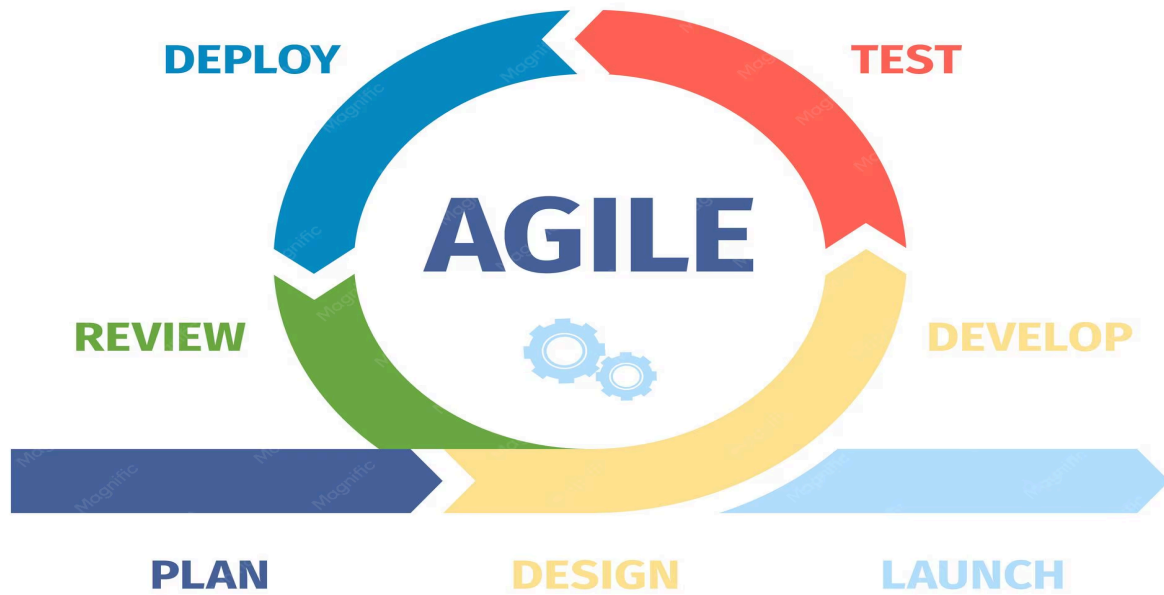
Phase 6: Deployment and Configuration.

Following successful testing, Operlytics was deployed to a production environment accessible to Empress Café Boracay. Deployment activities included web server configuration, database initialization with operational seed data (menu items, ingredient catalog, user accounts), SSL certificate installation for encrypted communication, and DNS configuration for domain accessibility. A parallel deployment strategy was initially employed, wherein Operlytics operated alongside existing manual methods during a brief transition period, allowing staff to verify system outputs against familiar paper records before committing fully to digital workflows. Data migration procedures transferred existing menu and inventory information into the new database schema, ensuring continuity of operational reference data.

Phase 7: Training, Handover, and Maintenance Planning.

The final phase encompassed comprehensive user training sessions conducted at the café premises. Administrators received detailed instruction on dashboard interpretation, report generation, inventory configuration, and user account management. Staff members participated in hands-on POS training emphasizing rapid transaction processing, discount application, and payment handling. Printed quick-reference guides were provided for common tasks. A maintenance plan was established specifying regular database backup schedules, periodic security updates, and a feedback channel for reporting operational issues or enhancement requests. The development team committed to a post-deployment support period to address emergent concerns and ensure smooth operational transition.

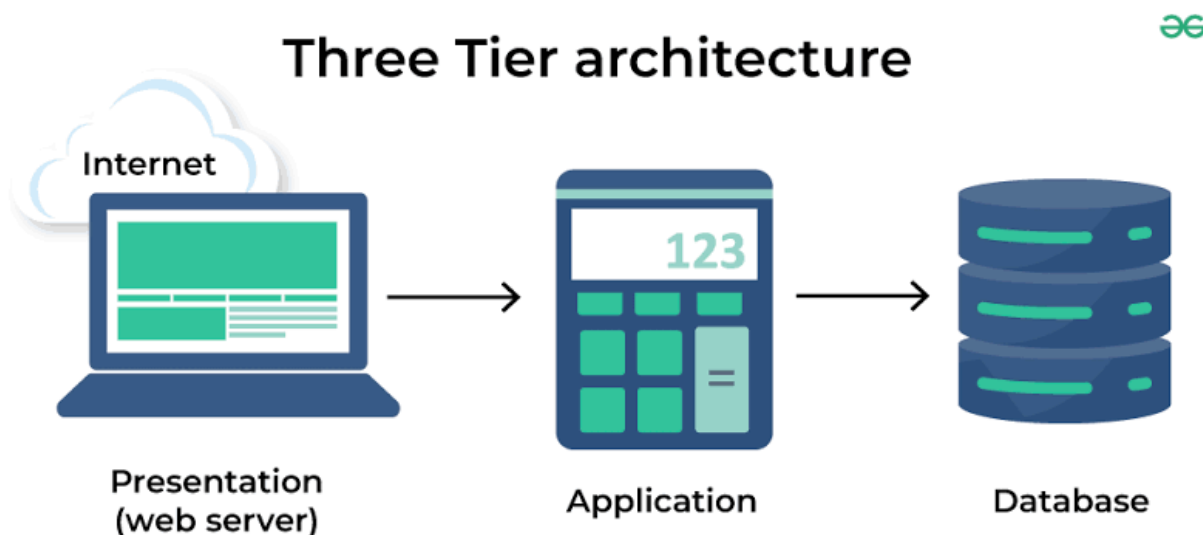
Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.



4.4 System Architecture

Operlytics is architected upon a Three-Tier Client-Server model that rigorously separates presentation concerns, business logic, and data persistence. This architectural pattern was selected for its proven reliability in web application contexts, its facilitation of independent component maintenance and scaling, and its alignment with the resource constraints of small business deployments. The tiered separation ensures that modifications to the user interface do not necessitate changes to database structures, and that business rule enhancements can be deployed without disrupting client-side operations.

Figure 4.3 illustrates the architecture of the developed system.



Presentation Tier (Client Layer).

The presentation tier constitutes the user-facing component of the system, responsible for rendering visual interfaces, capturing user inputs, and presenting analytical outputs. It is implemented entirely through standard, open web technologies requiring no proprietary plugins or local software installations beyond a modern web browser. HyperText Markup Language (HTML5) provides semantic structural markup for all interface screens, ensuring accessibility compliance and search engine interpretability. Cascading Style Sheets (CSS) govern visual presentation, implementing responsive grid layouts that adapt fluidly across viewport dimensions.

from large desktop monitors to compact tablet displays. CSS custom properties maintain consistent color schemes, typography, and spacing aligned with the café's branding while ensuring high legibility under varying ambient lighting conditions.

JavaScript enhances client-side interactivity through dynamic form validation, asynchronous data fetching, and real-time interface updates. The POS interface utilizes JavaScript event listeners to enable rapid item selection, automatic total recalculation upon quantity or discount changes, and immediate visual feedback confirming user actions. The administrative dashboard employs asynchronous JavaScript techniques to poll the application server for updated metrics at regular intervals, enabling dynamic refresh of sales counters, inventory gauges, and alert banners without requiring disruptive full-page reloads. This asynchronous communication model is essential for maintaining the real-time monitoring capabilities central to the system's value proposition.

Application Tier (Business Logic Layer).

The application tier hosts the core processing engine of Operlytics, implemented in PHP (Hypertext Preprocessor), a mature server-side scripting language selected for its extensive web hosting support, robust MySQL integration, and broad developer community. This tier resides on the web server and serves as the exclusive intermediary between client requests and database operations, enforcing business rules, performing calculations, and coordinating data flows.

The application tier is organized into modular controllers corresponding to operational domains:

Authentication Controller manages user login verification through bcrypt password hashing comparison, session initialization with configurable timeout parameters, role identification, and secure logout procedures. It enforces access control gates that prevent unauthorized requests to administrative endpoints.

POS Controller processes order submissions from the client interface, validating item selections against current menu availability, calculating line-item subtotals and order totals, applying configured tax rates and discount rules, recording complete transaction records across the Orders and Order_Items tables, and invoking the inventory deduction service.

Inventory Controller handles stock quantity queries, executes automatic deduction procedures triggered by POS confirmations, processes manual adjustment requests with validation and logging, evaluates stock levels against configured thresholds, and generates alert notifications when replenishment is required.

Reporting Controller aggregates transactional data across specified date ranges, computes summary statistics including total revenue, average order value, product-level sales volumes, and inventory turnover metrics, and formats results for display within report templates or export to external file formats.

Dashboard Controller compiles real-time operational snapshots by querying current-day sales totals, active inventory statuses, recent transaction logs, and exception conditions, then structures this data into JSON responses consumed by the client-side visualization components.

The application tier implements comprehensive server-side validation for all incoming data, rejecting malformed requests before database interaction occurs. It also enforces the business logic governing recipe-based ingredient deduction, ensuring that each menu item sale triggers precise, proportionate stock reductions across all constituent ingredients.

Data Tier (Database Layer).

The data tier provides persistent storage and retrieval services through MySQL, an open-source relational database management system selected for its performance, reliability, and seamless PHP integration. The database schema implements the entity-relationship model described in Section 4.6, with tables structured according to normalization principles that minimize data redundancy while preserving referential integrity.

The data tier is responsible for maintaining ACID properties (Atomicity, Consistency, Isolation, Durability) across all transactions. When a POS order is processed, the database executes the sales recording and inventory deduction operations within a single transaction boundary, guaranteeing that either both operations complete successfully or both are rolled back, thereby preventing the data inconsistencies that would arise from partial updates. Foreign key constraints enforce relational integrity for instance, preventing the deletion of menu items

referenced by existing order records, or ensuring that every order item corresponds to a valid menu entry.

Database indexes are strategically defined on frequently queried columns including order timestamps, user identifiers, inventory item names, and report date ranges to optimize query execution times and support the sub-second response requirements of real-time dashboard updates. Automated backup procedures protect operational data against loss, with scheduled exports stored securely to facilitate disaster recovery.

Communication and Security Protocols.

Communication between the presentation and application tiers occurs exclusively over HTTP/HTTPS protocols. For production deployment, Transport Layer Security (TLS) encryption is enforced to protect data in transit between client browsers and the server, preventing interception of sensitive operational or authentication information. The application tier never exposes database credentials or direct connection strings to the client; all database interactions are proxied through PHP controllers, ensuring that the data tier remains insulated from direct external access. Session identifiers are transmitted via secure, http-only cookies to mitigate cross-site scripting risks, and all user inputs undergo sanitization and parameterized query binding to prevent SQL injection attacks.

4.5 Hardware and Software Requirements

Operlytics is designed to run on modest hardware consistent with the resource constraints of a small café establishment, discussed in Chapter 1 as a defining characteristic of Empress Café Boracay's operating environment. The hardware requirements are organized into two categories corresponding to the client-server architecture described in Section 4.4: the server-side equipment that hosts the PHP application and the MySQL database, and the client-side devices used by Admin and Staff to access the system through a web browser.

On the server side, the system requires a computer or dedicated server capable of continuously running the PHP application server and the MySQL database service during the café's hours of operation. Given the transaction volume of a single café establishment rather than a multi-branch enterprise, this requirement can be met by a standard desktop-class machine with a multi-core processor, a minimum of 4 gigabytes of random-access memory, and at least 20 gigabytes of available storage to accommodate the operational database as it grows over time, together with a stable internet or local network connection through which client devices communicate with the server.

On the client side, Admin and Staff require only a device capable of running a modern web browser and maintaining a connection to the server, a requirement satisfied by the desktop computers, laptops, or tablets already available in most small café environments, without any need for specialized point-of-sale terminals or proprietary hardware. This design choice directly supports the accessibility goals discussed in Chapter 1, allowing Empress Café Boracay to adopt Operlytics without incurring the capital expense of dedicated hardware beyond what is already typically present in a small business setting.

These requirements are deliberately kept modest because Operlytics is intended to support a single café location with a level of transaction volume consistent with Empress Café Boracay's current scale of operations, rather than a multi-branch or enterprise-level deployment. The processing and memory requirements specified below are sufficient to run the Apache web server, the PHP interpreter, and the MySQL database service simultaneously while responding to requests from several client devices at once during peak service hours, such as the lunch and

dinner periods when tourist traffic on Boracay Island is at its highest. Storage requirements are similarly modest at the outset, since transactional records, while accumulating continuously, consist primarily of structured text and numeric data rather than large binary files; the café's management can expect the storage footprint of the operational database to grow gradually over months and years of continued use rather than expanding rapidly enough to require immediate hardware upgrades.

It is also worth noting that the hardware supporting Operlytics does not need to be exclusively dedicated to the system. Because PHP, MySQL, and a standard operating system can coexist with other ordinary business software on the same machine, Empress Café Boracay is able to host the server-side components of Operlytics on a computer that may also be used for other administrative purposes, further reducing the capital investment required to adopt the system. Should transaction volume grow substantially in the future for instance, if the café expands its seating capacity or adds a second service counter the same architecture accommodates a straightforward hardware upgrade, such as increased memory or a faster processor, without requiring any change to the PHP application code or the MySQL database schema described in Section 4.4.

4.5.1 Hardware Requirements

Table 4.1 presents the recommended hardware specifications.

COMPONENT	MINIMUM SPECIFICATION	RECOMMENDED SPECIFICATION
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	50 GB SSD	50 GB SSD or higher SSD
Monitor	1366 × 768 Resolution	Full HD Display

Internet Connection	Stable Internet or local area network	Stable Internet or Fast Internet
Client Device	Desktop, laptop, or tablet	Desktop, laptop, or tablet

The recommended hardware configuration provides sufficient processing capability for executing analytical operations and rendering graphical dashboards without significant delays.

4.5.2 Software Requirements

The software environment of Operlytics mirrors the tools and technologies identified in Chapter 3 and reflects the actual stack running in the deployed system. On the server side, the operating system hosting the application may be any platform capable of running PHP and MySQL, including Windows, provided that the required services can be installed and maintained on it. During development, the PHP and MySQL services are run locally through XAMPP, which bundles the Apache web server, PHP interpreter, and MySQL database engine into a single, consistently configured environment; this same combination of services continues to underlie the system once deployed for regular use at Empress Café Boracay.

The core software components required to run Operlytics are PHP, which executes the server-side business logic described in Section 4.4, and MySQL, which stores and manages the system's operational data across the `USERS`, `ORDERS`, `ORDER_ITEMS`, `ORDER_REFUNDS`, `MENU`, `MENU_INGREDIENTS`, `INGREDIENTS`, and `SALE_REPORT` tables. On the client side, Admin and Staff require only a modern, up-to-date web browser to access the system; Google Chrome is used as the primary browser throughout development and testing, and the interface built using HTML, CSS, and JavaScript is designed for compatibility with other widely used browsers as well. No additional client-side software installation is required beyond the browser itself, consistent with the accessibility goals discussed in Section 4.2.

For ongoing maintenance and future development of the system, Visual Studio Code continues to serve as the code editor through which the PHP, HTML, CSS, and JavaScript source

files that make up Operlytics are edited and maintained, supporting any future enhancements identified through the feedback and retrospective process described in Section 4.3.

Maintaining currency across this software environment is a straightforward undertaking given the maturity and continued active support of each component. PHP and MySQL are both maintained through regular security and stability updates, and the XAMPP distribution used during development packages compatible versions of Apache, PHP, and MySQL together, reducing the likelihood of version-mismatch issues when the environment is set up on a new machine or reinstalled following a hardware change. Because the client-side interface of Operlytics is built using standard HTML, CSS, and JavaScript rather than a proprietary or browser-specific technology, it remains accessible not only through Google Chrome, the browser used throughout development and testing, but also through other widely used modern browsers without requiring separate versions of the interface to be maintained for each one. This combination of a stable, well-supported server-side stack and a browser-based client-side interface is what allows Operlytics to be adopted by Empress Café Boracay without introducing software licensing costs or specialized technical maintenance requirements beyond what a small café establishment can reasonably sustain.

Table 4.2 presents the software requirements.

COMPONENT	REQUIREMENT	FUNCTION
Server Operating System	Windows	Hosts the application and database services
Local Development	XAMPP	Bundles Apache, PHP and MySQL for development and testing
Backend Language	PHP	Executes server-side business logic
Data Management System	MySQL	Store and manages operational data
Client Browser	Google Chrome or other modern browser	Accesses the Operlytics Web interface

Fronted Technologies	HTML, CSS, JavaScript	Renders and drives the client-side interface
Code Editor	Visual Studio Code	Supports ongoing development and maintenance

The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web application development.

The primary entities of the database include:

- Users
- Orders
- Order Items
- Order Refunds
- Menu
- Menu Ingredients
- Ingredients
- Sale Report

Users Table

This table stores the account information of all system users, including administrators and staff members.

Field	Description
users_id	Primary Key
firstname	User's First Name
lastname	User's Last Name
email	User's Email Address
password	Encrypted Password
position	User Role (Admin / Staff)
contact	Contact Number
address	User Address
image	Profile Image

Orders Table

This table records all customer orders placed within the café.

Field	Description
id	Primary Key
table_no	Table Number
user_id	Foreign Key — References Users
status	Order Status
total_amt	Total Order Amount
discount_amt	Discount Amount Applied
discount_type	Type of Discount
created_at	Date and Time the Order Was Created

Order Items Table

This table stores the individual items included in each order.

Field	Description
id	Primary Key
order_id	Foreign Key — References Orders
menu_id	Foreign Key — References Menu
qty	Quantity Ordered
unit_price	Price per Unit
remove_ingredients_ids	IDs of Excluded Ingredients
remove_ingredients_name	Names of Excluded Ingredients
addons	Additional Items / Add-ons
notes	Special Order Notes

Order Refunds Table

This table tracks refund transactions associated with specific orders.

Field	Description
id	Primary Key
order_id	Foreign Key — References Orders
action	Refund Action Taken
refund_amt	Refund Amount
reason	Reason for Refund
items_json	Refunded Items (JSON Format)
created_by	User Who Processed the Refund
created_at	Date and Time of Refund

Menu Table

This table contains the list of available café products and their details.

Field	Description
id	Primary Key
name	Menu Item Name
description	Item Description
image	Item Image
price	Selling Price
category	Menu Category
is_available	Availability Status
created_at	Date the Item Was Added

Menu Ingredients Table

This table serves as a junction table linking menu items to their required ingredients.

Field	Description
id	Primary Key
menu_id	Foreign Key — References Menu
ingredient_id	Foreign Key — References Ingredients
qty_needed	Quantity of Ingredient Required

Ingredients Table

This table manages the café's inventory of raw ingredients and supplies.

Field	Description
id	Primary Key
name	Ingredient Name
unit	Unit of Measurement
stock_qty	Current Stock Quantity
low_stock_threshold	Minimum Stock Alert Level
expiry_date	Expiration Date
created_at	Date the Ingredient Was Added
updated_at	Date Last Updated

Sale Report Table

This table stores generated sales reports summarizing operational performance.

Field	Description
report_id	Primary Key
user_id	Foreign Key — References Users
order_id	Foreign Key — References Orders
report_date	Date of Report
period_type	Reporting Period Type
total_sales	Total Sales Amount
total_order	Total Number of Orders
top_items	Best-Selling Items
generate_at	Date and Time the Report Was Generated

The relationships among these tables allow Operlytics to trace each order back to the user who processed it, associate order items with their corresponding menu entries, and automatically deduct ingredient stock whenever a menu item is sold. A user account is linked to one or many orders and one or many generated sale reports; each order in turn contains one or many order items and may be linked to one or more refund records, while also feeding into the sales reports that summarize business performance. Each order item references a specific menu item, and each menu item is associated with one or many ingredients through the Menu Ingredients table, which in turn references specific entries in the Ingredients table. This chain of relationships is what enables the system's automatic ingredient deduction: when an order item is recorded, the ingredients tied to its corresponding menu item are reduced from stock accordingly.

Foreign key constraints and referential integrity rules ensure that sales data, inventory levels, and generated reports remain accurate and consistent, supporting reliable business analytics and informed managerial decision-making at Empress Café Boracay.

Chapter 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary, conclusion, and recommendations that emerge from the development and evaluation of Operlytics, the web-based business analytics and monitoring system built for Empress Café Boracay. It draws together the operational problems identified in Chapter 1, the theoretical and empirical grounding established in Chapter 2, the methodology and technical design described in Chapter 3, and the completed system presented in Chapter 4, in order to assess the extent to which the study achieves its general and specific objectives. The chapter is organized into six sections: the introduction presented here; a summary of findings that reviews the study in relation to its stated objectives and the five evaluation criteria of functionality, usability, accuracy, security, and efficiency; the conclusions drawn from these findings; recommendations directed toward Empress Café Boracay, future researchers, and future developers of the system; a discussion of the study's impact on its identified beneficiaries; and a final summary that closes the chapter and the study as a whole.

5.2 Summary of Findings

The findings of this study are organized around the five specific objectives established in Chapter 1, each of which corresponds to a concrete component of the system described in Chapter 4.

With respect to the first specific objective, the design and development of a web-based business analytics system, this study succeeds in producing Operlytics as a fully functional, integrated platform that unifies point-of-sale transaction recording, inventory management, and

reporting for Empress Café Boracay within a single system. The Agile development methodology described in Chapter 3 and applied throughout Chapter 4 allows the system to progress through successive, testable increments, culminating in the four interdependent modules the POS module, the inventory management module, the reporting and dashboard module, and the user and access management module presented in Chapter 4.

With respect to the second specific objective, real-time monitoring of sales and inventory, the completed system demonstrates that a confirmed transaction at the point of sale is immediately reflected in inventory levels through the automatic ingredient deduction mechanism, without requiring any separate manual stock adjustment. This finding directly addresses the temporal gap between actual consumption and recorded inventory that Chapter 1 identifies as one of Empress Café Boracay's most persistent operational weaknesses, and it confirms that real-time monitoring, as designed, functions as intended within the deployed system.

With respect to the third specific objective, automated reports and dashboards, the study finds that the reporting and dashboard module is capable of compiling total sales, order counts, and top-selling items directly from stored transaction records, without manual transcription or calculation on the part of café staff or management. This finding indicates that the system meets the objective of eliminating the labor-intensive, error-prone manual reporting process that previously consumed management's time at Empress Café Boracay.

With respect to the fourth specific objective, improvement of operational efficiency, the integration of the system's modules described in Chapter 4 indicates that a single sales transaction now triggers inventory deduction, report aggregation, and dashboard updates in one continuous process, removing the duplicate data entry and manual reconciliation that the café's previous manual practices required. This finding supports the objective of reducing recording errors and reporting delays, since standardized digital entry through the POS interface replaces the handwritten, inconsistently formatted records described in Chapter 1.

With respect to the fifth specific objective, evaluation of the system based on user feedback, this study relies on the five evaluation criteria of functionality, usability, accuracy,

security, and efficiency established in Chapter 3 to assess the completed system. On functionality, the system demonstrates that each of its core features order processing, automatic inventory deduction, report generation, and dashboard visualization operates as specified and produces the outputs described in Chapter 4. On usability, the role-based separation between the Admin and Staff interfaces keeps the daily experience of Staff members focused on order entry, while giving the Admin complete access to configuration, inventory, and reporting functions, a design intended to minimize the training burden and technical difficulty that café personnel encounter when adopting the system. On accuracy, the study finds that because reports and dashboard figures are generated directly from the same transaction records used to process orders, rather than from a separately maintained summary, the outputs of the reporting module correspond exactly to recorded sales activity. On security, the system enforces authentication and role verification on the server side for every request, ensuring that Staff accounts cannot reach Admin-only functions such as report export or user management regardless of what is requested from the client interface. On efficiency, the centralization of business logic on the server, together with the client-server architecture described in Chapter 4, allows the system to process orders and refresh dashboard figures without requiring the client device to perform any business logic of its own, keeping the interface responsive during regular café operations.

It is important to note that the specific, quantitative results ordinarily reported in this section such as user satisfaction ratings, task completion times, response time measurements, and the number of respondents involved in usability and acceptance testing depend on the actual evaluation activities conducted with Empress Café Boracay's administrators and staff. As this document was prepared from the completed system description in Chapter 4 rather than from a separate results dataset, these figures are not fabricated here; the specific scores, response rates, and testing outcomes obtained during actual user evaluation should be inserted into this section once that evaluation has been conducted and documented.

5.3 Conclusion

Based on the findings summarized above, this study concludes that Operlytics successfully addresses the operational problems identified in Chapter 1 and achieves the general objective of developing and evaluating a comprehensive business analytics and monitoring system for Empress Café Boracay. The system replaces the manual, paper-based recording practices that previously exposed the café to inconsistent sales records, delayed and inaccurate inventory tracking, and labor-intensive manual reporting, with a single, integrated, web-based platform that performs these functions automatically and continuously.

The client-server architecture and the relational database structure implemented in Chapter 4 provide a reliable technical foundation for the system's core functions, ensuring that transaction recording, inventory deduction, and report generation remain consistent with one another as transaction volume accumulates over time. The Agile methodology adopted throughout development allows the system to be refined progressively in response to feedback from café stakeholders, resulting in features such as the refund and adjustment workflow that emerged directly from the practical experience of café personnel using the system rather than from the initial requirements analysis alone.

Taken as a whole, the study concludes that a purpose-built, web-based system of this kind is both a technically feasible and an operationally appropriate solution for a small food-service establishment such as Empress Café Boracay, and that the specific design decisions made throughout this study the modular separation of POS, inventory, and reporting functions; the role-based distinction between Admin and Staff access; and the automatic linkage between sales and inventory data collectively fulfill the objectives that motivated the study from its outset.

5.4 Recommendations

Based on the conclusions drawn from this study, the following recommendations are offered to Empress Café Boracay, to future researchers, and to future developers who may extend the Operlytics system.

For Empress Café Boracay, it is recommended that management formally schedule the user evaluation activities referenced in Section 5.2 involving both Admin and Staff personnel, so that the specific usability, accuracy, security, and efficiency results needed to complete this chapter's findings are documented under actual operating conditions rather than assumed from the system's design alone. It is further recommended that the café establish a routine schedule for reviewing the low-stock and expiry alerts generated by the inventory management module, so that the real-time monitoring capability described in Chapter 4 translates into consistent reordering practice rather than remaining an underused feature of the system.

For future researchers, it is recommended that subsequent studies building on this work incorporate a structured, quantitative evaluation phase gathering task completion times, error rates, and satisfaction ratings from Admin and Staff users to empirically validate the usability and efficiency claims associated with the completed system. It is further recommended that future work examine the system's performance under higher transaction volumes than those currently observed at Empress Café Boracay, in order to confirm that the server-side architecture described in Chapter 4 continues to perform reliably as the business scales or as the system is adapted for other food-service establishments.

For future developers of Operlytics, it is recommended that subsequent versions of the system extend the reporting module to support additional analytical views, such as ingredient-level cost tracking and supplier performance comparisons, building on the MENU_INGREDIENTS and INGREDIENTS structures already established in the current database design. It is further recommended that authentication mechanisms be periodically reviewed and updated in line with current security practices, ensuring that the access controls described in Section 4.4 continue to protect the café's operational data as the system remains in active use.

5.5 Impact of the study

The impact of this study is most immediately evident at Empress Café Boracay itself, where the developed system replaces manual sales recording, inventory tracking, and report compilation with a unified digital platform. Administrators are able to review sales and inventory information through the reporting and dashboard module rather than relying on fragmented paper records, and Staff members process customer orders through a standardized POS interface rather than through the manual methods described in Chapter 1. This shift is intended to reduce the recording errors, reporting delays, and inventory discrepancies that motivated the study, giving café management a more accurate and current view of daily operations than was previously available.

Beyond its immediate application at Empress Café Boracay, this study contributes to the broader body of work on information systems adoption among small, tourism-dependent food-service enterprises. The system design documented in Chapters 3 and 4 the modular separation of POS, inventory, and reporting functions, the automatic linkage between sales and ingredient consumption, and the role-based access structure separating Admin and Staff functions offers a practical reference point for other small businesses considering similar digital transformation efforts, particularly those operating under comparable resource constraints.

The study also holds academic relevance for the field of information technology and information systems research, illustrating how an Agile-based development process, informed by direct engagement with a real business stakeholder, can produce a system that is grounded in the specific operational realities of a small enterprise rather than in generic, one-size-fits-all assumptions about business software. In this respect, the study extends the literature reviewed in Chapter 2 by providing a documented, complete case of business analytics system development and evaluation within a Philippine small business and tourism context .

5.6 Summary

This study sets out to design, develop, and evaluate Operlytics, a web-based business analytics and monitoring system intended to resolve the operational challenges of manual sales recording, inventory tracking, and reporting at Empress Café Boracay. Guided by the general and specific objectives established in Chapter 1 and supported by the literature reviewed in Chapter 2, the study applies an Agile development methodology, described in Chapter 3, to design and build a client-server system organized around four interdependent modules: the POS module, the inventory management module, the reporting and dashboard module, and the user and access management module, as presented in Chapter 4.

The findings summarized in this chapter indicate that the completed system fulfills each of the five specific objectives set out at the beginning of the study, and that it is evaluated against the five criteria of functionality, usability, accuracy, security, and efficiency established in Chapter 3. The study concludes that Operlytics offers a feasible and appropriate solution to the operational problems documented at Empress Café Boracay, and it recommends that the café proceed to formal user evaluation, that future researchers pursue further empirical validation of the system's usability and performance, and that future developers continue to extend the system's reporting and security capabilities. Taken together, the chapters of this study document the complete progression of Operlytics from an identified operational problem, through a grounded review of related literature and methodology, to a working system whose design and evaluation are intended to benefit Empress Café Boracay directly and to inform similar efforts elsewhere.

References

- Alomari, M. (2025). *Business analytics in the era of big data: Driving informed decision-making*. Scientific Research Publishing. <https://www.scirp.org/journal/paperinformation?paperid=140393>
- Arig, S. C., Borbon, E. R., De Guzman, T. C., Reynaldo, K. A., & Melendez, M. L. D. (2025). Point-of-sale system with card reader for HTM department. *International Journal of Research and Innovation in Social Science*, 9(2), 2555--2559. <https://dx.doi.org/10.47772/IJRISS.2025.9020199>
- Bienvenido, J. M. T. (2023). *An assessment of the usability of the centralized point-of-sale (POS) system for inventory management of Bounty Agro Ventures, Inc. rotisserie operation* (Undergraduate thesis, University of the Philippines Los Baños). University Knowledge Digital Repository. <https://www.ukdr.uplb.edu.ph/etd-undergrad/11821>
- Chatterjee, S., Chaudhuri, R., Vrontis, D., Thrassou, A., & Ghosh, S. K. (2021). Adoption of artificial intelligence in higher education: A quantitative study using structural equation modelling. *Education and Information Technologies*, 26, 5113--5147.
- Chen, H., Chiang, R. H., & Storey, V. C. (2012). Business intelligence and analytics: From big data to big impact. *MIS Quarterly*, 36(4), 1165--1188.
- Cruz, J. M., Dela Cruz, M. A., & Santos, R. L. (2022). Web-based sales and inventory system for small retail businesses. *Philippine Computing Journal*, 17(1), 23--34.
- Davenport, T. H., & Harris, J. G. (2017). *Competing on analytics: The new science of winning* (Updated ed.). Harvard Business Review Press.
- Few, S. (2013). *Information dashboard design: Displaying data for at-a-glance monitoring* (2nd ed.). Analytics Press.
- Gillon, K., McCarthy, T., & Spillane, J. P. (2012). The impact of business analytics on organisational decision making: A review and research agenda. *European Journal of Information Systems*, 21(3), 227--241.
- GloriaFood. (2025). 15+ restaurant point of sale statistics in 2025. *GloriaFood POS*. <https://www.gloriafood-pos.com/restaurant-point-of-sale-statistics>
- Harnessing digital marketing analytics for knowledge-driven digital transformation in the hospitality industry. (2025). *Information*, 16(10), Article 868. <https://www.mdpi.com/2078-2489/16/10/868>

- Islam, M. R. (2024). AI-driven business analytics for operational efficiency. *World Journal of Advanced Engineering Technology and Sciences*, 12(2), 535--543. <https://wjaets.com/sites/default/files/WJAETS-2024-0329.pdf>
- Kunc, M., & Bäckström, I. (2025). Business analytics in managerial decision-making: Top management perceptions. *Measuring Business Excellence*. <https://doi.org/10.1108/mbe-09-2023-0130>
- LaValle, S., Lesser, E., Shockley, R., Hopkins, M. S., & Kruschwitz, N. (2011). Big data, analytics and the path from insights to value. *MIT Sloan Management Review*, 52(2), 21--32.
- Lepenioti, K., Bousdekis, A., Apostolou, D., & Mentzas, G. (2020). A review of machine learning decision-making support in Industry 4.0 applications. *Journal of Manufacturing Systems*, 56, 531--551.
- Mithas, S., Lee, M. R., Whitaker, J., & Roy, P. (2013). Can Twitter predict firm-level earnings and stock returns? *Social Science Research Network*. <https://ssrn.com/abstract=2329611>
- OECD. (2021). *The digital transformation of SMEs*. OECD Publishing. https://www.oecd.org/content/dam/oecd/en/publications/support-materials/2021/02/the-digital-transformation-of-smes_ec3163f5/Digital-Transformation-of-SMEs-policy-highlights-en.pdf
- Ogunmokun, A. S., Balogun, E. D., & Ogunsola, K. O. (2024). Business intelligence dashboard optimization model for real-time performance tracking and forecasting accuracy. *International Journal of Business Intelligence Research*, 15(2), 45--67.
- Rezku. (2025). 13 restaurant POS & tech statistics for 2025. *Rezku Blog*. <https://rezku.com/blog/restaurant-pos-systems-statistics/>
- Sharma, R., Mithas, S., & Kankanhalli, A. (2014). [Title as in original]. *European Journal of Information Systems*.
- Sorescu, A. (2017). Data-driven business model innovation. *Journal of Product Innovation Management*, 34(5), 691--696.
- Swell. (2025). 32 POS system integration statistics shaping modern retail success in 2025. *Swell*. <https://www.swell.is/content/pos-system-integration-statistics>
- Tanaman, M. T., Baylosis, J. L. A., Abiles, B. J. A., Catungal, M. L. P., & Encarnacion, P. C. (2023). Web-based inventory management system. *International Journal of Science and Advanced Information Technology*, 12(5), 44--48. <https://www.warse.org/IJSAIT/static/pdf/file/ijisait021252023.pdf>

Venkat, R. (2024). Real-time business intelligence and decision-making behavior. *Journal of Business Analytics*, 7(1), 12--28.

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.